

Hands-on Labs

Create a Bedrock Custom Model with Fine-tuning and inference

Test this lab: if you're a machine learning engineer, data scientist, or AI enthusiast, this lab will empower you with the knowledge and skills needed to leverage AWS Bedrock and unlock the full potential of Foundation Models.

Real-Time Scenario:

You are a Machine Learning Engineer working for a company that runs an online gardening and plant-care platform. Your task is to fine-tune a Foundation Model (FM) using Amazon Bedrock so it can accurately answer domain-specific gardening questions - topics like plant care, composting, pest control, and soil management - that the base model doesn't handle well out of the box. After fine-tuning the model on your curated gardening dataset, you will deploy it for on-demand inference and test it in the Bedrock Playground to confirm it generates relevant, accurate gardening advice for your users.

In addition to the primary gardening scenario, this lab also includes **six optional industry scenarios** - covering Support Ticket Triage, IT/DevOps Helpdesk, Pharmacovigilance, Patient Message Triage, Banking, and E-Commerce - so you can explore how the same fine-tuning workflow applies across a wide range of real-world business domains.

Description:

In this hands-on lab, you will learn how to fine-tune a Foundation Model (FM) in Amazon Bedrock to cater to a specific business use case. Amazon Bedrock provides you with a seamless platform to work with various Foundation Models and customize them based on your own data.

You will start with a primary use case - a gardening knowledge assistant - and walk through the complete workflow: uploading training data to S3, running a supervised fine-tuning job, deploying the model for on-demand inference, and validating it in the Bedrock Playground.

Once the core workflow is clear, the lab expands into **six optional industry scenarios** that many of our learners have specifically requested. Each scenario follows the exact same process but applies it to a different domain - from IT helpdesk automation to pharmaceutical adverse-event triage - so you can see how fine-tuning translates into practical value across industries. By the end of this lab, you will have a solid, repeatable framework for customizing Foundation Models for any specialized business application.

K21 Academy

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1. INTRODUCTION

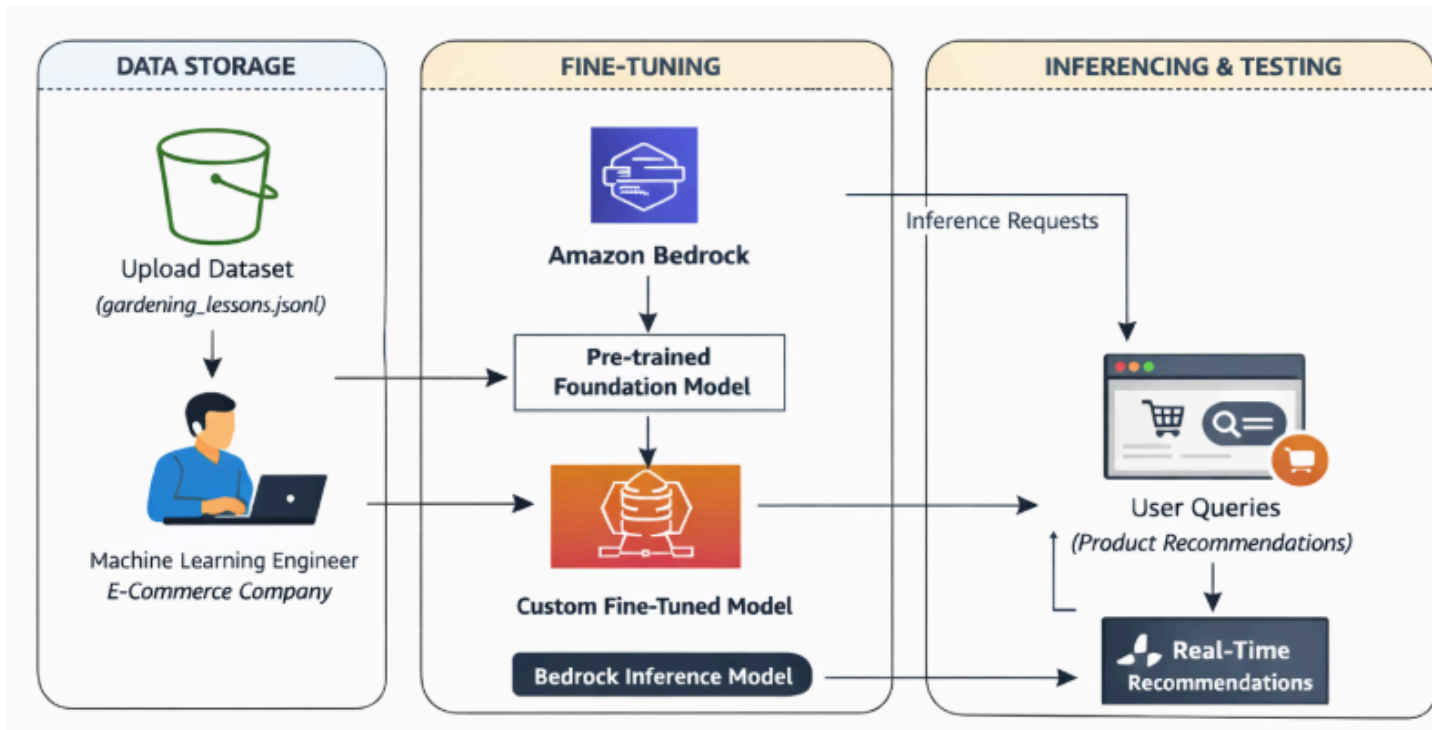
In today's AI-driven landscape, **Foundation Models (FMs)** enable organizations to build intelligent applications quickly without training models from scratch. While these pre-trained models are highly capable, many real-world business use cases require domain-specific knowledge to deliver accurate and relevant responses. This is where **fine-tuning** and **inference** become essential.

Amazon Bedrock provides a fully managed platform for accessing, customizing, and deploying Foundation Models using your own datasets, making it easier to create AI applications tailored to specific industries and use cases.

Fine-tuning allows you to adapt a pre-trained Foundation Model using a curated dataset so it performs better on specialized tasks. **Inference** is the process of using the fine-tuned model to generate accurate responses for new user queries in real time.

In this lab, you will fine-tune an Amazon Bedrock Foundation Model using a curated **gardening knowledge dataset** containing information about plant care, composting, pest control, and soil management. Once the model has been customized, you will deploy it for **on-demand inference** and validate its performance using the **Amazon Bedrock Playground** to ensure it provides accurate, domain-specific gardening advice.

By the end of this lab, you will understand how to customize Foundation Models for specialized business applications, deploy them for inference, and evaluate their performance in a production-ready environment. These skills will help you build intelligent AI assistants that deliver accurate, context-aware responses for industry-specific use cases using Amazon Bedrock.



This activity guide covers steps for:

1. Create an S3 Bucket to Store Data
2. Create and Customize a Foundation Model (FM) in Amazon Bedrock
3. Access and Test the Customized Foundation Model
4. Delete Resources (Clean Up)

2. OVERVIEW & KEY CONCEPTS

In this section, we'll cover key concepts used in this guide, including an overview of what's covered in this guide.

a. What is Amazon Bedrock?

Amazon Bedrock is a fully managed service that provides access to high-performance foundation models (FMs) from leading AI companies such as AI21 Labs, Anthropic, Cohere, Meta, Stability AI, and Amazon through a single API.

It offers extensive features for building generative AI applications, making development easier while ensuring privacy and security with Amazon Bedrock, you can experiment with various top FMs, customize them privately with your data using methods like fine-tuning and retrieval augmented generation (RAG), and create managed agents for complex business tasks such as booking travel, processing insurance claims, creating ad campaigns, and managing inventory—all without needing to write any code.

Being serverless, Amazon Bedrock eliminates the need to manage infrastructure, allowing you to securely integrate and deploy generative AI capabilities into your applications using familiar AWS services.

b. Amazon S3 (Simple Storage Service)

Amazon S3 is a scalable object storage service designed to store and retrieve large amounts of data.

In the Bloodbank project, S3 is used to store files such as images, documents, and backups.

The service provides durability and scalability, which makes it ideal for storing static assets like donor photographs, medical records, and reports, ensuring they are readily available and secure.

The S3 service also integrates with other AWS tools like Lambda and CloudFront, enabling automation of tasks such as data processing or distributing content globally via CDN.

Its pay-as-you-go pricing ensures cost-effectiveness even as the amount of data grows.

c. **Bedrock Custom Model**

A **custom model in Amazon Bedrock** is a machine learning model that is built and fine-tuned from a pre-trained foundation model (FM) to meet the specific needs of a given use case.

By uploading your own data, you can fine-tune the foundation model to enhance its accuracy and relevance for your specific task. After fine-tuning, the model can be tested and deployed for real-time inference or batch processing.

This allows businesses to create specialized AI solutions tailored to their unique requirements, while leveraging Amazon Bedrock's scalable infrastructure.

3. DOCUMENTATION LINKS

1. Getting Started with AWS Amazon Bedrock (Enable Foundation Model)

<https://docs.aws.amazon.com/bedrock/latest/userguide/getting-started.html>

2. BedRock Supported Regions

<https://docs.aws.amazon.com/bedrock/latest/userguide/bedrock-regions.html>

3. Amazon BedRock Pricing

<https://aws.amazon.com/bedrock/pricing/>

(**Note:** There is no charge to perform this lab.)

Basic Concepts or Key Definitions: Foundation model (FM), Base model, Model inference, Prompt, Token, Model parameters, Inference parameters, Playground, Embedding, etc...

<https://docs.aws.amazon.com/bedrock/latest/userguide/key-definitions.html>

4. PREREQUISITE

1. An AWS Account: If you don't have an AWS Account, then please check Activity **Guide: Create AWS Account** [Here](#).

AWS for Beginners 21%

Presentation: Introduction to AWS Cloud.

Getting Started on AWS Cloud

Module 1: AWS Services Overview

Module 2: Hands-On Labs

Lab 1: Create an AWS Cloud Account ✓

Upgrading To Paid Plan To Use All Servi...

AWS FREE TIER MAJOR UPDATE

AWS Pricing Fundamentals

Lab 2: Create CloudWatch Billing ... ✓

Lab 3: Managing AWS Costs, AWS Budg...

How to Check AWS Bill and Reach Out t...

How to Troubleshoot Billing Issues

Lab 1: Create an AWS Cloud Account

1. Step-by-Step Guide: [Create AWS Account & Access Console](#)

2. Related / References

- <https://aws.amazon.com/free/>

5. COST TO PERFORM THIS LAB

The **estimated cost** for this lab is **\$0.01**, as long as usage stays within the Free Tier limits. Please note that if you run more prompts, you may incur higher charges.

 **Important Note:** However, don't worry; any charges will be deducted from the **free credits** you received when you created your AWS account. As long as you have remaining credits, you won't need to pay out of pocket for these charges.

AWS Free Tier Credits Breakdown

- **\$100 Available Immediately:** When you sign up for an AWS Free Tier account, you receive \$100 in credits that are available for use right away.
- **\$100 After Completing Specific Activities:** After fulfilling certain conditions, such as completing AWS labs or certain tasks, you'll receive an additional \$100 in credits.

Billing and Cost Management > Bills

Filter by service name or region name

Description	Usage Quantity	Amount in USD
\$0.0005 per 1K token for Nova2.Lite-input-tokens-cross-region-global in US East (N. Virginia)	2.282 1K tokens	USD 0.00
Amazon Bedrock USE1-Nova2.Lite-output-tokens-cross-region-global		USD 0.01
\$0.0025 per 1K token for Nova2.Lite-output-tokens-cross-region-global in US East (N. Virginia)	2.405 1K tokens	USD 0.01
Amazon Bedrock USE1-NovaCanvas-T2I-2048-Standard		USD 0.06
0.06 USD of Nova Canvas in us-east-1	1 image	USD 0.06
Amazon Bedrock USE1-NovaLite-input-tokens		USD 0.00
\$0.00006 per 1K tokens for Nova Lite-input-tokens in us-east-1 (N. Virginia)	1.895 1K tokens	USD 0.00
Amazon Bedrock USE1-NovaLite-output-tokens		USD 0.00
\$0.00024 per 1K tokens for Nova Lite-output-tokens in us-east-1 (N. Virginia)	0.815 1K tokens	USD 0.00
Amazon Bedrock USE1-NovaMicro-Customization-Training		USD 0.01
\$0.001 per 1K tokens for Nova Micro-Customization-Training in US East (N. Virginia)	12.804 1K tokens	USD 0.01
Amazon Bedrock USE1-NovaMicro-input-tokens-custom-model		USD 0.00
\$0.000035 per 1K tokens for USE1-NovaMicro-input-tokens-custom-model in US East (N. Virginia)	0.007 1K tokens	USD 0.00
Amazon Bedrock USE1-NovaMicro-output-tokens-custom-model		USD 0.00
\$0.00014 per 1K tokens for USE1-NovaMicro-output-tokens-custom-model in US East (N. Virginia)	0.265 1K tokens	USD 0.00
Amazon Bedrock USE1-TitanEmbeddingsG1-Text-input-tokens		USD 0.00
\$0.0001 per 1K tokens for TitanEmbeddingsG1-Text-input-tokens in US East (N. Virginia)	2.95 1K tokens	USD 0.00

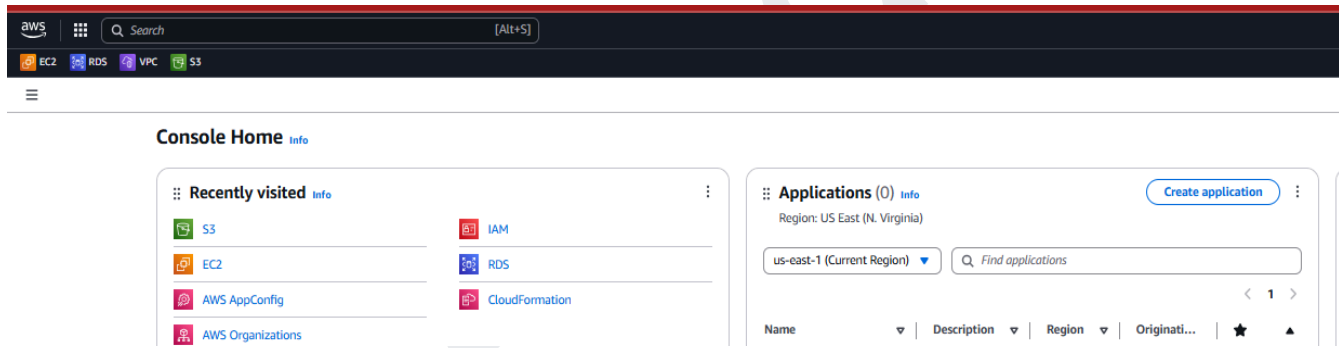
6. CREATE AN S3 BUCKET AND UPLOAD THE CODE

Creating an **S3 bucket** is a crucial step in this lab because it provides a secure and scalable location for storing your dataset, which is essential for fine-tuning a Foundation Model in Amazon Bedrock.

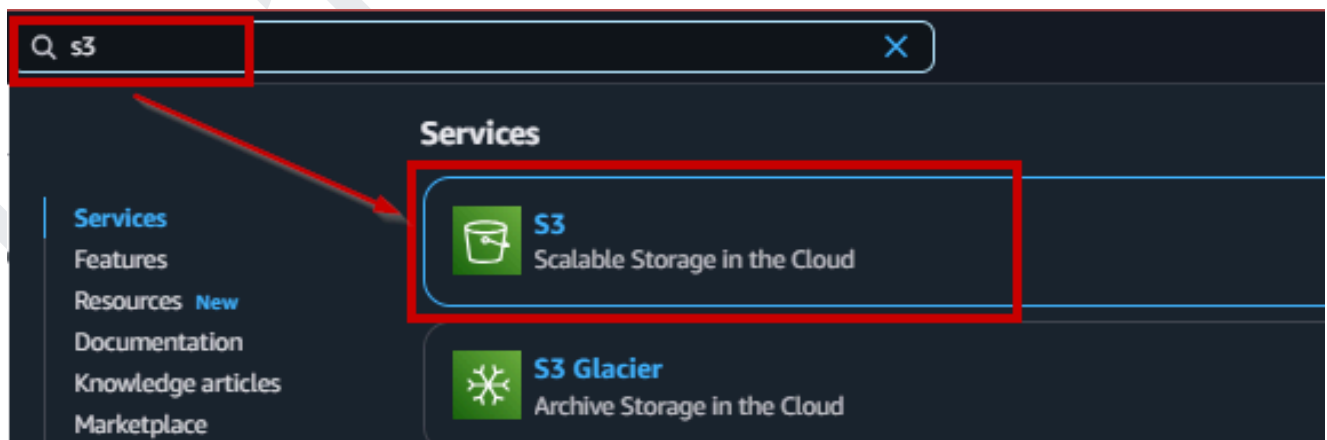
The S3 bucket allows for easy access to your data, ensuring it can be integrated seamlessly with Amazon Bedrock for model training. It also offers flexibility in handling different data types (e.g., text, images), and ensures the data remains secure and well-managed.

This step is foundational for preparing the data needed to train and test your customized model.

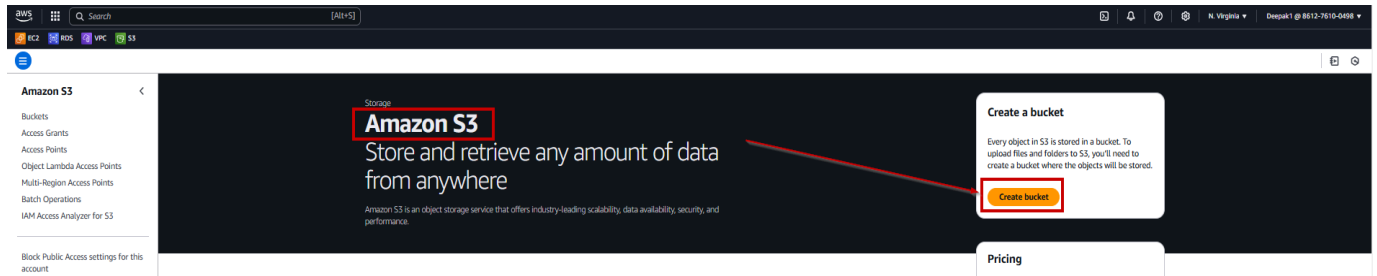
1. Open your AWS Console.



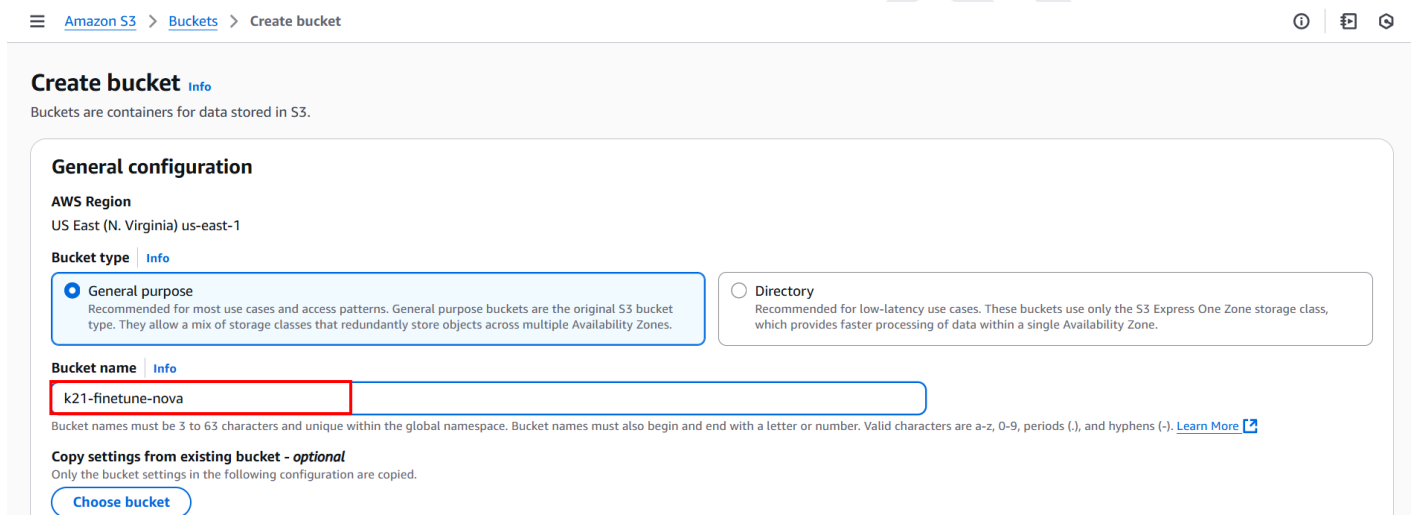
2. Click on the search bar, type **S3**, and click on **S3**.



3. Click on the Create **Bucket** button.



4. In the Name and Region screen, provide a unique **Bucket Name** (k21-finetune-nova) to your bucket and select a **Region** of your choice, then click **Next**.



Note: An Amazon S3 bucket name is **globally unique**, and the namespace is shared by all AWS accounts. This means that after a bucket is created, the name of that bucket cannot be used by another AWS account in any AWS Region until the bucket is deleted.

Note: Now you shall see in the S3 console it shows '**Global**', which means S3 is not region-specific. However, you must then ask why we are selecting a region in the above step. S3 is a Global service, which means that you can access any bucket from any region, but if you want that access to be fast, you should have the bucket in the same region. That is, you shall choose any region for your bucket that is geographically close to you to optimize latency and minimize costs.

5. Now, under object ownership, choose the **ACLs enabled** option and scroll down to the page.

Object Ownership [Info](#)

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

☐ **ACLs disabled (recommended)**
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☒ **ACLs enabled**
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

⚠ We recommend disabling ACLs, unless you need to control access for each object individually or to have the object writer own the data they upload. Using a bucket policy instead of ACLs to share data with users outside of your account simplifies permissions management and auditing.

Object Ownership

☒ **Bucket owner preferred**
If new objects written to this bucket specify the bucket-owner-full-control canned ACL, they are owned by the bucket owner. Otherwise, they are owned by the object writer.

☐ **Object writer**
The object writer remains the object owner.

ℹ If you want to enforce object ownership for new objects only, your bucket policy must specify that the bucket-owner-full-control canned ACL is required for object uploads. [Learn more](#)

6. Here, we allow public access to our bucket and tick the acknowledge option.

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☐ **Block all public access**
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☐ **Block public access to buckets and objects granted through new access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

☐ **Block public access to buckets and objects granted through any access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.

☐ **Block public access to buckets and objects granted through new public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☐ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

⚠ Turning off block all public access might result in this bucket and the objects within becoming public
AWS recommends that you turn on block all public access, unless public access is required for specific and verified use cases such as static website hosting.

☒ **I acknowledge that the current settings might result in this bucket and the objects within becoming public.**

7. Scroll down, let everything rest by default, and click on **Create Bucket**.

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

- ☒ Disable
☐ Enable

Tags - optional (0)

You can use bucket tags to track storage costs and organize buckets. [Learn more](#)

No tags associated with this bucket.

[Add tag](#)

Default encryption [Info](#)

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type [Info](#)

- ☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)
☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)
☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)
Secure your objects with two separate layers of encryption. For details on pricing, see DSSE-KMS pricing on the Storage tab of the [Amazon S3 pricing page](#).

Bucket Key

Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

- ☐ Disable
☒ Enable

► **Advanced settings**

① After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.

[Cancel](#)

[Create bucket](#)

8. In this screen, you shall see that the bucket has been created and is available for use.

Amazon S3 > Buckets

Successfully created bucket "k21-finetune-nova"
 To upload files and folders, or to configure additional bucket settings, choose [View details](#).

General purpose buckets All AWS Regions Directory buckets

General purpose buckets (46) [Info](#)

Buckets are containers for data stored in S3.

1 match

Name	AWS Region	Creation date
☐ k21-finetune-nova	US East (N. Virginia) us-east-1	September 2, 2025, 14:45:48 (UTC+05:30)

9. Here, we will create a Folder. You can click on the **bucket** → create folder, named as output.

Amazon S3 > Buckets > k21-finetune-nova

k21-finetune-nova [Info](#)

[Objects](#) | [Metadata](#) | [Properties](#) | [Permissions](#) | [Metrics](#) | [Management](#) | [Access Points](#)

Objects (0)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Name	Type	Last modified	Size	Storage class
No objects				

You don't have any objects in this bucket.

[Upload](#)

10. Now, provide the Name of the folder as output and click on create folder.

Create folder [Info](#)

Use folders to group objects in buckets. When you create a folder, S3 creates an object using the name that you specify followed by a slash (/). This object then appears as folder on the console. [Learn more](#)

Your bucket policy might block folder creation

If your bucket policy prevents uploading objects without specific tags, metadata, or access control list (ACL) grantees, you will not be able to create a folder using this configuration. Instead, you can use the [upload configuration](#) to upload an empty folder and specify the appropriate settings.

Folder

Folder name

output

Folder names can't contain "/". [See rules for naming](#)

Server-side encryption [Info](#)

Server-side encryption protects data at rest.

 The following encryption settings apply only to the folder object and not to sub-folder objects.

Server-side encryption

☒ Don't specify an encryption key

The bucket settings for default encryption are used to encrypt the folder object when storing it in Amazon S3.

☐ Specify an encryption key

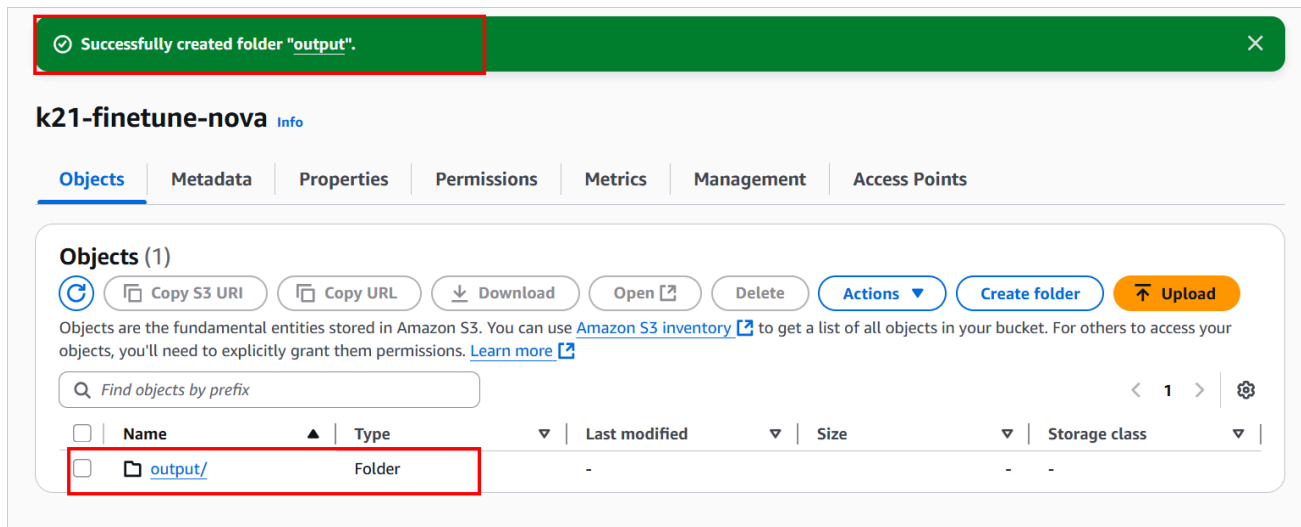
The specified encryption key is used to encrypt the folder object before storing it in Amazon S3.

 If your bucket policy requires objects to be encrypted with a specific encryption key, you must specify the same encryption key when you create a folder. Otherwise, folder creation will fail.

[Cancel](#)

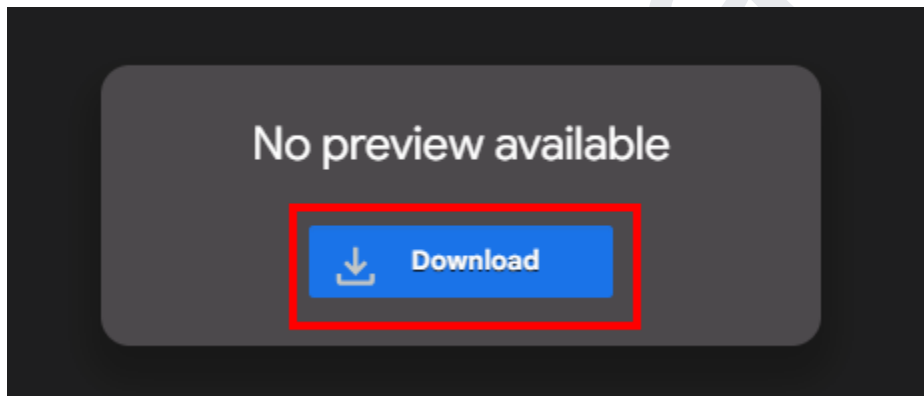
Create folder

11. We have successfully created a folder inside our bucket to store output.



12. Now let's upload a file "gardening_lessons.jsonl" inside our bucket. Download the file here:

[gardening_lessons.jsonl](#)



13. Now go to the bucket and click on Upload.

k21-finetune-nova [Info](#)

[Objects](#) | [Metadata](#) | [Properties](#) | [Permissions](#) | [Metrics](#) | [Management](#) | [Access Points](#)

Objects (1)

[Refresh](#) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

☐	Name	Type	Last modified	Size	Storage class
☐	output/	Folder	-	-	-

14. Now, we will upload the object gardening lesson to our bucket. Click on Add file, select the same file we downloaded earlier, and select upload.

[Amazon S3](#) > [Buckets](#) > [k21-finetune-nova](#) > Upload

Files and folders (1 total, 27.5 KB)

All files and folders in this table will be uploaded.

☐	Name	Folder	Type	Size
☐	gardening_lessons.json	-	-	27.5 KB

Destination [Info](#)

Destination
[s3://k21-finetune-nova](#)

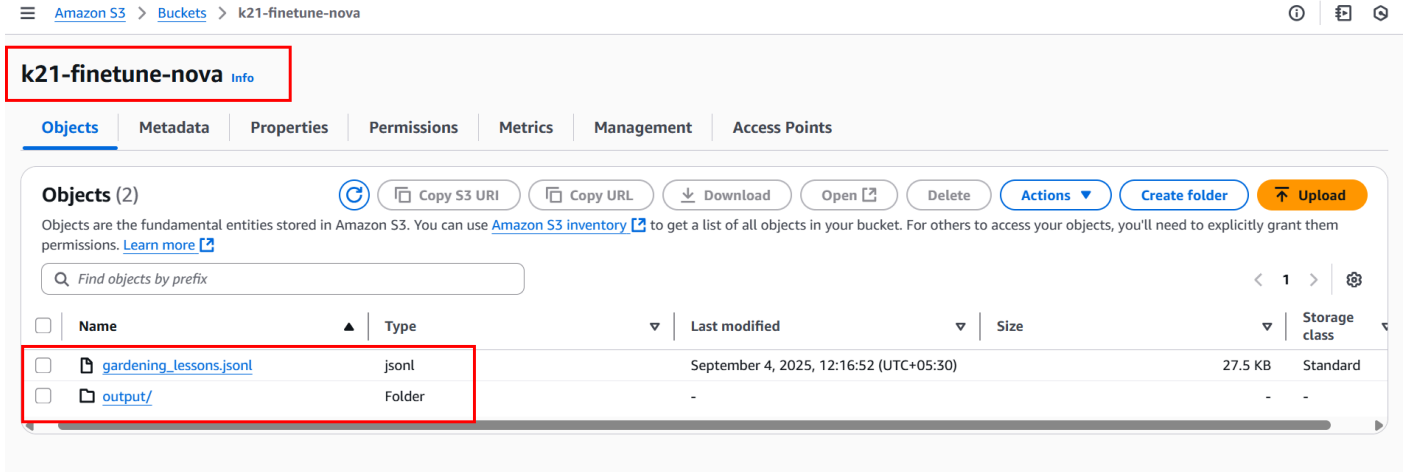
► **Destination details**
Bucket settings that impact new objects stored in the specified destination.

► **Permissions**
Grant public access and access to other AWS accounts.

► **Properties**
Specify storage class, encryption settings, tags, and more.

[Cancel](#) [Upload](#)

15. You can see the file successfully uploaded to your bucket.



The screenshot shows the Amazon S3 console interface for the bucket 'k21-finetune-nova'. The 'Objects' tab is active, displaying a list of objects. A red box highlights the 'gardening_lessons.jsonl' file and the 'output/' folder. The file 'gardening_lessons.jsonl' is a JSONL file, 27.5 KB in size, and was last modified on September 4, 2025, at 12:16:52 (UTC+05:30). The folder 'output/' is a standard folder.

Name	Type	Last modified	Size	Storage class
☐ gardening_lessons.jsonl	jsonl	September 4, 2025, 12:16:52 (UTC+05:30)	27.5 KB	Standard
☐ output/	Folder	-	-	-

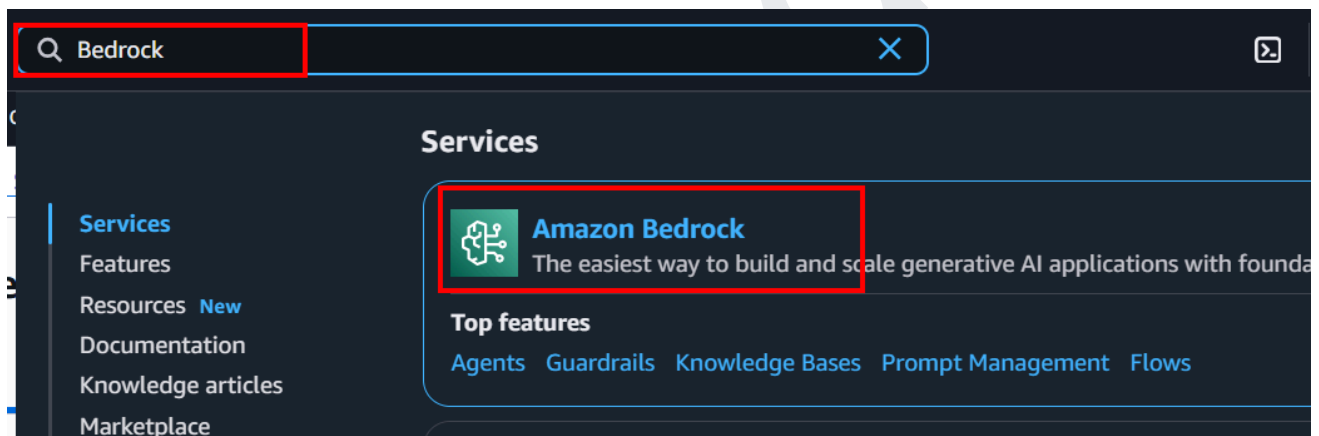
We have successfully uploaded an object and a folder to the bucket we will be using in the next steps.

7. CREATE A BEDROCK CUSTOM MODEL WITH FINE-TUNE

In this step, we'll fine-tune a custom model in Amazon Bedrock using a gardening example. This step will allow us to adapt a pre-trained foundation model (FM) to recognize and respond to specific queries related to gardening, such as plant care, pest control, and soil management.

Fine-tuning is essential because the **generic pre-trained models** may not be optimized to understand or provide accurate answers about gardening. By fine-tuning the model with **gardening-specific data**, we ensure it can provide more relevant and accurate responses tailored to gardening-related tasks, improving its usefulness for real-world applications.

1. In the **AWS Console**, in the search bar, type "**Amazon Bedrock**" and select it. (We are working in the Northern Virginia region)



2. From the left navigation pane, choose Custom models under Foundation models.

Amazon Bedrock > Custom models

Amazon Bedrock

- Discover
- Test
- Infer
- Tune
 - Custom models**
 - Prompt router models
 - Imported models
 - Marketplace model deployment
- Build
- Assess
- Configure and learn

Custom models Info

Customize a model with Distillation, Fine-tuning or Continued Pre-training. You can also share or copy custom models.

How it works

Step 1: Create custom model

Create a custom model using one of these methods.

- Distillation
- Fine-tuning
- Continued Pre-training

Create

Step 2: Set up inference

Choose an inference option for your custom model. For flexible, pay-per-use pricing, use on-demand inference with a custom model deployment. For dedicated, consistent performance, purchase provisioned throughput.

[Learn More](#)

3. In the Models tab, choose a **Supervised fine-tuning job** depending on the type of model you want to train, and then click on Create.

Amazon Bedrock > Custom models

API keys

- Test
 - Chat / Text playground
 - Image / Video playground
 - Watermark detection
 - Tokenizer [New](#)
- Infer
 - Cross-region inference
 - Batch inference
 - Provisioned Throughput
 - Custom model on-demand
- Tune
 - Custom models**
 - Prompt router models
 - Imported models
 - Marketplace model deployments
- Build

Custom models Info

Customize models using available customization processes. Access and manage custom models based on your region's capabilities.

Customization techniques

Models (0) Info

View all your custom models together in one unified table. Each model shows its training job, with easy setup for deployment.

Find model

Custom model name	Model Status	Job name	Job status	Creation time (UTC-07:00)
No models				

Create your first model to get started.

Create

Models shared with you (0)

4. Now provide the name of the Job, whatever you want, we are giving it the name “finetune-nova-K21”.

Create Fine-tuning job [Info](#)

Select the model you wish to fine-tune and submit your data location.

Job configuration

Job name

Enter a name to identify the training job necessary to pre-train and create a new model.

finetune-nova-K21

Maximum of 63 alphanumeric characters. Cannot include hyphens (-) and spaces. Must be unique within your account in AWS Region.

► Tags - optional

5. In the Model details section, we will choose the model that you want to customize with your own data and give your resulting model a name.

Model details

Source model

Choose from a list of models that you wish to customize with using your own data.

Select model

2

Fine-tuned model name

Enter a name to identify the new fine-tuned model.

finetune-nova-K21

1

Maximum of 63 alphanumeric characters. Cannot include hyphens (-) and spaces. Must be unique within your account in AWS Region.

☐ KMS key selection [Info](#)

► Tags - optional

6. From the list of AWS models, we will be selecting the Nova 2 Lite model and clicking on Apply.

Select model

Input

Choose opti... ▼

Output

Choose opti... ▼

1. Categories

Serverless model providers ▼

 Amazon

2. Models

Nova 2 Lite v1
Input: Text, Image, Video
Output: Text

Nova Pro 1.0
Input: Text, Image, Video
Output: Text

Nova Lite 1.0
Input: Text, Image, Video
Output: Text

Nova Micro 1.0
Input: Text
Output: Text

Titan Multimodal Embeddings G1
Input: Text, Image
Output: Embedding

Cancel

Apply

7. In the Hyperparameters section, keep everything as it is.

Hyperparameters [Info](#)

Epochs

The maximum number of iterations through the entire training dataset.

Enter an integer between 1 and 5.

Batch size

The number of samples processed before updating model parameters.

Learning rate

Multiplier that influences the learning rate at which model parameters are updated after each batch.

Enter a float value between 0.000001 and 0.0001

Learning rate warmup steps

Number of iterations over which learning rate is gradually increased to the initial rate specified.

Enter an integer between 0 and 20.

K21

8. In the Input data section, select the S3 bucket we created in the previous step and select the location of the training dataset file (gardening lessons) and the validation dataset file.

Input data [Info](#)

Choose a file in the S3 location. The files you choose must be in the [dataset format](#) that the model needs for training. You can check the data format for your specified model for any potential errors using simple python [script](#). You can also use Sagemaker Ground Truth to create and label training datasets. [Learn more](#)

S3 location

[View](#) [Browse S3](#)

Validation dataset S3 location (optional)

[View](#) [Browse S3](#)

Output data [Info](#)

Choose S3 location to store the model validation outputs.

S3 location

[View](#) [Browse S3](#)

Browse S3 to select the S3 bucket prefix where the output will be stored.

Choose an archive in S3

[S3 buckets](#) > k21-finetune-nova

Objects (1/2)

Key	Last modified	Size
output/	-	-
gardening_lessons.jsonl	September 04, 2025, 12:16 (UTC+05:30)	27.55 KB

[Cancel](#) [Choose](#)

9. In the Output data section, enter the Amazon S3 location where Amazon Bedrock should save the output of the job. Amazon Bedrock stores the training loss metrics and validation loss metrics for each epoch in separate files in the location that you specify.

Input data [Info](#)

Choose a file in the S3 location. The files you choose must be in the [dataset format](#) that the model needs for training. You can check the data format for your specified model for any potential errors using simple python [script](#). You can also use Sagemaker Ground Truth to create and label training datasets. [Learn more](#)

S3 location

[View](#)
[Browse S3](#)

Validation dataset S3 location (optional)

[View](#)
[Browse S3](#)

Output data [Info](#)

Choose S3 location to store the model validation outputs.

S3 location

[View](#)
[Browse S3](#)

Browse S3 to select the S3 bucket prefix where the output will be stored.

Choose an archive in S3

[S3 buckets](#) > k21-finetune-nova

Objects (1/2)

Key	Last modified	Size
☒ output/	-	-
☐ gardening_lessons.json	September 04, 2025, 12:16 (UTC+05:30)	27.55 KB

[Cancel](#)
[Choose](#)

10. In the Service access section, create and use a new service role – Enter a name for the service role. Now choose Create job to begin the job.

Service access [Info](#)



To run batch inference, you must use a service role with permissions to access and write to S3 on your behalf.

Choose a method to authorize Bedrock

- ☐ Use an existing service role
- ☒ Create and use a new service role

Service role name

k21-finetune-nova

Maximum 64 characters. Use alphanumeric and '+=, @-_' characters.

[View permission details](#)

Set up inference for your fine-tuned model

After you create your fine-tuned model, you must set up inference for it. To learn more about inference options, [See Here](#)

[Cancel](#)

[Create job](#)

Service access [Info](#)



To run batch inference, you must use a service role with permissions to access and write to S3 on your behalf.

 You need to create a new service role because there are no suitable roles currently available in the account.

Service role name

k21-finetune-nova

Maximum 64 characters. Use alphanumeric and '+=, @-_' characters.

[View permission details](#)

Set up inference for your fine-tuned model

After you create your fine-tuned model, you must set up inference for it. To learn more about inference options, [See Here](#)

[Cancel](#)

[Create job](#)

11. Now our job is in the Progress state, it will take a few minutes to hours to Active status.

✕

Training setup & model creation in progress
Preparing your training environment and building your custom model through validation, setup, training, and optimization steps. This process typically takes 2-24 hours to complete.

✕

Fine-tuning job **k21-nova-finetune** creation is in progress. Once the job creates successfully the model will be available.

K21-nova-finetune Info

↻
Setup inference ▼
Actions ▼

Model details

Model ARN - Customization type Fine-tuning	Job ARN arn:aws:bedrock:us-east-1:005185643085:model-customization-job/amazon.nova-micro-v1:0:128k/d32cda7hdv5x Source Model ARN arn:aws:bedrock:us-east-1::foundation-model/amazon.nova-micro-v1:0:128k Source model name Nova Micro	Custom model encryption KMS key Bedrock owned KMS key Model creation time January 19, 2026, 16:56 (UTC+05:30)
--	---	--

12. It will take from a few hours to 2 days.

Amazon Bedrock > Custom models
🔍

Chat / Text playground

Image / Video playground

Watermark detection

▼ Infer

Cross-region inference

Batch inference

Provisioned Throughput

Custom model on-demand New

▼ Tune

Custom models

Prompt router models

Imported models

Marketplace model deployment

Continued Pre-training

Create ▼

throughput.

[Learn More](#)

Models
Jobs

Models (1)

Choose a model to set up inference. You can set up On-demand inference or purchase provisioned throughput for the model.

< 1 >
⚙️

Custom model name	Model Status	Customized model	Type	Inference set up	Share status	Creation time (UTC-07:
☐ K21-nova-finetune	✔ Active	Nova Micro	Fine-tuning	🔒 Required	-	September 04, 2025, 15

Note: For us, it took 4 hours to be in Active Status.

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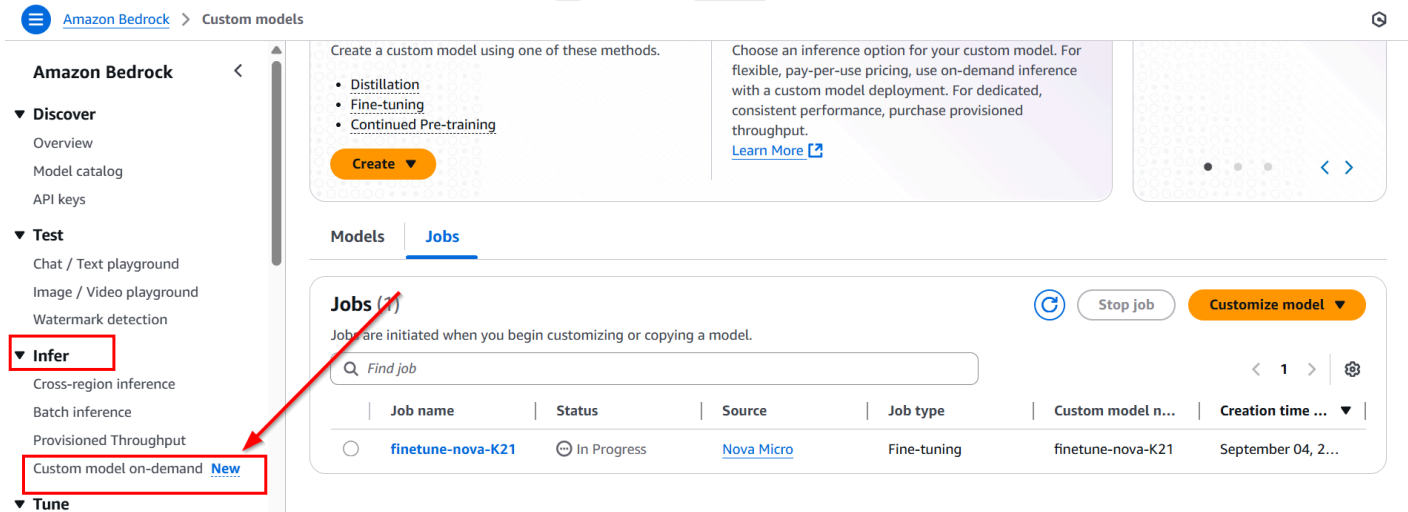
8. CREATE A CUSTOM MODEL ON DEMAND WITH INFERENCE

This step involves **creating a custom AI model** by fine-tuning a pre-trained **foundation model** (FM) using **Amazon Bedrock** and performing **inference**. Inference refers to the process by which the model makes predictions or generates outputs based on new data or queries.

We need this step to adapt a pre-trained model to our **specific use case** (e.g., gardening, customer service). Fine-tuning with your own data ensures that the model performs accurately and is tailored to your unique needs.

Inference allows you to test and use the model in real-world scenarios, making it practical for actual tasks like answering queries, generating content, or automating workflows. This step is essential for creating **relevant, high-performance models** that can deliver meaningful results.

1. Now in the same Bedrock Dashboard, scroll down and under the infer select the custom model on demand.



Amazon Bedrock > Custom models

Amazon Bedrock

- Discover
 - Overview
 - Model catalog
 - API keys
- Test
 - Chat / Text playground
 - Image / Video playground
 - Watermark detection
- Infer**
 - Cross-region inference
 - Batch inference
 - Provisioned Throughput
 - Custom model on-demand **New**
- Tune

Create a custom model using one of these methods.

- Distillation
- Fine-tuning
- Continued Pre-training

Create

Choose an inference option for your custom model. For flexible, pay-per-use pricing, use on-demand inference with a custom model deployment. For dedicated, consistent performance, purchase provisioned throughput. [Learn More](#)

Models | **Jobs**

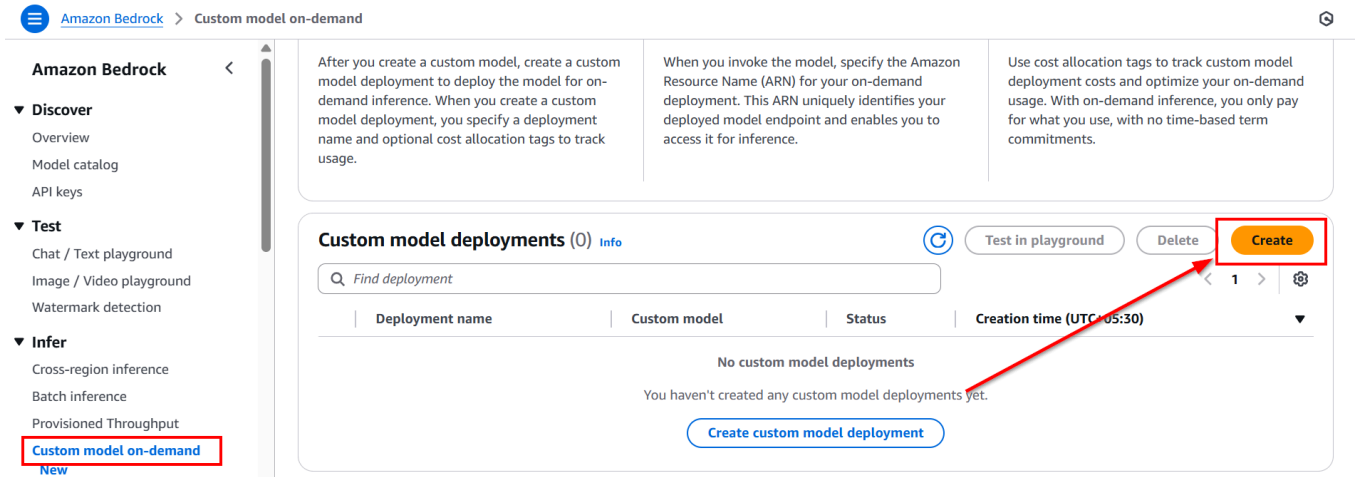
Jobs (1) Stop job Customize model

Jobs are initiated when you begin customizing or copying a model.

Find job

	Job name	Status	Source	Job type	Custom model n...	Creation time ...
	finetune-nova-K21	In Progress	Nova Micro	Fine-tuning	finetune-nova-K21	September 04, 2...

2. Now we will be creating a custom model for inference.



The screenshot shows the Amazon Bedrock console interface. On the left, the navigation menu includes 'Discover', 'Test', and 'Infer'. Under 'Infer', 'Custom model on-demand' is highlighted with a red box. The main content area shows 'Custom model deployments (0)' with a search bar and a table. The 'Create' button is highlighted with a red box and a red arrow. The table has columns for 'Deployment name', 'Custom model', 'Status', and 'Creation time (UTC+05:30)'. Below the table, it says 'No custom model deployments' and 'You haven't created any custom model deployments yet.' with a 'Create custom model deployment' button.

3. Provide the name of your deployment, “**K21-nova-inferenceondemand**,” and click on the select model.

Create deployment for on-demand [Info](#)

Set up on-demand inference by deploying the custom model. The information you specify enables you to manage ar

Deployment details

Name

k21-nova-inferenceondemand **1**

Valid characters are a-z, A-Z, 0-9, _(underscore) and -(hyphen).The name can have up to 63 characters.

Description - optional

Enter deployment description (optional)

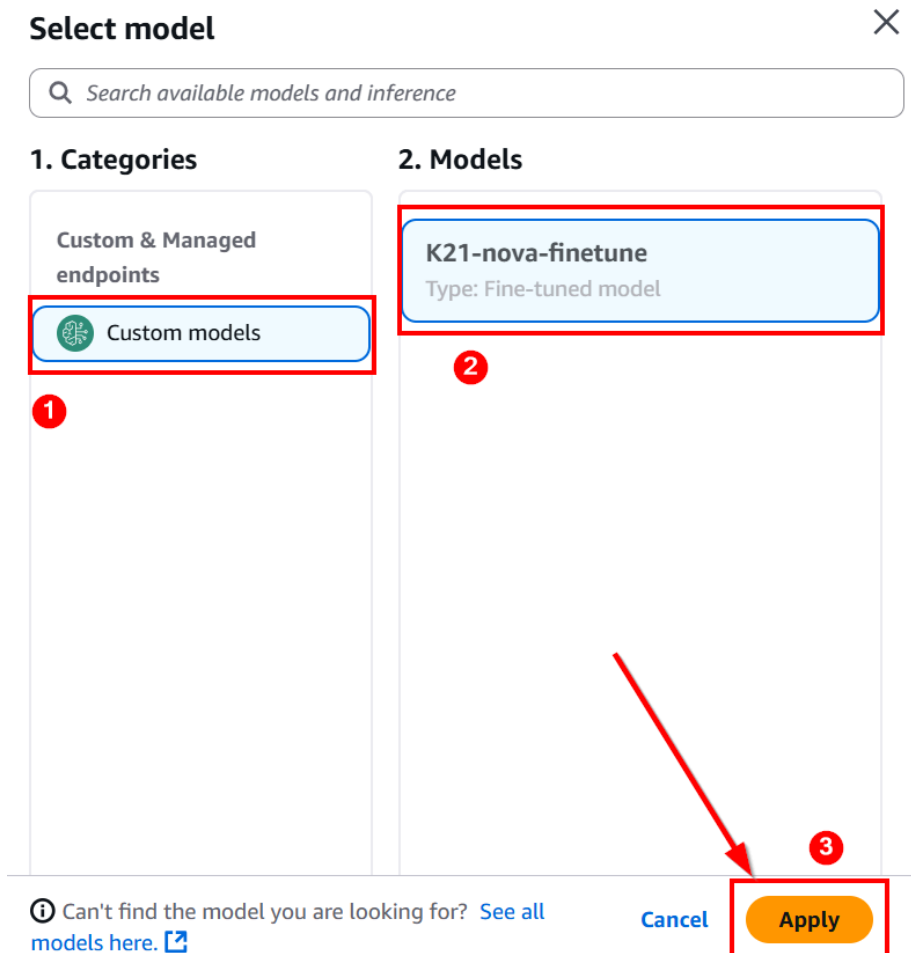
The description can have up to 2048 characters

Select model

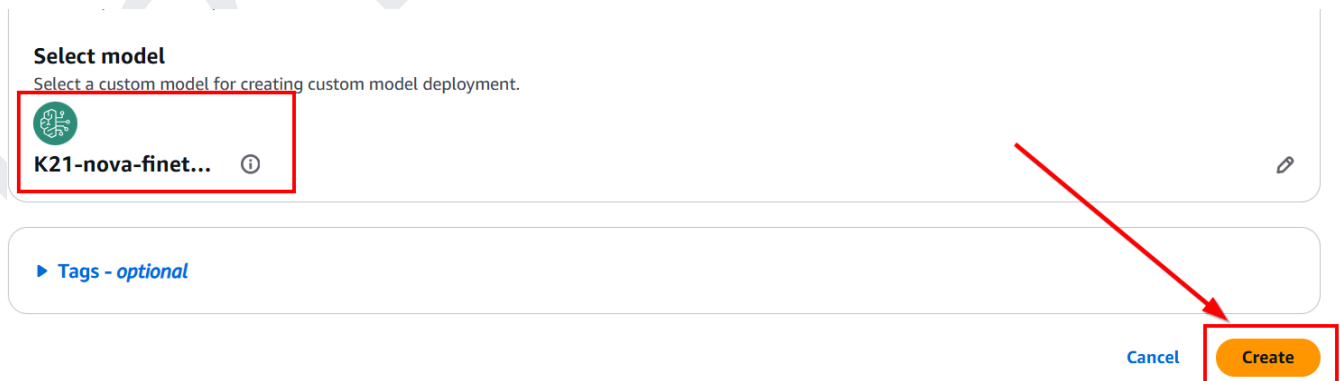
Select a custom model for creating custom model deployment.

Select model **2**

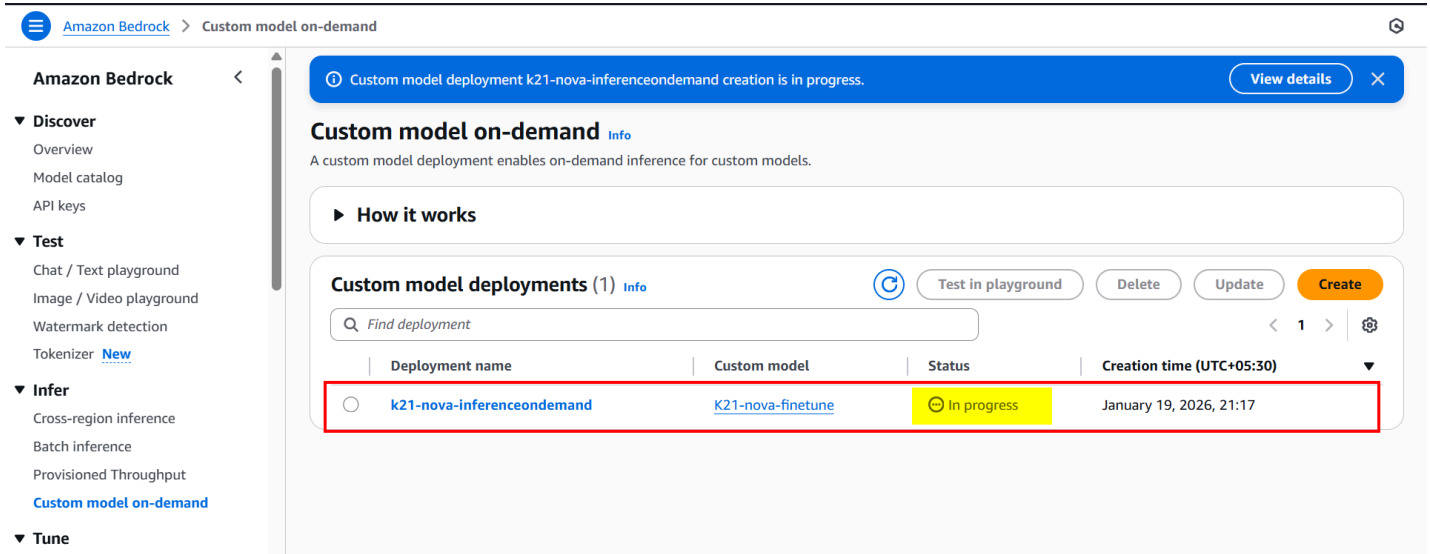
4. In this step, we will be selecting the custom model that we created in the last step and clicking on Apply.



5. Now our model is selected, click on Create.



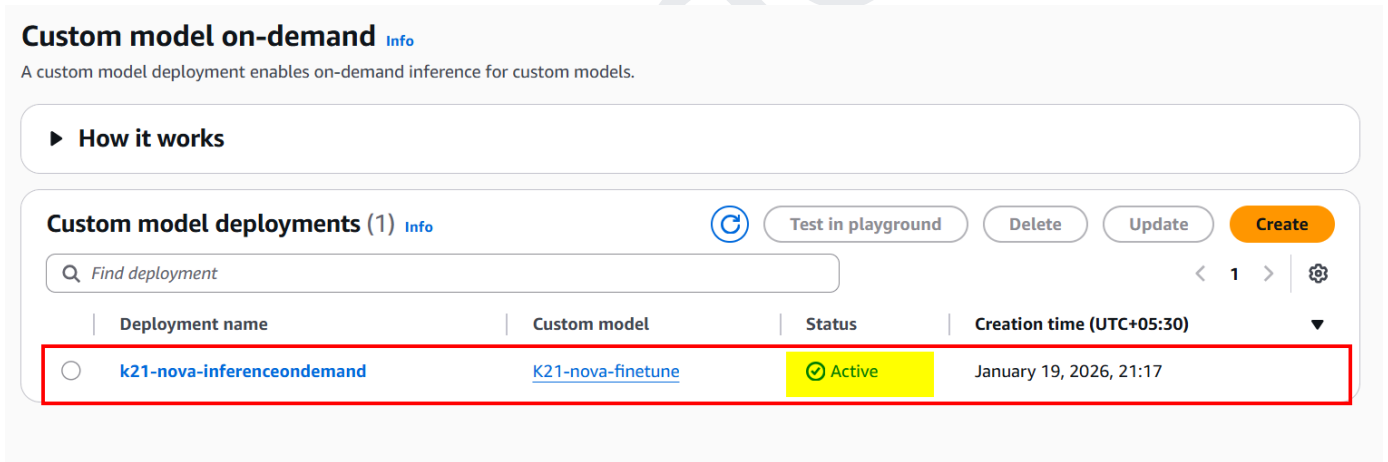
6. Our custom model is in the Progress state. It will take some time to be active.



The screenshot shows the Amazon Bedrock console for the 'Custom model on-demand' section. A blue banner at the top indicates that the custom model deployment 'k21-nova-inferenceondemand' is in progress. The left sidebar lists navigation options under 'Amazon Bedrock', including 'Discover', 'Test', 'Infer', and 'Tune'. The main content area shows 'Custom model deployments (1)' with a table listing the deployment. The deployment 'k21-nova-inferenceondemand' is associated with the custom model 'K21-nova-finetune' and is currently in the 'In progress' state, as indicated by a yellow status box. The creation time is listed as January 19, 2026, 21:17.

Deployment name	Custom model	Status	Creation time (UTC+05:30)
k21-nova-inferenceondemand	K21-nova-finetune	In progress	January 19, 2026, 21:17

7. Now our Custom Model is Active; let's move on to the next Step.



This screenshot shows the same Amazon Bedrock console page, but the deployment 'k21-nova-inferenceondemand' is now in the 'Active' state, indicated by a green status box. The rest of the interface, including the navigation sidebar and the deployment table, remains the same.

Deployment name	Custom model	Status	Creation time (UTC+05:30)
k21-nova-inferenceondemand	K21-nova-finetune	Active	January 19, 2026, 21:17

Note: For us, it took 5-10 minutes to be in Active Status.

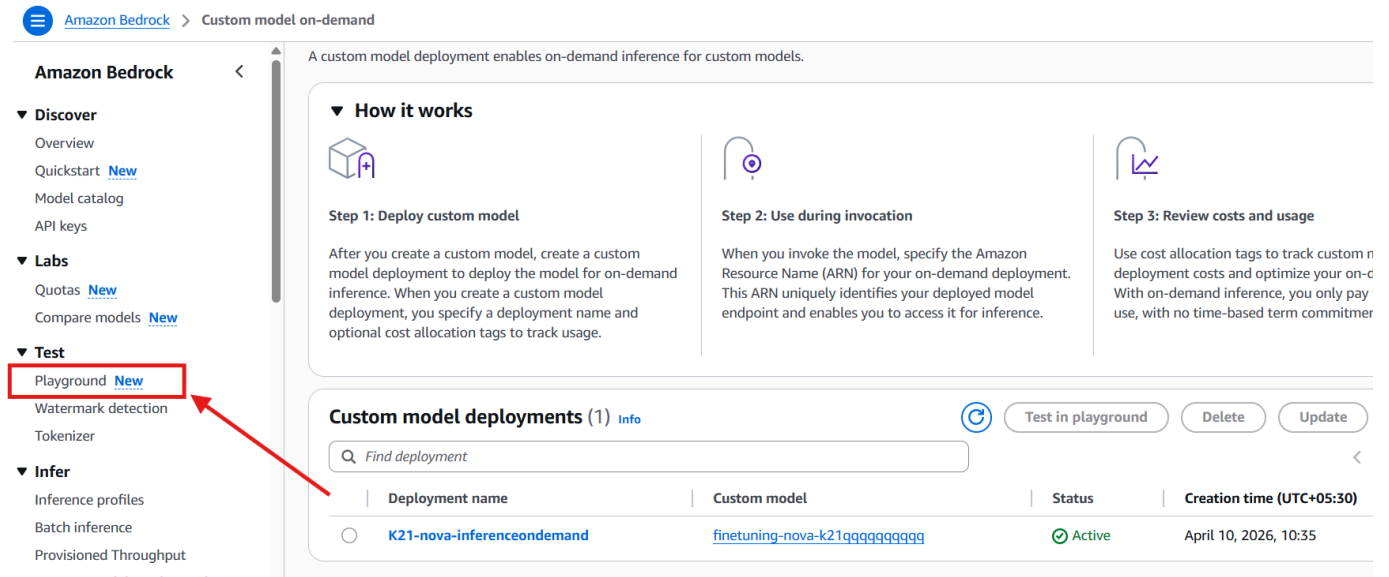
9. TESTING YOUR CUSTOM MODEL IN AMAZON BEDROCK

After fine-tuning your custom model in Amazon Bedrock, testing it is a crucial step to ensure it performs as expected.

Testing involves providing sample input data and evaluating the model's output for accuracy, relevance, and efficiency. This helps verify that the model can handle real-world data and generate correct results.

During testing, you may need to run inference tasks through Amazon Bedrock's APIs and review the results, making adjustments or retraining the model if necessary to improve its performance before deployment.

1. In the Same Bedrock Dashboard, go to test and click on Playground.



The screenshot shows the Amazon Bedrock console interface. On the left sidebar, under the 'Test' section, the 'Playground' link is highlighted with a red box and a red arrow. The main content area displays 'Custom model deployments (1)' with a table listing a deployment named 'K21-nova-inferenceondemand' with a status of 'Active'. Above the table, there are three steps: 'Step 1: Deploy custom model', 'Step 2: Use during invocation', and 'Step 3: Review costs and usage'. The 'Test in playground' button is visible in the top right of the deployment section.

2. Select Model

≡ [Amazon Bedrock](#) > Chat / Text playground

Mode Chat ▼

Select model

3. We need to choose from the custom model, the model we created earlier, and then inference.

Select model ✕

🔍 Find available models and inference

Input/output filter is not available for custom and managed endpoints.

1. Categories	2. Models	3. Inference
Custom & managed endpoi... ▼		On demand ▼
 Custom models	finetuning-nova-k21qqqqqqqqqq Type: Fine-tuned model	K21-nova-inferenceondemand

Cancel **Apply**

4. Now you can ask any questions about Gardening in this model.

Q: Why use Compost in the Garden?

The screenshot shows the Amazon Bedrock Playground interface. On the left is a sidebar with navigation options: Discover (Overview, Quickstart, Model catalog, API keys), Labs (Quotas, Compare models), Test (Playground, Watermark detection, Tokenizer), and Infer (Inference profiles, Batch inference, Provisioned Throughput, Custom model on-demand). The main area shows a chat with 'Nova 2 Lite'. The prompt is 'Why use Compost in the Garden?'. The response is 'Compost is one of the most valuable resources for any garden—whether you're growing vegetables, flowers, fruits, or ornamentals. Here's why it's essential:'. The interface includes a top bar with 'Mode: Chat' and a 'Build' button. The input and output tokens are 103 and 1346 respectively, with a latency of 11608 ms.

5. Now, let's ask one more question: Provide Gardening tips?

The screenshot shows the Amazon Bedrock Playground interface. On the left is a sidebar with navigation options: Discover (Overview, Quickstart, Model catalog, API keys), Labs (Quotas, Compare models), Test (Playground, Watermark detection, Tokenizer), and Infer (Inference profiles, Batch inference, Provisioned Throughput, Custom model on-demand). The main area shows a chat with 'Nova 2 Lite'. The prompt is 'Provide Gardening tips?'. The response is 'Essential Gardening Tips for a Thriving Garden. Whether you're a beginner or an experienced gardener, these practical tips will help you grow healthier plants, avoid common mistakes, and enjoy a productive, beautiful garden.' The interface includes a top bar with 'Mode: Chat' and a 'Build' button. The input and output tokens are 1015 and 2245 respectively, with a latency of 16033 ms.

Our custom model is answering questions as expected. You can ask more questions about it as well.

In this lab, we worked with a use case where the model was required to respond to gardening-related queries. Initially, the foundation model did not have the necessary knowledge. To address this, we fine-tuned the model using our custom sample data. After fine-tuning, we deployed the model and successfully obtained the expected outputs for gardening queries. This process demonstrated how to enhance a foundation model's ability to handle domain-specific tasks.

10. OPTIONAL (ADDITIONAL SCENARIOS FOR FURTHER EXPLORATION)

Fine-tuning is strongest when you're baking in behavior, format, tone, or classification — not facts (facts are a RAG job). Each dataset below teaches a repeatable enterprise behavior, so learners get a clear before/after: the base model rambles, the fine-tuned model returns the exact structured output. AWS's own flagship example is intent classification (Nova Micro 41% → 97% accuracy).

This follows the same flow as the original lab (S3 → Fine-tune job → Custom model on-demand → Playground test → Clean up), repeated once per scenario. Steps 1–2 are done once; repeat Steps 3–6 for each dataset you want to fine-tune.

Note: The following steps are optional and are not mandatory to complete this lab. These additional scenarios are provided for learners who would like to explore more custom model use cases.

As we already created an S3 bucket in the previous steps, you can reuse the same bucket for these scenarios. The detailed steps for creating and configuring a custom model have already been covered earlier, so in this section, we will only provide the high-level steps.

You can follow the same process used in the previous custom model creation exercise and apply it to different scenarios based on your requirements. Each custom model may take approximately 2 hours to 24 hours to create, after which you can test and validate the model.

The purpose of the previous steps was to show how to create a custom model for one sample scenario. You are not limited to that scenario; you can use the same approach to work with other scenarios as well.

Note on Additional Scenarios

The six scenarios below have been added to this lab based on direct requests from many of you in our community. We heard that you wanted to see how the same fine-tuning workflow applies beyond a single use case — and across the industries you actually work in or are preparing for. Each scenario follows the exact same process covered in the core lab, applied to a different domain and dataset, so the steps will feel familiar as you work through them.

We hope these additional scenarios give you a broader view of what is possible with Amazon Bedrock fine-tuning and help you connect this skill to your own professional context. We will continue updating this lab as more use cases are requested - so keep the feedback coming in the community.

11. CREATE THE S3 BUCKET (ONE TIME)

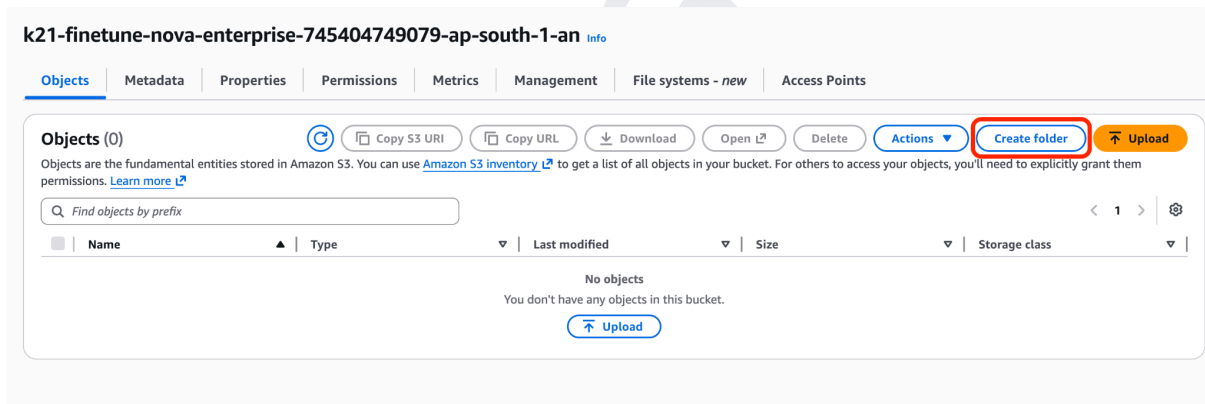
Note: Since we already created the S3 bucket in the previous step, you can reuse it for this exercise. Create separate folders within the existing bucket to organize the input data and output results required for the custom model workflow, or create a new bucket as well.

Download all the `.jsonl` files from the GitHub folder:

https://github.com/k21academyuk/AWS-GenAI_AIML/tree/Custom-Model-jsonl-files

1. Inside your new bucket, create these folders (bucket → **Create folder**, type the name, **Create folder**):

- training-data/
- output/



2. Enter **training-data** as the folder name, keep the default encryption settings, and click **Create folder**.

Folder

Folder name

training-data /

Folder names can't contain "/" - [See rules for naming](#)

Server-side encryption [Info](#)

Server-side encryption protects data at rest.

The following encryption settings apply only to the folder object and not to sub-folder objects.

Server-side encryption

☒ **Don't specify an encryption key**
The bucket settings for default encryption are used to encrypt the folder object when storing it in Amazon S3.

☐ **Specify an encryption key**
The specified encryption key is used to encrypt the folder object before storing it in Amazon S3.

⚠ If your bucket policy requires objects to be encrypted with a specific encryption key, you must specify the same encryption key when you create a folder. Otherwise, folder creation will fail.

Cancel **Create folder**

3. Create a second folder named Output.

Folder

Folder name

output /

Folder names can't contain "/" - [See rules for naming](#)

Server-side encryption [Info](#)

Server-side encryption protects data at rest.

The following encryption settings apply only to the folder object and not to sub-folder objects.

Server-side encryption

☒ **Don't specify an encryption key**
The bucket settings for default encryption are used to encrypt the folder object when storing it in Amazon S3.

☐ **Specify an encryption key**
The specified encryption key is used to encrypt the folder object before storing it in Amazon S3.

⚠ If your bucket policy requires objects to be encrypted with a specific encryption key, you must specify the same encryption key when you create a folder. Otherwise, folder creation will fail.

Cancel **Create folder**

4. Open the **training-data** folder, click **Upload**, and upload your training dataset files.

training-data/ Copy S3 URI

Objects | Properties

Objects (0) Refresh Copy S3 URI Copy URL Download Open Delete Actions Create folder **Upload**

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

☐ **Name** ☐ **Type** ☐ **Last modified** ☐ **Size** ☐ **Storage class**

No objects
You don't have any objects in this folder.

Upload

5. Click **Add files**, then select all the **.jsonl** files and upload them here, one file at a time.

Upload info

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose **Add files** or **Add folder**.

Files and folders (6 total, 766.8 KB) Remove **Add files** Add folder

All files and folders in this table will be uploaded.

☐	Name	Folder	Type	Size
☐	banking_assistant.jsonl	-	-	143.3 KB
☐	support_ticket_triage.jsonl	-	-	129.5 KB
☐	ecommerce_product_copy.jsonl	-	-	130.7 KB
☐	pharma_adverse_event_triage.jsonl	-	-	144.7 KB
☐	patient_message_triage.jsonl	-	-	93.5 KB
☐	it_helpdesk_t1.jsonl	-	-	124.9 KB

6. Then click Upload.

Files and folders (6 total, 766.8 KB)
All files and folders in this table will be uploaded.

Remove

Add files

Add folder

Find by name

< 1 >

☐	Name	Folder	Type	Size
☐	...	...	...	...

Destination [Info](#)

Destination
[s3://k21-finetune-nova-enterprise-745404749079-ap-south-1-an/training-data/](#)

► Destination details

Bucket settings that impact new objects stored in the specified destination.

► Permissions

Grant public access and access to other AWS accounts.

► Properties

Specify storage class, encryption settings, tags, and more.

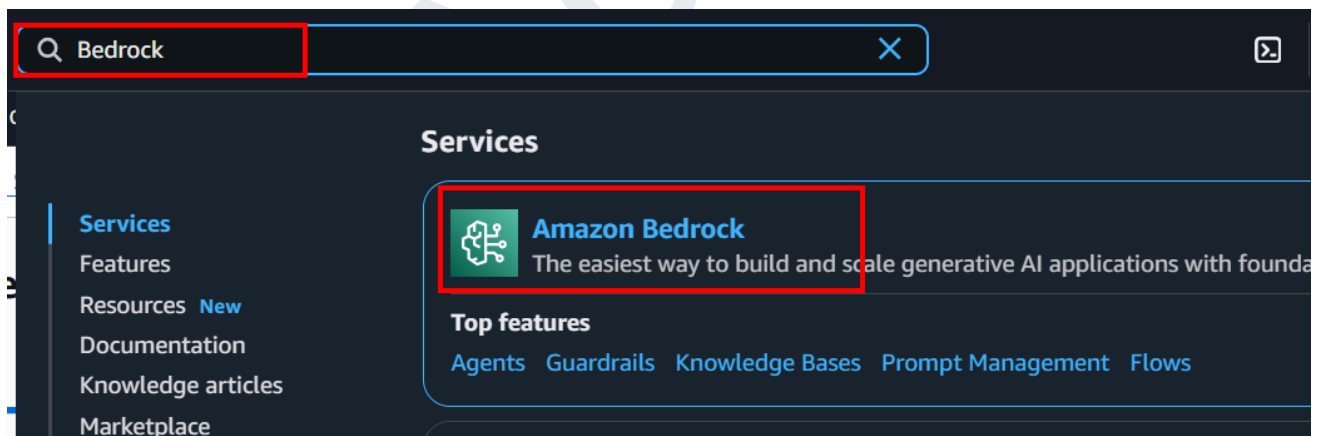
Cancel

Upload

12. SCENARIO 1: SUPPORT TICKET TRIAGE (*RECOMMENDED STARTING SCENARIO*)

This scenario trains the model to automatically classify and route incoming customer support tickets based on their content and urgency. In real-world support operations, teams receive hundreds of tickets daily covering billing issues, technical outages, account problems, and general inquiries. Without automation, agents spend significant time reading and manually sorting these tickets before any resolution work begins. By fine-tuning the model on labeled support ticket examples, it learns to instantly identify the ticket category, assign a priority level, and recommend the right team — cutting response time and reducing agent workload. This is one of the most common and high-impact enterprise use cases for fine-tuning, making it the recommended starting point.

1. In the AWS Management Console, search for **Amazon Bedrock** using the search bar and open the **Amazon Bedrock** service. From here, you can access and manage the AI resources created during the lab, including knowledge bases, guardrails, agents, and other Bedrock components.



- Go to **Custom models** under the **Tune** section and verify that no custom models remain. Delete any listed models if necessary.

Custom models

Create a model

Purchase provisioned throughput to use the custom model

Select a model(s) to customize using training and validation (if applicable) datasets in an Amazon S3 bucket. Share your custom model with other accounts if you have an [AWS Organizations](#) account. You can copy a model that you own or that has been shared with you to another AWS Region.

Purchase provisioned throughput to use a custom fine-tuned model, a continued pre-trained model or a copied model. You can then use your model in the Amazon Bedrock playgrounds or in your applications.

Models (0)

Customized model and their jobs successfully completed will appear here.

Find model

Custom model name	Model Status	Customized model	Type	Share status	Creation time (UTC-07:00)
No custom models					

There are currently no resources.

[Fine-tune model](#)

- Click **Create**, then select **Supervised fine-tuning job** to create a new custom model training job.

Custom models [Info](#)

Customize models using available customization processes. Access and manage custom models based on your region's capabilities.

Customization techniques

Reinforcement fine-tuning - new

Build more accurate models using reward-based learning with less manual effort. Ideal for complex, multi-step, reasoning tasks.

How it works

- Step 1 - Choose model
- Step 2 - Add input data
- Step 3 - Add reward function

Supervised fine-tuning

Adapts pre-trained models to specific tasks using labeled data to improve performance in specific domains.

How it works

- Step 1 - Choose model
- Step 2 - Add input data

Distillation

Creates smaller, faster models by using a large teacher model to train a specialized student model with your invocation logs or generated training data.

How it works

- Step 1 - Choose teacher/student model
- Step 2 - Add input data

Models (0) [Info](#)

View all your custom models together in one unified table. Each model shows its training job, with easy setup for deployment a

Find model

Custom model name	Model Status	Job name	Job status	Creation time (UTC-07:00)
-------------------	--------------	----------	------------	---------------------------

No models

Create your first model to get started



Stop Job

Setup inference

Actions

Create

- Reinforcement fine tuning job - new
- Supervised fine-tuning job**
- Distillation job

4. Enter a unique **Job name**, then click **Select model** to choose the foundation model you want to fine-tune.

Create Fine-tuning job [Info](#)

Select the model you wish to fine-tune and submit your data location.

Job configuration

Job name

Enter a name to identify the training job necessary to pre-train and create a new model.

finetune-support-triage

Maximum of 63 alphanumeric characters. Cannot include hyphens (-) and spaces. Must be unique within your account in AWS Region.

► Tags - optional

Model details

Source model

Choose from a list of models that you wish to customize with using your own data.

Select model

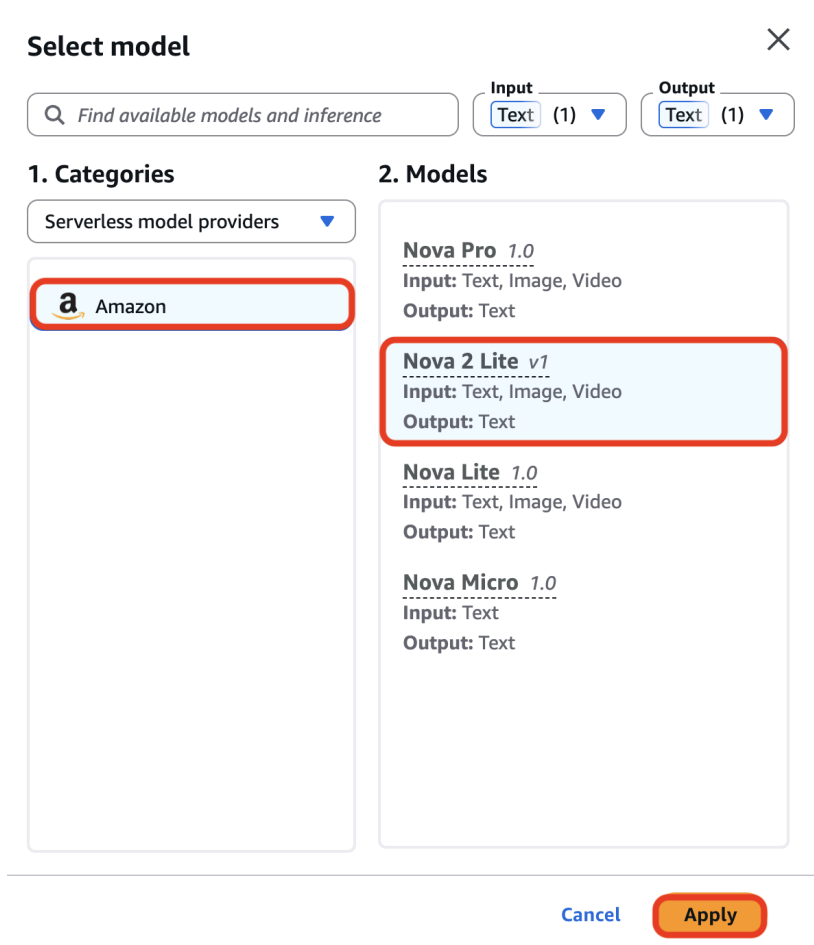
Fine-tuned model name

Enter a name to identify the new fine-tuned model.

fine-tuned-model-d02b175d

Maximum of 63 alphanumeric characters. Cannot include hyphens (-) and spaces. Must be unique within your account in AWS Region.

5. Select **Amazon** as the model provider, choose **Nova 2 Lite v1** as the source model, and click **Apply**.



Select model ✕

Find available models and inference

Input: Text (1) ▼ Output: Text (1) ▼

1. Categories

Serverless model providers ▼

Amazon

2. Models

Nova Pro 1.0
Input: Text, Image, Video
Output: Text

Nova 2 Lite v1
Input: Text, Image, Video
Output: Text

Nova Lite 1.0
Input: Text, Image, Video
Output: Text

Nova Micro 1.0
Input: Text
Output: Text

Cancel Apply

6. Hyperparameters → leave defaults.
7. Click **Browse S3** under **Input data** to select the training dataset from your Amazon S3 bucket. Then, choose an S3 location under **Output data** where the fine-tuning job results and model artifacts will be stored.

Input data [Info](#)

Choose a file in the S3 location. The files you choose must be in the [dataset format](#) that the model needs for training. You can check the data format for your specified model for any potential errors using simple python [script](#). You can also use Sagemaker Ground Truth to create and label training datasets. [Learn more](#)

S3 location

[View](#)
[Browse S3](#)

Output data [Info](#)

Choose S3 location to store validation results, metrics, logs, and training data.

S3 location

[View](#)
[Browse S3](#)

Browse S3 to select the S3 bucket prefix where the output will be stored.

► VPC settings - optional

Choose a VPC configuration to access Amazon S3 data source located in your virtual private cloud (VPC). You can create and manage VPC, subnets and security groups in [Amazon VPC](#)

8. Select the **support_ticket_triage.jsonl** training dataset from the S3 bucket, then click **Choose** to use it as the input data for the fine-tuning job.

Choose an archive in S3

[S3 buckets](#) > [k21-finetune-nova-enterprise-745404749079-us-east-1-an](#) > training-data/

Objects (1/6)

	Key	Last modified	Size
☐	banking_assistant.jsonl	July 10, 2026, 13:41 (UTC+05:30)	143.33 KB
☐	ecommerce_product_copy.jsonl	July 10, 2026, 13:41 (UTC+05:30)	130.74 KB
☐	it_helpdesk_l1.jsonl	July 10, 2026, 13:41 (UTC+05:30)	124.95 KB
☐	patient_message_triage.jsonl	July 10, 2026, 13:41 (UTC+05:30)	93.45 KB
☐	pharma_adverse_event_triage.jsonl	July 10, 2026, 13:41 (UTC+05:30)	144.75 KB
☒	support_ticket_triage.jsonl	July 10, 2026, 13:41 (UTC+05:30)	129.54 KB

[Cancel](#)
[Choose](#)

9. For the **Output data** location, click **Browse S3** and select the S3 folder where the fine-tuning outputs, logs, and model artifacts will be stored (for example, `s3://k21-finetune-nova-enterprise/output/support-triage/`).

Input data [Info](#)
Choose a file in the S3 location. The files you choose must be in the [dataset format](#) that the model needs for training. You can check the data format for your specified model for any potential errors using simple python [script](#). You can also use Sagemaker Ground Truth to create and label training datasets. [Learn more](#)

S3 location
Q s3://k21-finetune-nova-enterprise-745404749079-us-east-1-an/training-data/support_ticket_triage.jsonl X View [View](#) [Browse S3](#)

Output data [Info](#)
Choose S3 location to store validation results, metrics, logs, and training data.

S3 location
Q s3://bucket/prefix/object View [View](#) [Browse S3](#)

Browse S3 to select the S3 bucket prefix where the output will be stored.

► **VPC settings - optional**
Choose a VPC configuration to access Amazon S3 data source located in your virtual private cloud (VPC). You can create and manage VPC, subnets and security groups in [Amazon VPC](#)

10. Select the **output/** folder from the S3 bucket, then click **Choose** to use it as the destination for storing the fine-tuning job outputs.

Choose an archive in S3

[S3 buckets](#) > k21-finetune-nova-enterprise-745404749079-us-east-1-an

Objects (1/2)

Find object by prefix			
	Key	Last modified	Size
☒	output/	-	-
☐	training-data/	-	-

Cancel

Choose

11. Under **Service access**, choose **Create and use a new service role**, enter **support-role** as the service role name, and then click **Create job** to start the fine-tuning job.

Service access [Info](#)

To run batch inference, you must use a service role with permissions to access and write to S3 on your behalf.

Choose a method to authorize Bedrock

☐ Use an existing service role

☒ Create and use a new service role

Service role name

support-role

Maximum 64 characters. Use alphanumeric and '+=, @, _' characters.

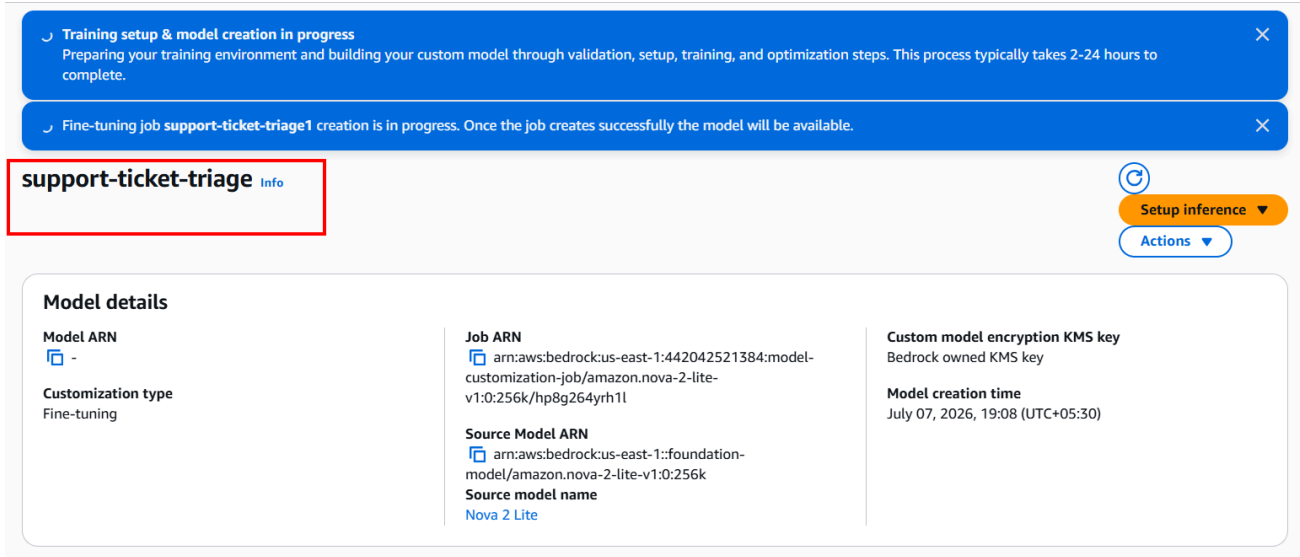
[View permission details](#)

Set up inference for your fine-tuned model

After you create your fine-tuned model, you must set up inference for it. To learn more about inference options, [See Here](#)

Cancel **Create job**

12. Job status starts at **In progress**. Wait until it shows **Active** (a few hours typically).

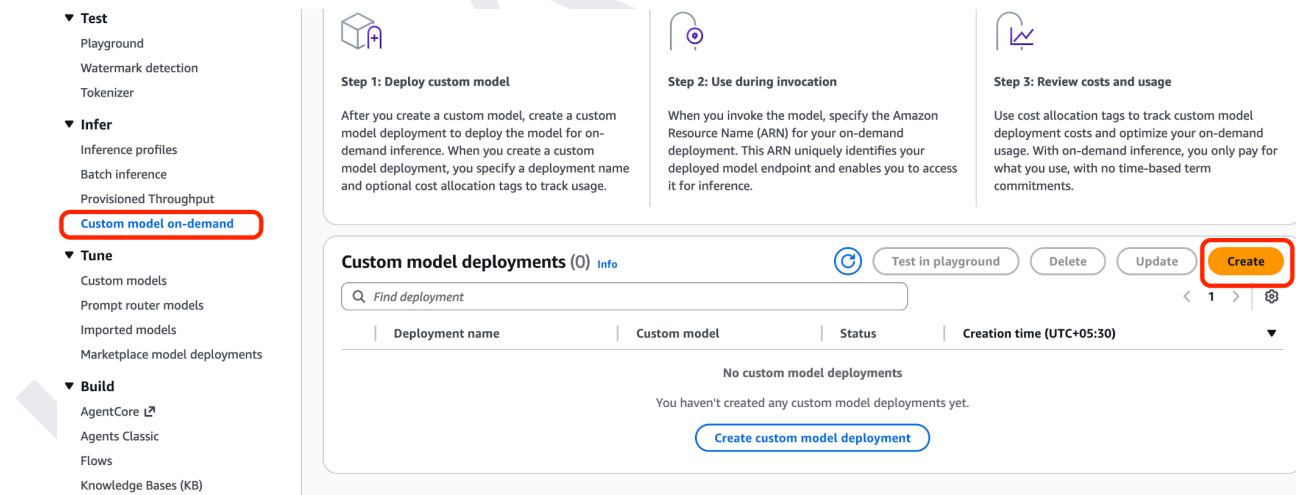


The screenshot shows the Amazon Bedrock console interface. At the top, there are two blue status bars: "Training setup & model creation in progress" and "Fine-tuning job support-ticket-triage1 creation is in progress. Once the job creates successfully the model will be available." Below these, the job name "support-ticket-triage" is highlighted with a red box. To the right, there are buttons for "Setup inference" and "Actions". The "Model details" section is visible, showing fields for Model ARN, Customization type (Fine-tuning), Job ARN, Source Model ARN, Source model name (Nova 2 Lite), Custom model encryption KMS key, and Model creation time (July 07, 2026, 19:08 (UTC+05:30)).

Note: For us, it took 4 hours to be in Active Status.

13. Deploy for on-demand inference

14. In the Bedrock dashboard, under **Infer**, choose **Custom model on demand**.



The screenshot shows the Amazon Bedrock console interface. On the left sidebar, the "Infer" tab is selected, and "Custom model on-demand" is highlighted with a red box. The main content area shows three steps: "Step 1: Deploy custom model", "Step 2: Use during invocation", and "Step 3: Review costs and usage". Below these steps, there is a section for "Custom model deployments (0)". The "Create" button is highlighted with a red box. The table below shows no custom model deployments yet.

Deployment name	Custom model	Status	Creation time (UTC+05:30)
No custom model deployments			

15. In the **Deployment details** section, enter **support-triage-inference** as the deployment name, then click **Select model** to choose the fine-tuned model for deployment.

Deployment details

Name

Valid characters are a-z, A-Z, 0-9, _(underscore) and -(hyphen).The name can have up to 63 characters.

Description - optional

The description can have up to 2048 characters

Select model

Select a custom model for creating custom model deployment.

Select model

16. In the **Select model** dialog, choose **Custom models**, select your newly fine-tuned model, and click **Apply** to use it for the deployment.

Select model

Find available models and inference

Input/output filter is not available for custom and managed endpoints.

1. Categories

Custom & managed endpoi... ▼

Custom models

2. Models

patient-triage-model
Type: Fine-tuned model

ecommerce-copy-model
Type: Fine-tuned model

banking-assistant-model
Type: Fine-tuned model

pharma-triage-model
Type: Fine-tuned model

it-helpdesk-model
Type: Fine-tuned model

fine-tuned-model-df81d92b
Type: Fine-tuned model

Cancel

Apply

17. Verify the deployment name and selected fine-tuned model, then click **Create** to deploy the custom model for inference.

Deployment details

Name

Valid characters are a-z, A-Z, 0-9, _(underscore) and -(hyphen).The name can have up to 63 characters.

Description - optional

The description can have up to 2048 characters

Select model

Select a custom model for creating custom model deployment.


fine-tuned-mo...
ⓘ
✎

► **Tags - optional**

Cancel
Create

18. Wait for the status to change from **In progress** to **Active** (usually 5–10 min).

19. Wait until the deployment status changes to **Active**, then click **Test in playground** to verify the fine-tuned model by sending sample prompts.

support-triage-inference

Test in playground
Actions ▼

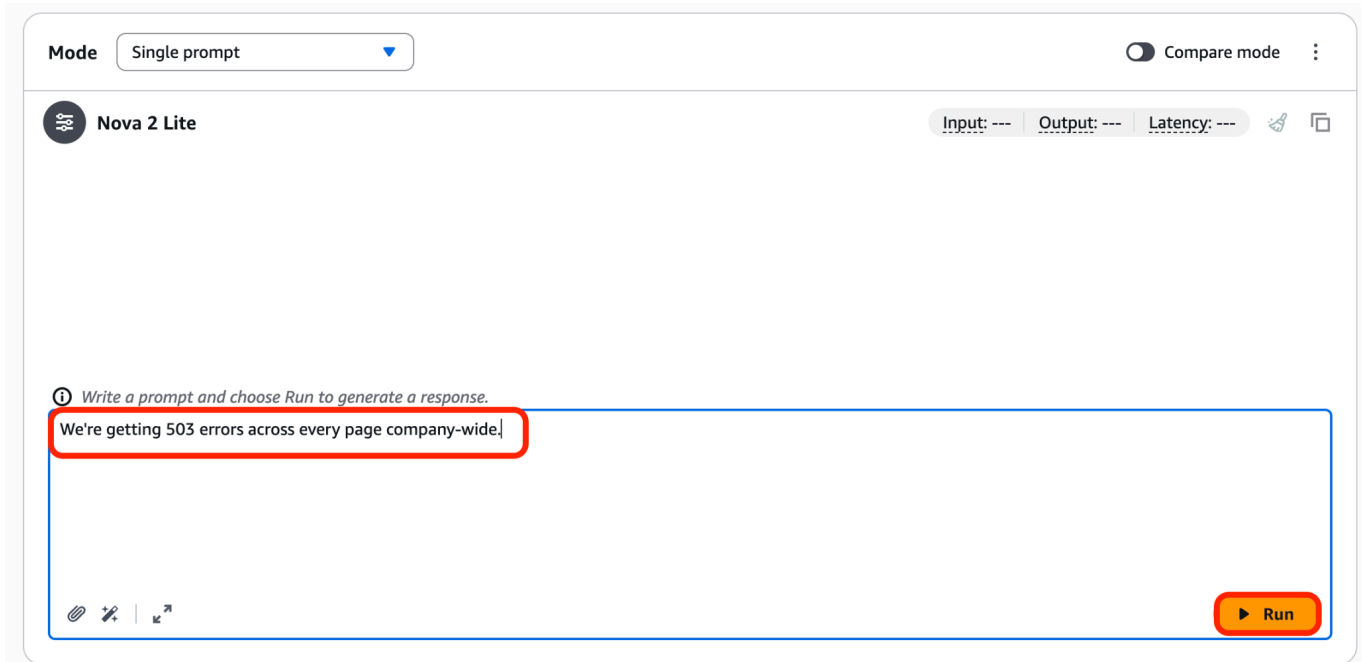
Custom model deployment overview

Deployment name support-triage-inference	Deployment ARN  arn:aws:bedrock:us-east-1:745404749079:custom-model-deployment/ycrc420abvmu	Creation time July 12, 2026, 10:21 (UTC+05:30)
Description -	Custom model fine-tuned-model-df81d92b	Last updated July 12, 2026, 10:23 (UTC+05:30)
Status  Active		

► **Job tags**

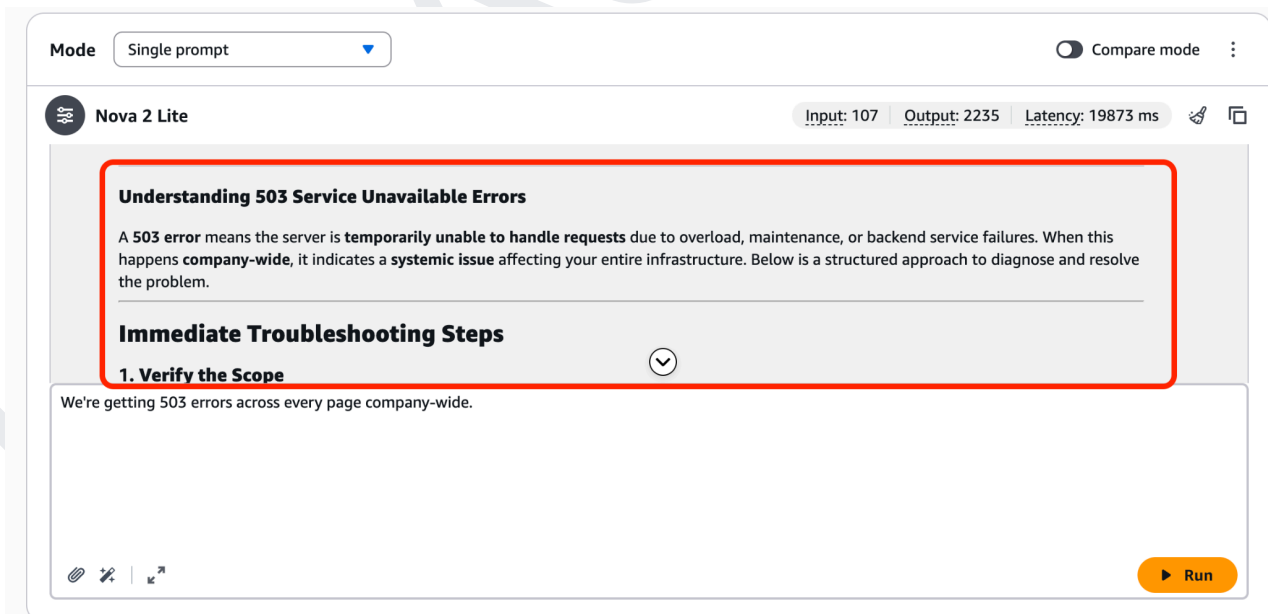
A tag is a label that you assign to an AWS resource. You can use tags to search and filter your resources or track your AWS costs.

20. Enter a sample prompt, such as **"We're getting 503 errors across every page company-wide."**, and click **Run** to test the deployed fine-tuned model and review its response.



The screenshot shows the K21Academy playground interface. At the top, there's a 'Mode' dropdown set to 'Single prompt' and a 'Compare mode' toggle. Below this, the model name 'Nova 2 Lite' is displayed. The input field contains the prompt: "We're getting 503 errors across every page company-wide.". A red box highlights this input field. At the bottom right, there is an orange 'Run' button. The interface also shows placeholders for 'Input: ---', 'Output: ---', and 'Latency: ---'.

21. Review the model's response in the playground to verify that the fine-tuned model correctly understands the prompt and provides an accurate, domain-specific troubleshooting recommendation.



The screenshot shows the K21Academy playground interface with the model's response. The 'Run' button has been clicked, and the output is displayed. The response is structured as follows:

- Understanding 503 Service Unavailable Errors**
A **503 error** means the server is **temporarily unable to handle requests** due to overload, maintenance, or backend service failures. When this happens **company-wide**, it indicates a **systemic issue** affecting your entire infrastructure. Below is a structured approach to diagnose and resolve the problem.
- Immediate Troubleshooting Steps**
 - 1. Verify the Scope**

The input prompt "We're getting 503 errors across every page company-wide." is visible at the bottom of the response area. A red box highlights the first two sections of the response. The interface also shows 'Input: 107', 'Output: 2235', and 'Latency: 19873 ms'.

22. Enter another sample prompt, such as **"I was charged twice this month, please refund the duplicate."**, and click **Run** to verify that the fine-tuned model generates an appropriate,

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domain-specific customer support response.

Mode

Single prompt

Compare mode

Nova 2 Lite

Input: 108 Output: 1105 Latency: 9293 ms

Reasoning

[REDACTED].

We're Here to Help with Your Duplicate Charge!

I'm sorry to hear about the duplicate charge — that can be frustrating! I'll assist you in getting this resolved as quickly as possible. To process your refund request efficiently, I'll need a few details from you.

I was charged twice this month, please refund the duplicate.

Run

13. SCENARIO 2: IT / DEVOPS L1 HELPDESK

This scenario teaches the model to act as a first-line IT support assistant capable of diagnosing and resolving common employee technology issues such as VPN disconnections, slow system performance, application crashes, and Microsoft Teams audio problems. L1 helpdesk teams handle a high volume of repetitive, predictable issues that follow known troubleshooting paths. Fine-tuning the model on real helpdesk examples means it can respond with accurate, step-by-step guidance without human intervention, allowing IT staff to focus on more complex escalations. For organizations with large workforces, this directly reduces ticket backlogs and improves employee productivity and satisfaction

1. From the **Create** menu, select **Supervised fine-tuning job** to start creating a new fine-tuning job using your labeled training dataset.

Custom models [Info](#)

Customize models using available customization processes. Access and manage custom models based on your region's capabilities.

▼ Customization techniques

Reinforcement fine-tuning - new

Build more accurate models using reward-based learning with less manual effort. Ideal for complex, multi-step, reasoning tasks.

How it works

- Step 1 - Choose model
- Step 2 - Add input data
- Step 3 - Add reward function

Supervised fine-tuning

Adapts pre-trained models to specific tasks using labeled data to improve performance in specific domains.

How it works

- Step 1 - Choose model
- Step 2 - Add input data

Distillation

Creates smaller, faster models by using a large teacher model to train a specialized student model with your invocation logs or generated training data.

How it works

- Step 1 - Choose teacher/student model
- Step 2 - Add input data

Models (0) [Info](#)

View all your custom models together in one unified table. Each model shows its training job, with easy setup for deployment and inference.

Custom model name	Model Status	Job name	Job status	Creation time (UTC-07:00)	
No models					

No models

Create your first model to get started



Stop Job

Setup inference ▼

Actions ▼

Create ▲

Reinforcement fine tuning job - new

Supervised fine-tuning job

Distillation job

2. Enter **finetune-it-helpdesk** as the **Job name**, then click **Select model** to choose the foundation model that you want to fine-tune.

Create Fine-tuning job [Info](#)

Select the model you wish to fine-tune and submit your data location.

Job configuration

Job name
Enter a name to identify the training job necessary to pre-train and create a new model.

finetune-it-helpdesk

Maximum of 63 alphanumeric characters. Cannot include hyphens (-) and spaces. Must be unique within your account in AWS Region.

► **Tags - optional**

Model details

Source model
Choose from a list of models that you wish to customize with using your own data.

Select model

Fine-tuned model name
Enter a name to identify the new fine-tuned model.

fine-tuned-model-8f6f929f

Maximum of 63 alphanumeric characters. Cannot include hyphens (-) and spaces. Must be unique within your account in AWS Region.

☐ **KMS key selection** [Info](#)

3. Select **Amazon** as the model provider, choose **Nova 2 Lite v1** as the source model, and click **Apply**.

Select model ×

Find available models and inference

Input **Output**
Text (1) **Text** (1)

1. Categories

Serverless model providers

Amazon

2. Models

Nova Pro 1.0
Input: Text, Image, Video
Output: Text

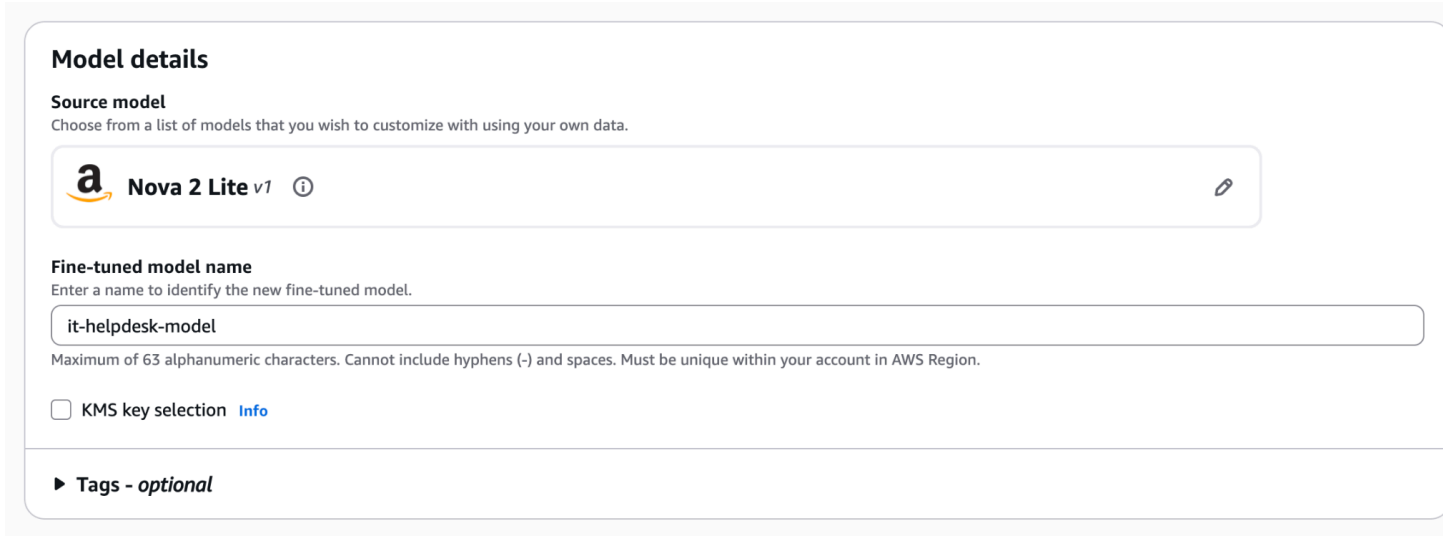
Nova 2 Lite v1
Input: Text, Image, Video
Output: Text

Nova Lite 1.0
Input: Text, Image, Video
Output: Text

Nova Micro 1.0
Input: Text
Output: Text

[Cancel](#) [Apply](#)

- Verify that **Nova 2 Lite v1** is selected as the **Source model**, enter **it-helpdesk-model** as the **Fine-tuned model name**, and leave the remaining settings as default.



Model details

Source model
Choose from a list of models that you wish to customize with using your own data.

 **Nova 2 Lite v1** 

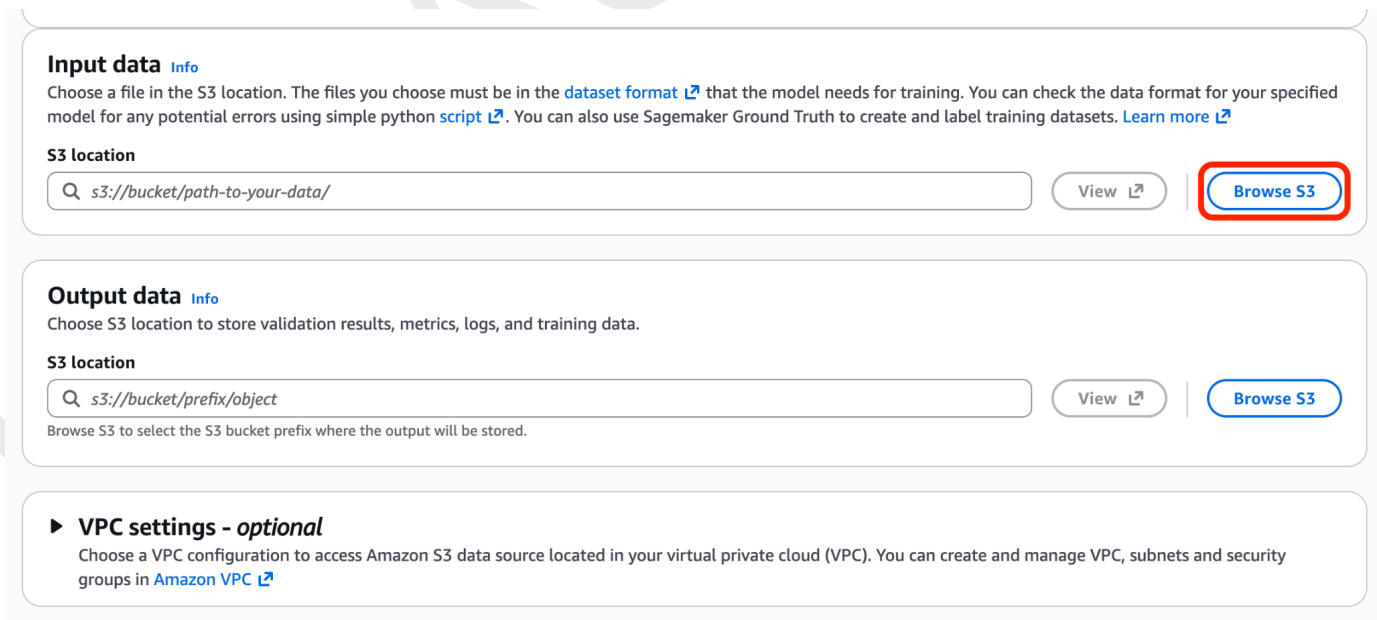
Fine-tuned model name
Enter a name to identify the new fine-tuned model.

Maximum of 63 alphanumeric characters. Cannot include hyphens (-) and spaces. Must be unique within your account in AWS Region.

☐ KMS key selection [Info](#)

► **Tags - optional**

- Hyperparameters: defaults.
- Click **Browse S3** under **Input data** to select the training dataset from your Amazon S3 bucket. Then, choose an S3 location under **Output data** where the fine-tuning job results and model artifacts will be stored.



Input data [Info](#)
Choose a file in the S3 location. The files you choose must be in the [dataset format](#) that the model needs for training. You can check the data format for your specified model for any potential errors using simple python [script](#). You can also use SageMaker Ground Truth to create and label training datasets. [Learn more](#)

S3 location

[View](#) [Browse S3](#)

Output data [Info](#)
Choose S3 location to store validation results, metrics, logs, and training data.

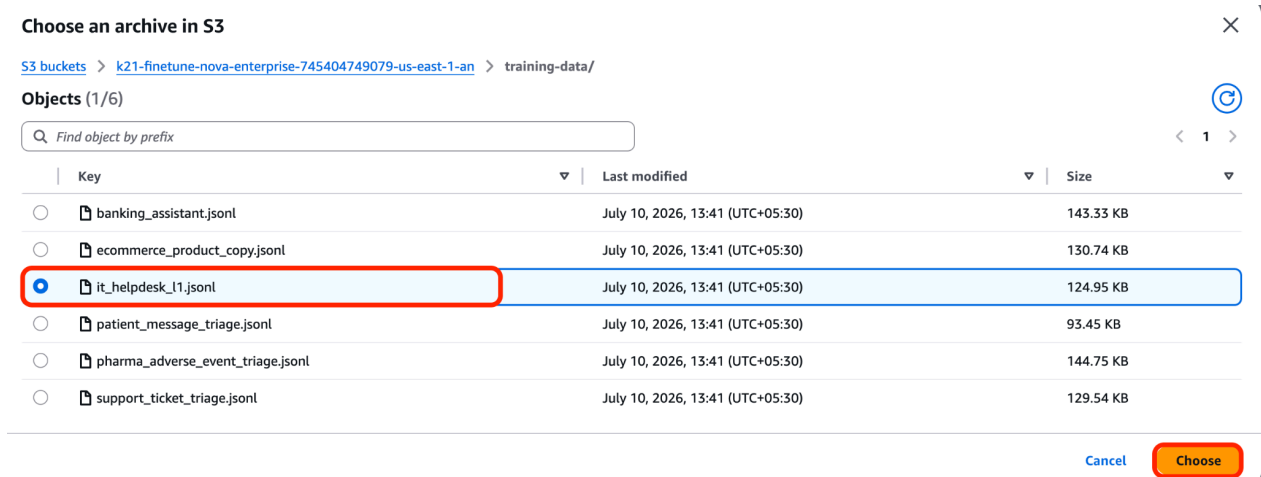
S3 location

[View](#) [Browse S3](#)

Browse S3 to select the S3 bucket prefix where the output will be stored.

► **VPC settings - optional**
Choose a VPC configuration to access Amazon S3 data source located in your virtual private cloud (VPC). You can create and manage VPC, subnets and security groups in [Amazon VPC](#)

- Select **it_helpdesk_[1].jsonl**, then click **Choose** to use it as the training dataset for the fine-tuning job.



8. For the **Output data** location, click **Browse S3** and select the S3 folder where the fine-tuning outputs, logs, and model artifacts will be stored (for example, `s3://k21-finetune-nova-enterprise/output/support-triage/`).

Input data [Info](#)

Choose a file in the S3 location. The files you choose must be in the [dataset format](#) that the model needs for training. You can check the data format for your specified model for any potential errors using simple python [script](#). You can also use Sagemaker Ground Truth to create and label training datasets. [Learn more](#)

S3 location

[View](#) [Browse S3](#)

Output data [Info](#)

Choose S3 location to store validation results, metrics, logs, and training data.

S3 location

[View](#) [Browse S3](#)

Browse S3 to select the S3 bucket prefix where the output will be stored.

► **VPC settings - optional**

Choose a VPC configuration to access Amazon S3 data source located in your virtual private cloud (VPC). You can create and manage VPC, subnets and security groups in [Amazon VPC](#)

9. Select the **output/** folder from the S3 bucket, then click **Choose** to use it as the destination for storing the fine-tuning job outputs.

Choose an archive in S3

[S3 buckets](#) > k21-finetune-nova-enterprise-745404749079-us-east-1-an

Objects (1/2)

Find object by prefix

Key	Last modified	Size
☒ output/	-	-
☐ training-data/	-	-

[Cancel](#)
[Choose](#)

10. Select **Create and use a new service role**, enter **it-helpdesk-model-role** as the **Service role name**, and click **Create job** to start the fine-tuning job.

Service access [Info](#)

To run batch inference, you must use a service role with permissions to access and write to S3 on your behalf.

Choose a method to authorize Bedrock

☐ Use an existing service role
☒ Create and use a new service role

Service role name

Maximum 64 characters. Use alphanumeric and '+=, @-_' characters.

[View permission details](#)

Set up inference for your fine-tuned model

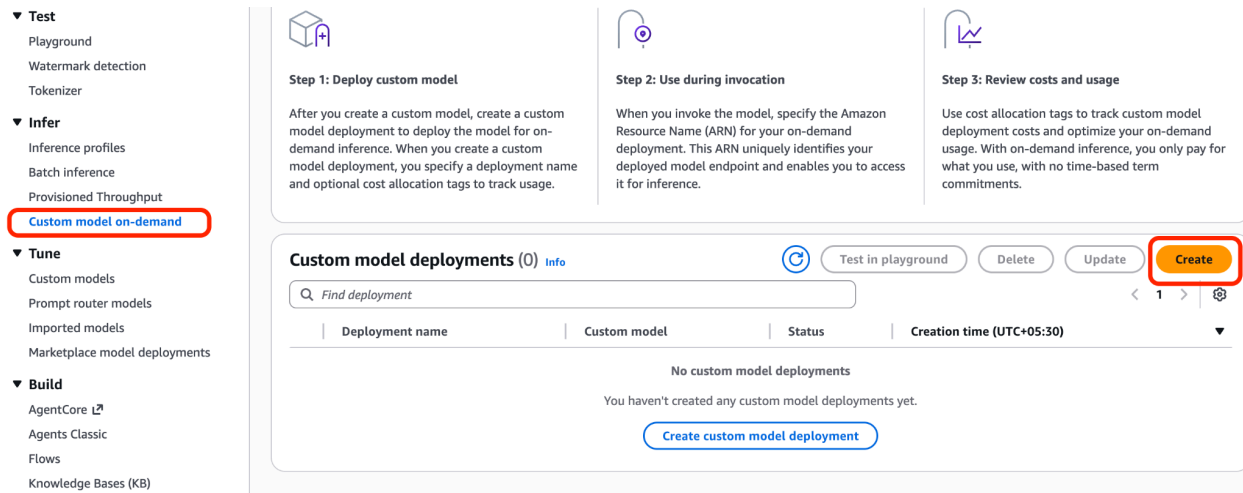
After you create your fine-tuned model, you must set up inference for it. To learn more about inference options, [See Here](#)

[Cancel](#)
[Create job](#)

Note: For us, it took 4 hours to be in Active Status.

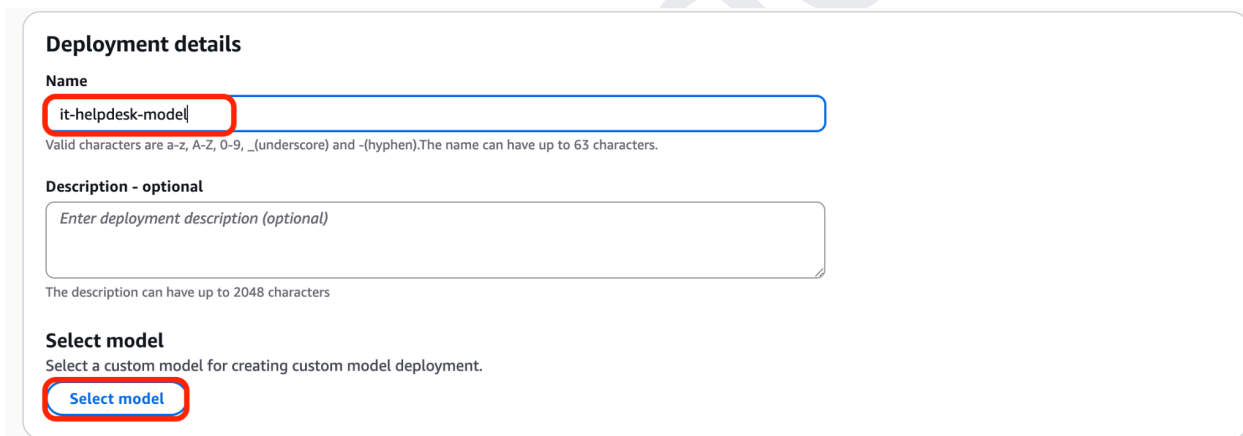
11. Deploy for on-demand inference

12. From the left navigation pane, select **Custom model on-demand**, then click **Create** to deploy your fine-tuned model for inference.



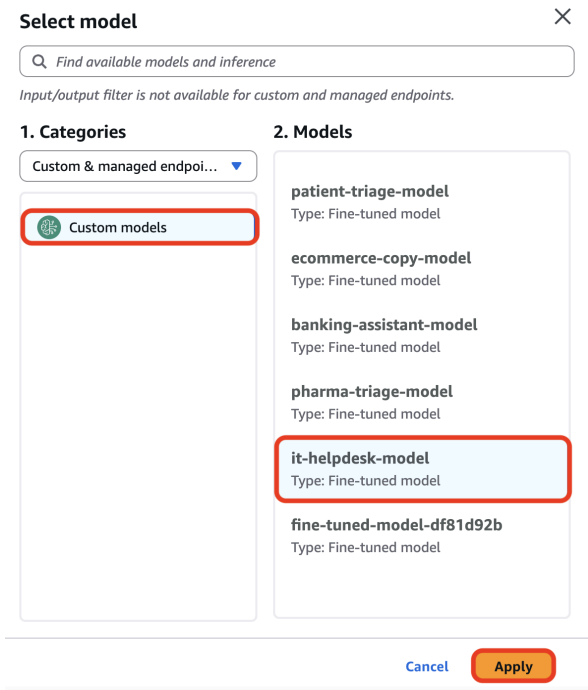
The screenshot shows the AWS SageMaker console interface. On the left, the navigation pane is expanded to 'Infer', and 'Custom model on-demand' is highlighted with a red box. The main content area shows three steps: 'Step 1: Deploy custom model', 'Step 2: Use during invocation', and 'Step 3: Review costs and usage'. Below these steps, the 'Custom model deployments (0)' section is visible, with the 'Create' button highlighted in a red box. The 'Create custom model deployment' button is also visible at the bottom of the section.

13. Enter **it-helpdesk-model** as the **Deployment name**, then click **Select model** to choose the fine-tuned model for deployment.



The screenshot shows the 'Deployment details' form in the AWS SageMaker console. The 'Name' field is filled with 'it-helpdesk-model' and is highlighted with a red box. Below the name field, there is a description field and a 'Select model' button, which is also highlighted with a red box. The form includes instructions for valid characters in the name and the maximum length for the description.

14. Select **Custom models**, choose **it-helpdesk-model**, and then click **Apply** to use it for the deployment.



Select model

Find available models and inference

Input/output filter is not available for custom and managed endpoints.

1. Categories

Custom & managed endpoi...

Custom models

2. Models

patient-triage-model
Type: Fine-tuned model

ecommerce-copy-model
Type: Fine-tuned model

banking-assistant-model
Type: Fine-tuned model

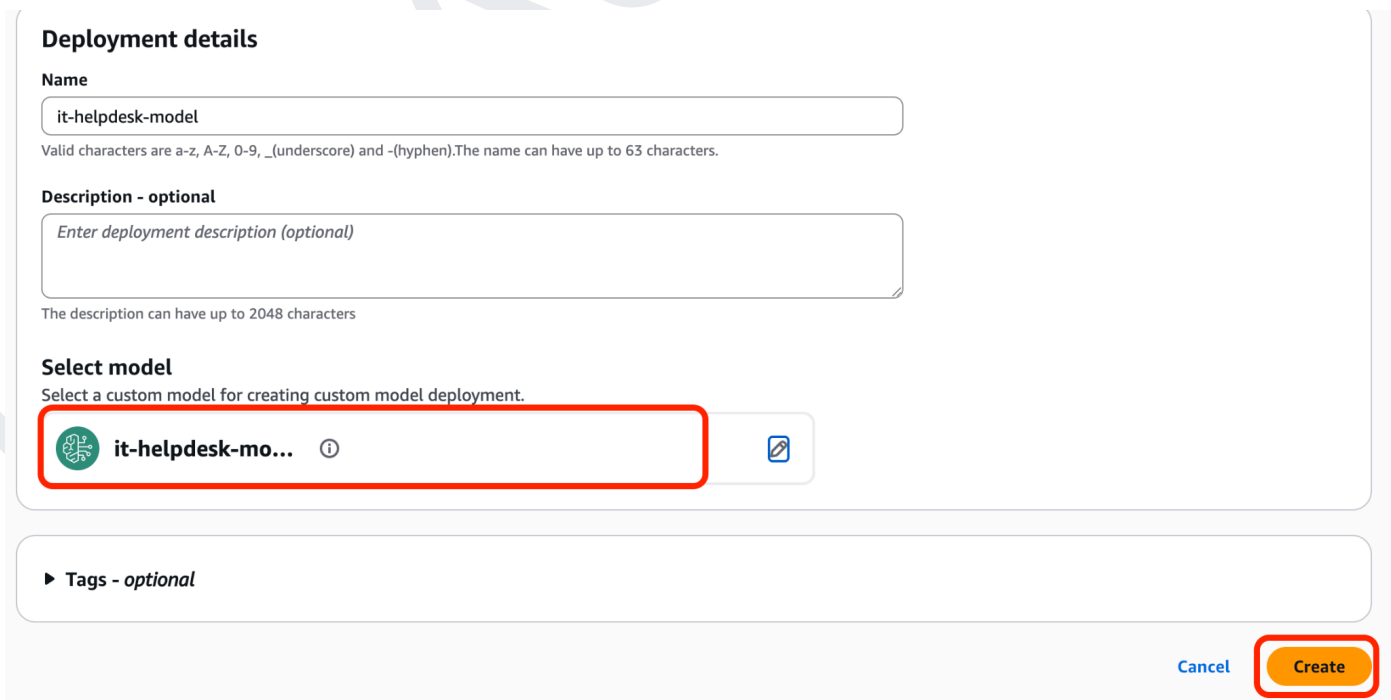
pharma-triage-model
Type: Fine-tuned model

it-helpdesk-model
Type: Fine-tuned model

fine-tuned-model-df81d92b
Type: Fine-tuned model

Cancel Apply

15. Verify that **it-helpdesk-model** is selected as the deployment model, then click **Create** to deploy the fine-tuned model for inference.



Deployment details

Name

it-helpdesk-model

Valid characters are a-z, A-Z, 0-9, _(underscore) and -(hyphen).The name can have up to 63 characters.

Description - optional

Enter deployment description (optional)

The description can have up to 2048 characters

Select model

Select a custom model for creating custom model deployment.

it-helpdesk-mo... ⓘ

Tags - optional

Cancel Create

16. Wait until the deployment status changes to **Active**, then click **Test in playground** to verify your fine-tuned model with sample prompts.

it-helpdesk-model Test in playground Actions

Custom model deployment overview

Deployment name it-helpdesk-model	Deployment ARN arn:aws:bedrock:us-east-1:745404749079:custom-model-deployment/wiydrd61amah	Creation time July 12, 2026, 10:25 (UTC+05:30)
Description -	Custom model it-helpdesk-model	Last updated July 12, 2026, 10:28 (UTC+05:30)
Status Active		

Job tags
A tag is a label that you assign to an AWS resource. You can use tags to search and filter your resources or track your AWS costs.

17. Enter the prompt **My VPN keeps disconnecting every few minutes**, then click **Run** to test the fine-tuned model's response.

Mode: Single prompt Compare mode

Nova 2 Lite Input: --- Output: --- Latency: ---

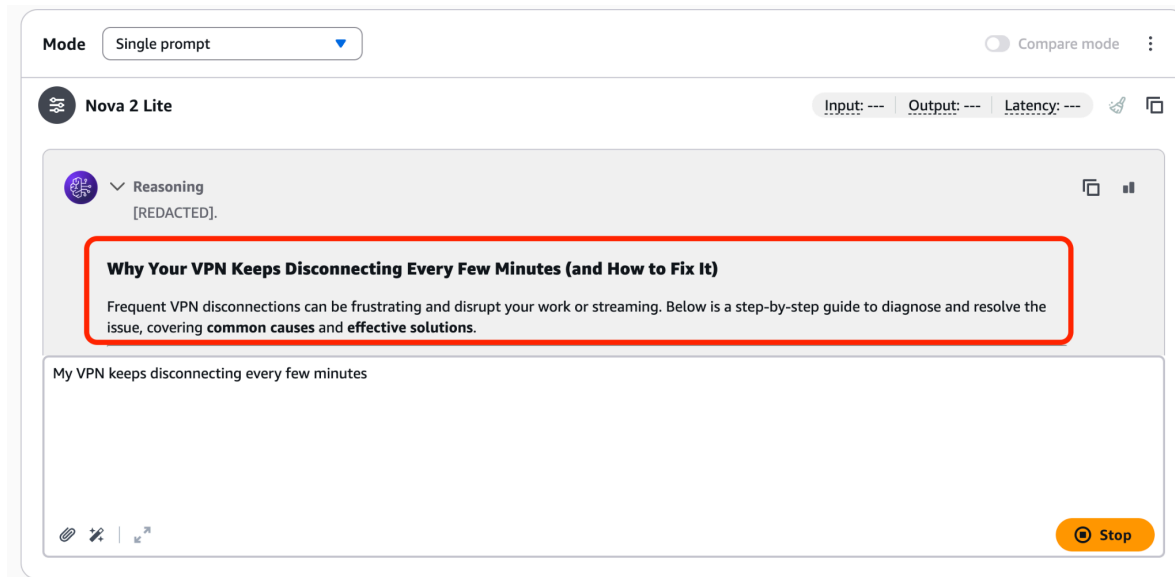
Write a prompt and choose Run to generate a response.

My VPN keeps disconnecting every few minutes

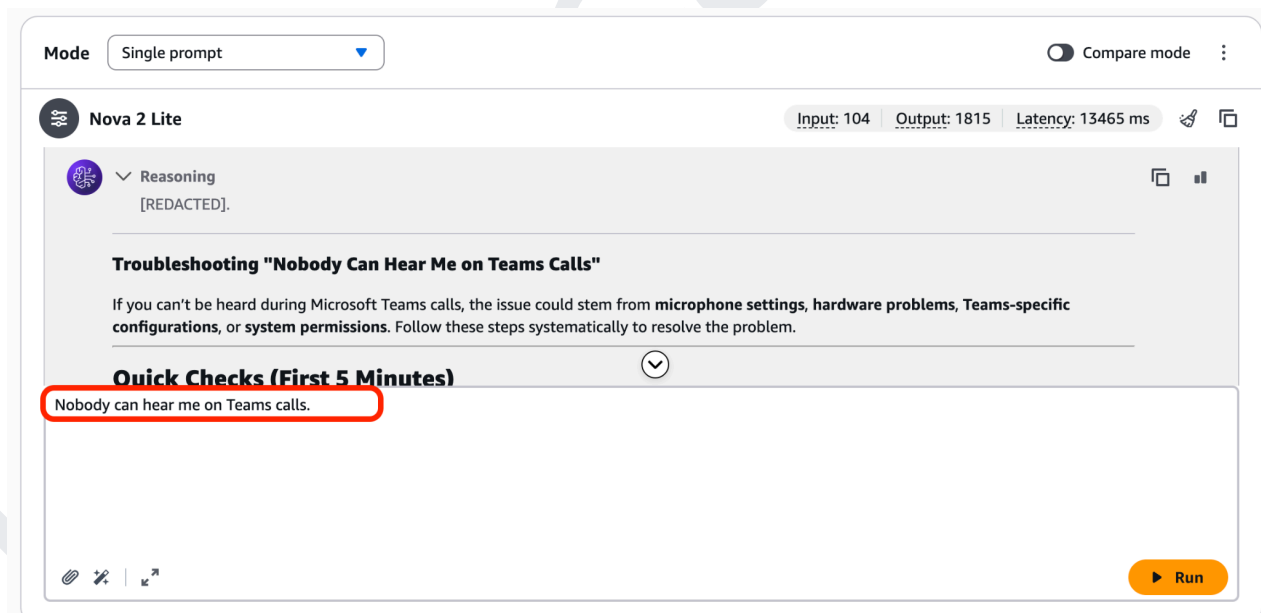
Run

18. Review the generated response to confirm that the fine-tuned model provides relevant troubleshooting guidance for the VPN connectivity issue, indicating that the customization was

successful.



19. Review the generated response to verify that the fine-tuned model provides accurate and relevant troubleshooting steps for the Microsoft Teams audio issue, confirming that the model has been successfully customized.



14. SCENARIO 3: PHARMACOVIGILANCE ADVERSE-EVENT TRIAGE

This scenario fine-tunes the model to analyze patient safety reports and classify them as either Serious Adverse Events (SAEs) or non-serious events, following standard pharmacovigilance guidelines. In the pharmaceutical and clinical trial industry, every adverse event reported by a patient or healthcare provider must be reviewed, classified, and escalated according to strict regulatory timelines — missing a serious event can have legal and patient safety consequences. Training the model on adverse-event examples enables it to quickly surface critical cases that require immediate reporting, while routing non-serious events through standard documentation workflows. This reduces the manual burden on safety teams and helps ensure regulatory compliance at scale.

1. Click **Create** and select **Supervised fine-tuning job** to start creating a new fine-tuning job for your custom model.

Custom models Info
 Customize models using available customization processes. Access and manage custom models based on your region's capabilities.

▼ Customization techniques

Reinforcement fine-tuning - new
 Build more accurate models using reward-based learning with less manual effort. Ideal for complex, multi-step, reasoning tasks.

How it works
 Step 1 - Choose model
 Step 2 - Add input data
 Step 3 - Add reward function

Supervised fine-tuning
 Adapts pre-trained models to specific tasks using labeled data to improve performance in specific domains.

How it works
 Step 1 - Choose model
 Step 2 - Add input data

Distillation
 Creates smaller, faster models by using a large teacher model to train a specialized student model with your invocation logs or generated training data.

How it works
 Step 1 - Choose teacher/student model
 Step 2 - Add input data

Models (0) Info

Stop Job

Setup inference ▼

Actions ▼

Create ▲

View all your custom models together in one unified table. Each model shows its training job, with easy setup for deployment a

Custom model name	Model Status	Job name	Job status	Creation time (UTC-07:00) ▼	
No models					

Reinforcement fine tuning job - new
Supervised fine-tuning job
 Distillation job

2. In the **Job name** field, enter **finetune-pharma-triage**, then click **Select model** under **Source model**.

Create Fine-tuning job [Info](#)
Select the model you wish to fine-tune and submit your data location.

Job configuration
Job name
Enter a name to identify the training job necessary to pre-train and create a new model.

Maximum of 63 alphanumeric characters. Cannot include hyphens (-) and spaces. Must be unique within your account in AWS Region.

► **Tags - optional**

Model details
Source model
Choose from a list of models that you wish to customize with using your own data.

Fine-tuned model name
Enter a name to identify the new fine-tuned model.

Maximum of 63 alphanumeric characters. Cannot include hyphens (-) and spaces. Must be unique within your account in AWS Region.

3. Select **Amazon**, choose **Nova 2 Lite v1**, and click **Apply**.

Select model ✕

Input (1) **Output** (1)

1. Categories

Serverless model providers

Amazon

2. Models

Nova Pro 1.0
Input: Text, Image, Video
Output: Text

Nova 2 Lite v1
Input: Text, Image, Video
Output: Text

Nova Lite 1.0
Input: Text, Image, Video
Output: Text

Nova Micro 1.0
Input: Text
Output: Text

[Cancel](#) [Apply](#)

4. Verify that **Nova 2 Lite v1** is selected, enter **pharma-triage-model** as the **Fine-tuned model name**, then scroll down to **Input data**.

Model details

Source model
Choose from a list of models that you wish to customize with using your own data.

Nova 2 Lite v1 ⓘ

Fine-tuned model name
Enter a name to identify the new fine-tuned model.

pharma-triage-model

Maximum of 63 alphanumeric characters. Cannot include hyphens (-) and spaces. Must be unique within your account in AWS Region.

☐ KMS key selection [Info](#)

► **Tags - optional**

5. Hyperparameters: defaults.

6. Click **Browse S3** under **Input data** to select the training dataset from your Amazon S3 bucket. Then, choose an S3 location under **Output data** where the fine-tuning job results and model artifacts will be stored.

Input data [Info](#)

Choose a file in the S3 location. The files you choose must be in the [dataset format](#) that the model needs for training. You can check the data format for your specified model for any potential errors using simple python [script](#). You can also use Sagemaker Ground Truth to create and label training datasets. [Learn more](#)

S3 location

[View](#) [Browse S3](#)

Output data [Info](#)

Choose S3 location to store validation results, metrics, logs, and training data.

S3 location

[View](#) [Browse S3](#)

Browse S3 to select the S3 bucket prefix where the output will be stored.

► **VPC settings - optional**

Choose a VPC configuration to access Amazon S3 data source located in your virtual private cloud (VPC). You can create and manage VPC, subnets and security groups in [Amazon VPC](#)

7. Select **pharma_adverse_event_triage.jsonl**, then click **Choose**.

Choose an archive in S3

[S3 buckets](#) > [k21-finetune-nova-enterprise-745404749079-us-east-1-an](#) > [training-data/](#)

Objects (1/6)

	Key	Last modified	Size
☐	banking_assistant.jsonl	July 10, 2026, 13:41 (UTC+05:30)	143.33 KB
☐	ecommerce_product_copy.jsonl	July 10, 2026, 13:41 (UTC+05:30)	130.74 KB
☐	it_helpdesk_l1.jsonl	July 10, 2026, 13:41 (UTC+05:30)	124.95 KB
☐	patient_message_triage.jsonl	July 10, 2026, 13:41 (UTC+05:30)	93.45 KB
☒	pharma_adverse_event_triage.jsonl	July 10, 2026, 13:41 (UTC+05:30)	144.75 KB
☐	support_ticket_triage.jsonl	July 10, 2026, 13:41 (UTC+05:30)	129.54 KB

[Cancel](#) [Choose](#)

8. For the **Output data** location, click **Browse S3** and select the S3 folder where the fine-tuning outputs, logs, and model artifacts will be stored (for example,

s3://k21-finetune-nova-enterprise/output/support-triage/).

Input data [Info](#)

Choose a file in the S3 location. The files you choose must be in the [dataset format](#) that the model needs for training. You can check the data format for your specified model for any potential errors using simple python [script](#). You can also use Sagemaker Ground Truth to create and label training datasets. [Learn more](#)

S3 location



[View](#)

[Browse S3](#)

Output data [Info](#)

Choose S3 location to store validation results, metrics, logs, and training data.

S3 location

[View](#)

[Browse S3](#)

Browse S3 to select the S3 bucket prefix where the output will be stored.

► VPC settings - optional

Choose a VPC configuration to access Amazon S3 data source located in your virtual private cloud (VPC). You can create and manage VPC, subnets and security groups in [Amazon VPC](#)

- Select the **output/** folder from the S3 bucket, then click **Choose** to use it as the destination for storing the fine-tuning job outputs.

Choose an archive in S3

[S3 buckets](#) > k21-finetune-nova-enterprise-745404749079-us-east-1-an

Objects (1/2)

< 1 >

Key	Last modified	Size
☒  output/	-	-
☐  training-data/	-	-

[Cancel](#)

[Choose](#)

10. Select **Create and use a new service role**, enter **pharma-triage-model-role** as the service role name, and click **Create job**.

Service access Info

To run batch inference, you must use a service role with permissions to access and write to S3 on your behalf.

Choose a method to authorize Bedrock

☐ Use an existing service role
☒ **Create and use a new service role**

Service role name

Maximum 64 characters. Use alphanumeric and '+=, @-_' characters.

[View permission details](#)

Set up inference for your fine-tuned model

After you create your fine-tuned model, you must set up inference for it. To learn more about inference options, [See Here](#)

[Cancel](#) [Create job](#)

Note: For us, it took 4 hours to be in Active Status.

11. Deploy for on-demand inference
12. In the Bedrock dashboard, under **Infer**, choose **Custom model on demand**.

Test
Playground
Watermark detection
Tokenizer

Infer
Inference profiles
Batch inference
Provisioned Throughput
Custom model on-demand

Tune
Custom models
Prompt router models
Imported models
Marketplace model deployments

Build
AgentCore
Agents Classic
Flows
Knowledge Bases (KB)

Step 1: Deploy custom model
After you create a custom model, create a custom model deployment to deploy the model for on-demand inference. When you create a custom model deployment, you specify a deployment name and optional cost allocation tags to track usage.

Step 2: Use during invocation
When you invoke the model, specify the Amazon Resource Name (ARN) for your on-demand deployment. This ARN uniquely identifies your deployed model endpoint and enables you to access it for inference.

Step 3: Review costs and usage
Use cost allocation tags to track custom model deployment costs and optimize your on-demand usage. With on-demand inference, you only pay for what you use, with no time-based term commitments.

Custom model deployments (0) Info

[Test in playground](#) [Delete](#) [Update](#) [Create](#)

Deployment name	Custom model	Status	Creation time (UTC+05:30)
No custom model deployments			
You haven't created any custom model deployments yet.			

[Create custom model deployment](#)

13. In **Deployment details**, enter **pharma-triage-model** as the deployment name, then click **Select model**.

Deployment details

Name

pharma-triage-model

Valid characters are a-z, A-Z, 0-9, _(underscore) and -(hyphen).The name can have up to 63 characters.

Description - optional

Enter deployment description (optional)

The description can have up to 2048 characters

Select model

Select a custom model for creating custom model deployment.

Select model

14. Select **Custom models**, choose **pharma-triage-model**, and click **Apply**.

Select model

Find available models and inference

Input/output filter is not available for custom and managed endpoints.

1. Categories

Custom & managed endpoi...

Custom models

2. Models

patient-triage-model
Type: Fine-tuned model

ecommerce-copy-model
Type: Fine-tuned model

banking-assistant-model
Type: Fine-tuned model

pharma-triage-model
Type: Fine-tuned model

it-helpdesk-model
Type: Fine-tuned model

fine-tuned-model-df81d92b
Type: Fine-tuned model

Cancel

Apply

15. Verify that **pharma-triage-model** is selected, then click **Create** to deploy the custom model.

Deployment details

Name

Valid characters are a-z, A-Z, 0-9, _(underscore) and -(hyphen).The name can have up to 63 characters.

Description - optional

The description can have up to 2048 characters

Select model

Select a custom model for creating custom model deployment.

 **pharma-triage-...** 

Tags - optional

[Cancel](#) [Create](#)

16. Once the deployment status changes to **Active**, click **Test in playground** to test your fine-tuned model.

pharma-triage-model

[Test in playground](#) [Actions](#)

Custom model deployment overview

Deployment name pharma-triage-model	Deployment ARN  arn:aws:bedrock:us-east-1:745404749079:custom-model-deployment/3v8mboy2alko	Creation time July 12, 2026, 10:28 (UTC+05:30)
Description -	Custom model pharma-triage-model	Last updated July 12, 2026, 10:29 (UTC+05:30)
Status  Active		

Job tags

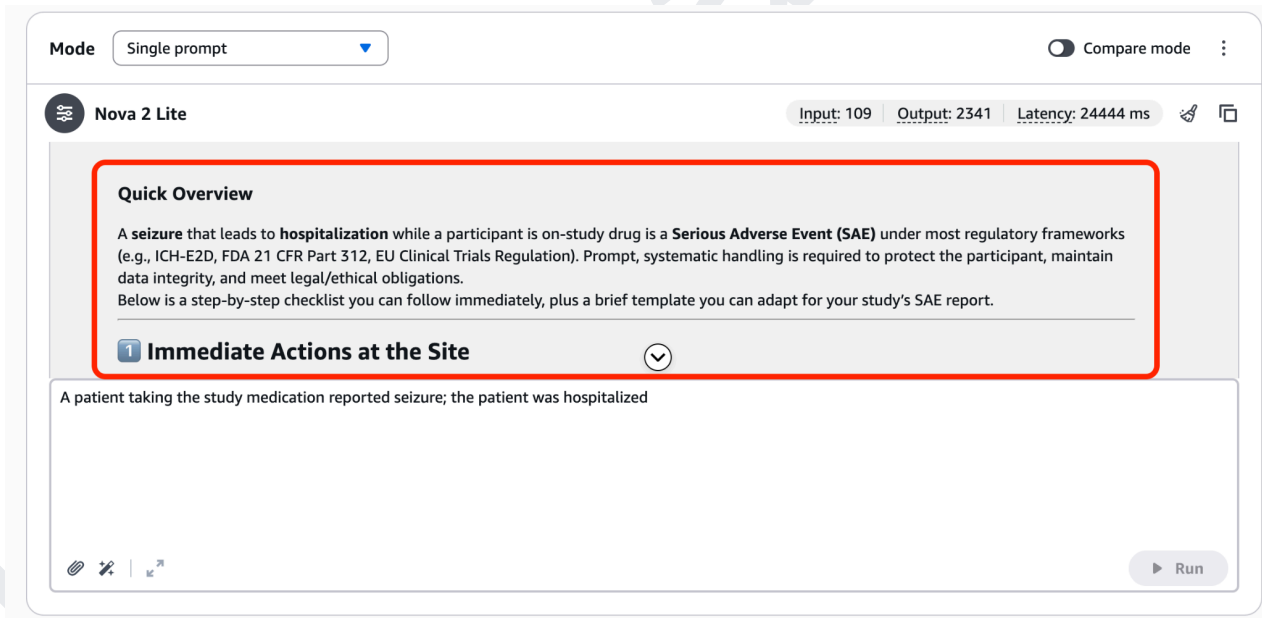
A tag is a label that you assign to an AWS resource. You can use tags to search and filter your resources or track your AWS costs.

17. In the playground, enter the prompt **"A patient taking the study medication reported seizure; the patient was hospitalized"**, then click **Run**.



The screenshot shows the K21Academy playground interface. At the top, there's a 'Mode' dropdown set to 'Single prompt' and a 'Compare mode' toggle. Below this, the model 'Nova 2 Lite' is selected. The input field contains the prompt: "A patient taking the study medication reported seizure; the patient was hospitalized". A red box highlights this input field. At the bottom right, there is a yellow 'Run' button.

18. Verify that the model correctly identifies the event as a **Serious Adverse Event (SAE)** and provides appropriate guidance for handling and reporting it.



The screenshot shows the K21Academy playground interface after the prompt has been run. The model's response is displayed in a box with a red border. The response is titled "Quick Overview" and contains the following text: "A seizure that leads to hospitalization while a participant is on-study drug is a **Serious Adverse Event (SAE)** under most regulatory frameworks (e.g., ICH-E2D, FDA 21 CFR Part 312, EU Clinical Trials Regulation). Prompt, systematic handling is required to protect the participant, maintain data integrity, and meet legal/ethical obligations. Below is a step-by-step checklist you can follow immediately, plus a brief template you can adapt for your study's SAE report." Below this text, there is a section titled "1 Immediate Actions at the Site" with a dropdown arrow. The input field at the bottom still contains the original prompt. The 'Run' button is now disabled.

19. Verify that the model correctly identifies the adverse event as **non-serious**, summarizes the key details, and recommends appropriate documentation and follow-up actions.

ModeSingle prompt

Compare mode

Nova 2 Lite

Input: 108Output: 1843Latency: 15546 ms

Analysis of the Patient Report: Nausea with Product A (Full Recovery)

1. Key Facts from the Report

- **Product:** Product A (identity unspecified; could be a medication, supplement, or diagnostic agent).
- **Adverse Event:** Nausea reported by the patient.
- **Outcome:** Full recovery (no lasting effects).

2. Immediate Actions & Documentation

A patient taking Product A reported nausea; the patient recovered fully

Run

15. SCENARIO 4: PATIENT MESSAGE TRIAGE (CLINIC ROUTING)

This scenario trains the model to read incoming patient messages and route them to the correct care pathway based on symptom urgency - distinguishing between medical emergencies, urgent care needs, routine appointments, and general administrative inquiries. In busy clinics and telehealth platforms, care coordinators are often overwhelmed by the volume of patient messages and the need to correctly prioritize responses. A fine-tuned triage model can instantly flag life-threatening symptoms like chest pain for emergency escalation, while directing routine appointment requests to scheduling staff. This improves patient safety, reduces triage errors, and helps clinical teams allocate their attention where it matters most.

1. Click **Create**, then select **Supervised fine-tuning job** to create a new custom model training job.

Custom models Info

Customize models using available customization processes. Access and manage custom models based on your region's capabilities.

▼ Customization techniques

Reinforcement fine-tuning - new

Build more accurate models using reward-based learning with less manual effort. Ideal for complex, multi-step, reasoning tasks.

How it works

Step 1 - Choose model
Step 2 - Add input data
Step 3 - Add reward function

Supervised fine-tuning

Adapts pre-trained models to specific tasks using labeled data to improve performance in specific domains.

How it works

Step 1 - Choose model
Step 2 - Add input data

Distillation

Creates smaller, faster models by using a large teacher model to train a specialized student model with your invocation logs or generated training data.

How it works

Step 1 - Choose teacher/student model
Step 2 - Add input data

Models (0) Info

View all your custom models together in one unified table. Each model shows its training job, with easy setup for deployment and inference.

Find model

Custom model name	Model Status	Job name	Job status	Creation time (UTC-07:00)	
No models					

No models

Continue your first model to get started



Stop Job

Setup inference ▼

Actions ▼

Create ▲

Reinforcement fine tuning job - new

Supervised fine-tuning job

Distillation job

2. In the **Job name** field, enter **finetune-patient-triage**, then click **Select model** to choose the foundation model for fine-tuning.

Create Fine-tuning job [Info](#)

Select the model you wish to fine-tune and submit your data location.

Job configuration

Job name

Enter a name to identify the training job necessary to pre-train and create a new model.

finetune-patient-triage

Maximum of 63 alphanumeric characters. Cannot include hyphens (-) and spaces. Must be unique within your account in AWS Region.

► Tags - optional

Model details

Source model

Choose from a list of models that you wish to customize with using your own data.

Select model

Fine-tuned model name

Enter a name to identify the new fine-tuned model.

fine-tuned-model-6d95326c

Maximum of 63 alphanumeric characters. Cannot include hyphens (-) and spaces. Must be unique within your account in AWS Region.

☐ KMS key selection [Info](#)

3. Select **Amazon** as the model provider, choose **Nova 2 Lite v1** as the source model, and click **Apply**.

Select model

Find available models and inference

InputText (1)

OutputText (1)

1. Categories

Serverless model providers

Amazon

2. Models

Nova Pro 1.0

Input: Text, Image, Video

Output: Text

Nova 2 Lite v1

Input: Text, Image, Video

Output: Text

Nova Lite 1.0

Input: Text, Image, Video

Output: Text

Nova Micro 1.0

Input: Text

Output: Text

Cancel

Apply

4. Confirm that **Nova 2 Lite v1** is selected as the source model. In the **Fine-tuned model name** field, enter **patient-triage-model1**, then continue to the training data configuration.

Model details

Source model
Choose from a list of models that you wish to customize with using your own data.

 **Nova 2 Lite v1** ⓘ

Fine-tuned model name
Enter a name to identify the new fine-tuned model.

patient-triage-model

Maximum of 63 alphanumeric characters. Cannot include hyphens (-) and spaces. Must be unique within your account in AWS Region.

☐ KMS key selection ⓘ

► **Tags - optional**

5. Hyperparameters: defaults.
6. Click **Browse S3** under **Input data** to select the training dataset from your Amazon S3 bucket. Then, choose an S3 location under **Output data** where the fine-tuning job results and model artifacts will be stored.

Input data ⓘ

Choose a file in the S3 location. The files you choose must be in the [dataset format](#) that the model needs for training. You can check the data format for your specified model for any potential errors using simple python [script](#). You can also use SageMaker Ground Truth to create and label training datasets. [Learn more](#)

S3 location

Q s3://bucket/path-to-your-data/ View ⓘ **Browse S3**

Output data ⓘ

Choose S3 location to store validation results, metrics, logs, and training data.

S3 location

Q s3://bucket/prefix/object View ⓘ **Browse S3**

Browse S3 to select the S3 bucket prefix where the output will be stored.

► **VPC settings - optional**

Choose a VPC configuration to access Amazon S3 data source located in your virtual private cloud (VPC). You can create and manage VPC, subnets and security groups in [Amazon VPC](#)

7. Under **Training data**, select **Choose** and browse to your S3 bucket. Select the **patient_message_triage.jsonl** training dataset, then click **Choose** to use it for

fine-tuning.

Choose an archive in S3

[S3 buckets](#) > [k21-finetune-nova-enterprise-745404749079-us-east-1-an](#) > training-data/

Objects (1/6)

Find object by prefix

	Key	Last modified	Size
☐	banking_assistant.jsonl	July 10, 2026, 13:41 (UTC+05:30)	143.33 KB
☐	ecommerce_product_copy.jsonl	July 10, 2026, 13:41 (UTC+05:30)	130.74 KB
☐	it_helpdesk_l1.jsonl	July 10, 2026, 13:41 (UTC+05:30)	124.95 KB
☒	patient_message_triage.jsonl	July 10, 2026, 13:41 (UTC+05:30)	93.45 KB
☐	pharma_adverse_event_triage.jsonl	July 10, 2026, 13:41 (UTC+05:30)	144.75 KB
☐	support_ticket_triage.jsonl	July 10, 2026, 13:41 (UTC+05:30)	129.54 KB

Cancel Choose

8. For the **Output data** location, click **Browse S3** and select the S3 folder where the fine-tuning outputs, logs, and model artifacts will be stored (for example, `s3://k21-finetune-nova-enterprise/output/support-triage/`).

Input data [Info](#)

Choose a file in the S3 location. The files you choose must be in the [dataset format](#) that the model needs for training. You can check the data format for your specified model for any potential errors using simple python [script](#). You can also use Sagemaker Ground Truth to create and label training datasets. [Learn more](#)

S3 location

Q s3://k21-finetune-nova-enterprise-745404749079-us-east-1-an/training-data/support_ticket_triage.jsonl X View [Browse S3](#)

Output data [Info](#)

Choose S3 location to store validation results, metrics, logs, and training data.

S3 location

Q s3://bucket/prefix/object View [Browse S3](#)

Browse S3 to select the S3 bucket prefix where the output will be stored.

► **VPC settings - optional**

Choose a VPC configuration to access Amazon S3 data source located in your virtual private cloud (VPC). You can create and manage VPC, subnets and security groups in [Amazon VPC](#)

- Select the **output/** folder from the S3 bucket, then click **Choose** to use it as the destination for storing the fine-tuning job outputs.

Choose an archive in S3

[S3 buckets](#) > k21-finetune-nova-enterprise-745404749079-us-east-1-an

Objects (1/2)

Find object by prefix

Key	Last modified	Size
☒ output/	-	-
☐ training-data/	-	-

[Cancel](#)
[Choose](#)

- Under **Service access**, select **Create and use a new service role**, enter **patient-triage-model** as the service role name, then click **Create job** to start the fine-tuning process.

Service access [Info](#)

To run batch inference, you must use a service role with permissions to access and write to S3 on your behalf.

Choose a method to authorize Bedrock

☐ Use an existing service role
☒ Create and use a new service role

Service role name

patient-triage-model

Maximum 64 characters. Use alphanumeric and '+', '=', '@', '-', '_' characters.

[View permission details](#)

Set up inference for your fine-tuned model

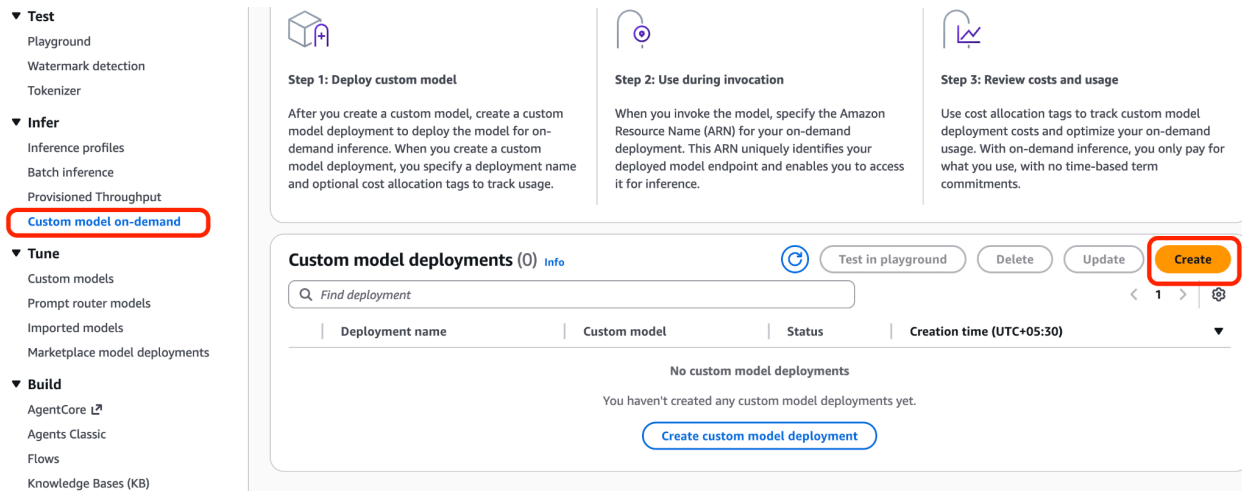
After you create your fine-tuned model, you must set up inference for it. To learn more about inference options, [See Here](#)

[Cancel](#)
[Create job](#)

Note: For us, it took 4 hours to be in Active Status.

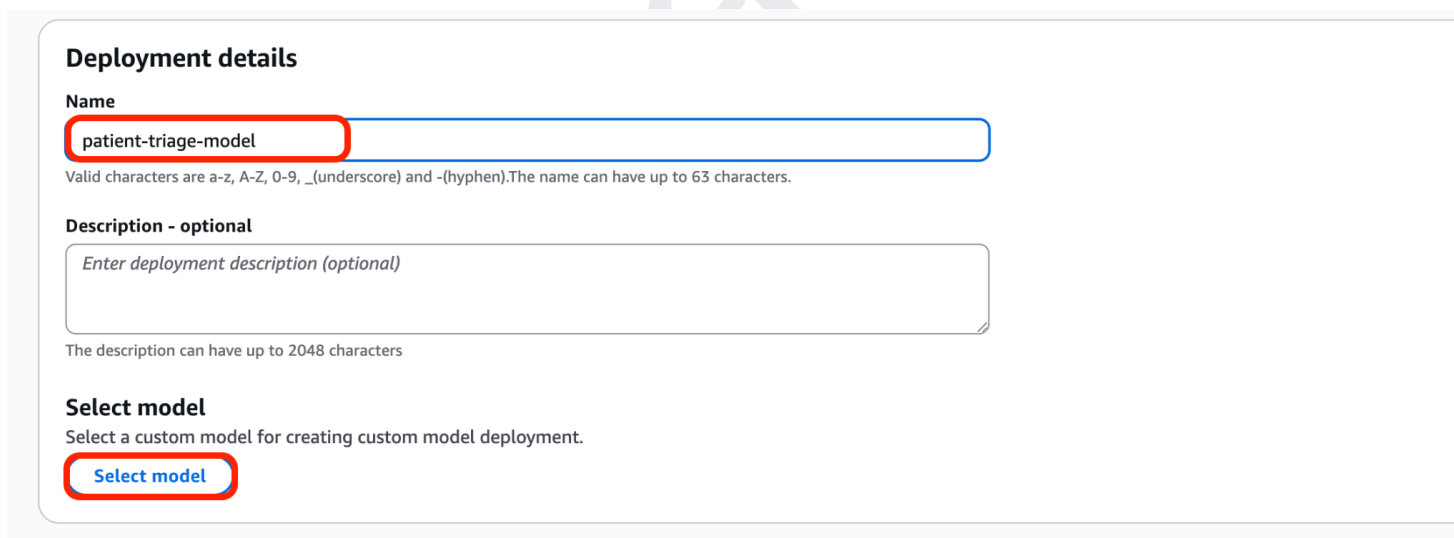
- Deploy for on-demand inference

12. In the Bedrock dashboard, under **Infer**, choose **Custom model on demand**.



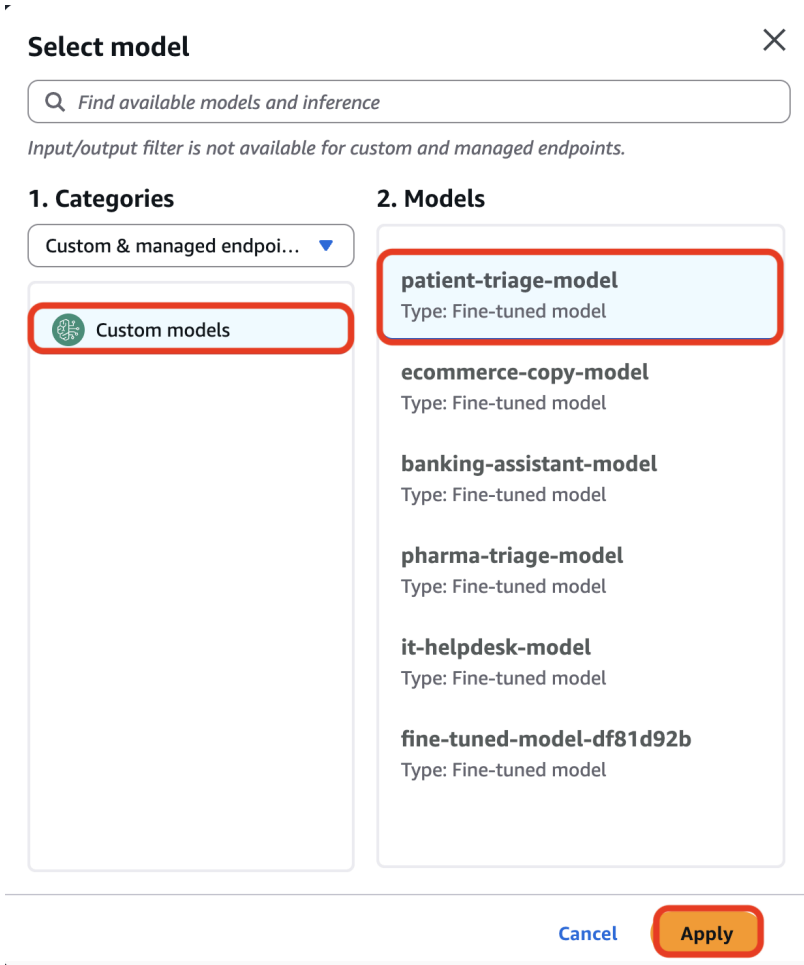
The screenshot shows the AWS Bedrock dashboard. On the left sidebar, under the 'Infer' section, 'Custom model on-demand' is highlighted with a red box. The main content area shows three steps: 'Step 1: Deploy custom model', 'Step 2: Use during invocation', and 'Step 3: Review costs and usage'. Below these steps, there is a section titled 'Custom model deployments (0) Info'. In this section, the 'Create' button is highlighted with a red box. Below the button, there is a table with columns: 'Deployment name', 'Custom model', 'Status', and 'Creation time (UTC+05:30)'. The table is currently empty, and a message states 'No custom model deployments. You haven't created any custom model deployments yet.' with a 'Create custom model deployment' button.

13. Navigate to **Custom model on-demand** under the **Infer** section. In **Deployment details**, enter **patient-triage-model** as the deployment name, then click **Select model** to choose the fine-tuned model for deployment.



The screenshot shows the 'Deployment details' form. The 'Name' field is filled with 'patient-triage-model' and is highlighted with a red box. Below the name field, there is a note: 'Valid characters are a-z, A-Z, 0-9, _(underscore) and -(hyphen).The name can have up to 63 characters.' The 'Description - optional' field is empty, with a placeholder text 'Enter deployment description (optional)'. Below the description field, there is a note: 'The description can have up to 2048 characters'. The 'Select model' section is titled 'Select model' and has a subtitle 'Select a custom model for creating custom model deployment.' The 'Select model' button is highlighted with a red box.

14. In the **Select model** dialog, choose **Custom models**, select **patient-triage-model** from the list of fine-tuned models, and click **Apply** to use it for deployment.



Select model ✕

Find available models and inference

Input/output filter is not available for custom and managed endpoints.

1. Categories

Custom & managed endpoi... ▼

Custom models

2. Models

patient-triage-model
Type: Fine-tuned model

ecommerce-copy-model
Type: Fine-tuned model

banking-assistant-model
Type: Fine-tuned model

pharma-triage-model
Type: Fine-tuned model

it-helpdesk-model
Type: Fine-tuned model

fine-tuned-model-df81d92b
Type: Fine-tuned model

Cancel Apply

15. Verify that **patient-triage-model** is selected under **Select model**, then click **Create** to deploy the fine-tuned model for on-demand inference.

Deployment details

Name
patient-triage-model
Valid characters are a-z, A-Z, 0-9, _(underscore) and -(hyphen).The name can have up to 63 characters.

Description - optional
Enter deployment description (optional)
The description can have up to 2048 characters

Select model
Select a custom model for creating custom model deployment.

patient-triage-... ⓘ

► **Tags - optional**

[Cancel](#) [Create](#)

16. Test in the Playground

17. Wait until the deployment status changes to **Active**, then click **Test in playground** to verify that the deployed fine-tuned model is ready for inference.

patient-triage-model [Test in playground](#) [Actions ▼](#)

Custom model deployment overview

Deployment name patient-triage-model	Deployment ARN arn:aws:bedrock:us-east-1:745404749079:custom-model-deployment/14qtwnixirh3	Creation time July 12, 2026, 10:31 (UTC+05:30)
Description -	Custom model patient-triage-model	Last updated July 12, 2026, 10:32 (UTC+05:30)
Status ✓ Active		

► **Job tags**
A tag is a label that you assign to an AWS resource. You can use tags to search and filter your resources or track your AWS costs.

18. In the playground, enter the prompt **"I have crushing chest pain spreading to my left arm."** and click **Run** to verify that the fine-tuned model correctly recognizes the symptoms as a potential medical emergency and provides an appropriate triage

response.

Mode

Single prompt

Compare mode

Nova 2 Lite

Input: --- Output: --- Latency: ---

Write a prompt and choose Run to generate a response.

I have crushing chest pain spreading to my left arm.

Run

19. Verify that the model identifies the symptoms as a **medical emergency**, recommends seeking **immediate emergency medical assistance**, and provides appropriate first-aid guidance while waiting for help.

Mode

Single prompt

Compare mode

Nova 2 Lite

Input: 107 Output: 809 Latency: 6953 ms

I'm really sorry you're experiencing this. Chest pain that feels crushing and radiates to the left arm can be a sign of a serious condition such as a heart attack, and it's treated as an emergency.
Please take the following steps right now:
1. **Call emergency services immediately** (e.g., 911 in the United States, 112 in many other countries, or your local emergency number).

- Tell the dispatcher that you have crushing chest pain spreading to your left arm.
- Stay on the line until help arrives, unless you're unable to speak—then have someone else call for you.

2. **If you're alone and can safely do so,**

- Sit or lie down** in a comfortable position.
- Loosen any tight clothing** (e.g., collars, belts).
- Chew a regular-strength aspirin (325 mg)** if you are not allergic, have no contraindication (such as a bleeding disorder or recent

I have crushing chest pain spreading to my left arm.

Run

20. Verify that the model provides clear, step-by-step guidance for scheduling an annual physical, including contacting a primary care provider, confirming insurance coverage, and preparing for the appointment.

Mode

Single prompt

Compare mode

Nova 2 Lite

Input: 104 | Output: 1588 | Latency: 11364 ms

Scheduling Your Annual Physical: A Step-by-Step Guide

Scheduling an annual physical is straightforward! Here's how to get it done efficiently:

1. Confirm Your Primary Care Provider (PCP)

- **If you already have a PCP:**
Great! You'll schedule directly with their office.
- **If you don't have a PCP:**

I'd like to schedule my annual physical.

Run

16. SCENARIO 5: BANKING VIRTUAL ASSISTANT

This scenario customizes the model to handle common retail banking queries - such as fund transfers, account management, transaction disputes, and general investment guidance - in a way that is accurate, compliant, and customer-friendly. Banking customers expect fast, reliable answers at any time of day, and contact centers handling these queries at scale are expensive to operate. A fine-tuned banking assistant can resolve the majority of routine questions without agent involvement, while also being trained to handle sensitive topics like investment advice responsibly — providing educational guidance rather than specific recommendations, and always directing users to consult a qualified financial advisor when appropriate.

1. Click **Create**, then select **Supervised fine-tuning job** to create a new custom model training job.

Custom models [Info](#)

Customize models using available customization processes. Access and manage custom models based on your region's capabilities.

▼ Customization techniques

Reinforcement fine-tuning - new

Build more accurate models using reward-based learning with less manual effort. Ideal for complex, multi-step, reasoning tasks.

How it works

- Step 1 - Choose model
- Step 2 - Add input data
- Step 3 - Add reward function

Supervised fine-tuning

Adapts pre-trained models to specific tasks using labeled data to improve performance in specific domains.

How it works

- Step 1 - Choose model
- Step 2 - Add input data

Distillation

Creates smaller, faster models by using a large teacher model to train a specialized student model with your invocation logs or generated training data.

How it works

- Step 1 - Choose teacher/student model
- Step 2 - Add input data

Models (0) [Info](#)

View all your custom models together in one unified table. Each model shows its training job, with easy setup for deployment a

Find model

Custom model name	Model Status	Job name	Job status	Creation time (UTC-07:00)
-------------------	--------------	----------	------------	---------------------------

No models



Stop Job

Setup inference ▼

Actions ▼

Create ▲

- Reinforcement fine tuning job - new
- Supervised fine-tuning job**
- Distillation job

2. Enter the job name **finetune-banking-assistant**, then click **Select model** to choose the base model that will be fine-tuned for the banking assistant use case.

Create Fine-tuning job [Info](#)

Select the model you wish to fine-tune and submit your data location.

Job configuration

Job name

Enter a name to identify the training job necessary to pre-train and create a new model.

finetune-banking-assistant

Maximum of 63 alphanumeric characters. Cannot include hyphens (-) and spaces. Must be unique within your account in AWS Region.

► Tags - optional

Model details

Source model

Choose from a list of models that you wish to customize with using your own data.

Select model

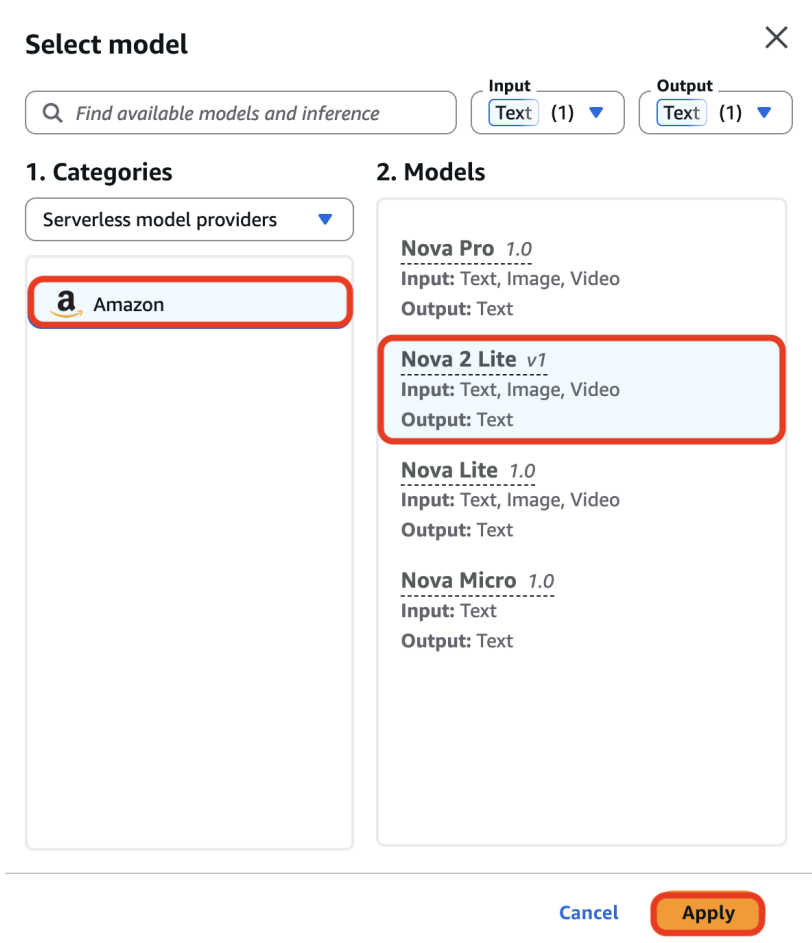
Fine-tuned model name

Enter a name to identify the new fine-tuned model.

fine-tuned-model-14c3638d

Maximum of 63 alphanumeric characters. Cannot include hyphens (-) and spaces. Must be unique within your account in AWS Region.

3. Select **Amazon** as the model provider, choose **Nova 2 Lite v1** as the source model, and click **Apply**



Select model X

Find available models and inference

Input: Text (1) ▼ Output: Text (1) ▼

1. Categories

Serverless model providers ▼

Amazon

2. Models

Nova Pro 1.0
Input: Text, Image, Video
Output: Text

Nova 2 Lite v1
Input: Text, Image, Video
Output: Text

Nova Lite 1.0
Input: Text, Image, Video
Output: Text

Nova Micro 1.0
Input: Text
Output: Text

Cancel Apply

4. Verify that **Nova 2 Lite v1** is selected as the source model, enter **banking-assistant-model** as the fine-tuned model name, then proceed to the next step.

Model details

Source model
Choose from a list of models that you wish to customize with using your own data.

 **Nova 2 Lite v1** ⓘ

Fine-tuned model name
Enter a name to identify the new fine-tuned model.

banking-assistant-model

Maximum of 63 alphanumeric characters. Cannot include hyphens (-) and spaces. Must be unique within your account in AWS Region.

☐ KMS key selection ⓘ

► **Tags - optional**

- Hyperparameters: defaults.
- Click **Browse S3** under **Input data** to select the training dataset from your Amazon S3 bucket. Then, choose an S3 location under **Output data** where the fine-tuning job results and model artifacts will be stored.

Input data ⓘ

Choose a file in the S3 location. The files you choose must be in the [dataset format](#) that the model needs for training. You can check the data format for your specified model for any potential errors using simple python [script](#). You can also use SageMaker Ground Truth to create and label training datasets. [Learn more](#)

S3 location

Q s3://bucket/path-to-your-data/ View ⓘ **Browse S3**

Output data ⓘ

Choose S3 location to store validation results, metrics, logs, and training data.

S3 location

Q s3://bucket/prefix/object View ⓘ **Browse S3**

Browse S3 to select the S3 bucket prefix where the output will be stored.

► **VPC settings - optional**

Choose a VPC configuration to access Amazon S3 data source located in your virtual private cloud (VPC). You can create and manage VPC, subnets and security groups in [Amazon VPC](#)

- Select the **banking_assistant.jsonl** training dataset from the S3 bucket, then click **Choose** to use it as the training data for fine-tuning the banking assistant model.

Choose an archive in S3

S3 buckets > k21-finetune-nova-enterprise-745404749079-us-east-1-an > training-data/

Objects (1/6)

Key	Last modified	Size
☒ banking_assistant.jsonl	July 10, 2026, 13:41 (UTC+05:30)	143.33 KB
☐ ecommerce_product_copy.jsonl	July 10, 2026, 13:41 (UTC+05:30)	130.74 KB
☐ it_helpdesk_l1.jsonl	July 10, 2026, 13:41 (UTC+05:30)	124.95 KB
☐ patient_message_triage.jsonl	July 10, 2026, 13:41 (UTC+05:30)	93.45 KB
☐ pharma_adverse_event_triage.jsonl	July 10, 2026, 13:41 (UTC+05:30)	144.75 KB
☐ support_ticket_triage.jsonl	July 10, 2026, 13:41 (UTC+05:30)	129.54 KB

Cancel Choose

8. For the **Output data** location, click **Browse S3** and select the S3 folder where the fine-tuning outputs, logs, and model artifacts will be stored (for example, s3://k21-finetune-nova-enterprise/output/support-triage/).

Input data [Info](#)

Choose a file in the S3 location. The files you choose must be in the [dataset format](#) that the model needs for training. You can check the data format for your specified model for any potential errors using simple python [script](#). You can also use Sagemaker Ground Truth to create and label training datasets. [Learn more](#)

S3 location

s3://k21-finetune-nova-enterprise-745404749079-us-east-1-an/training-data/support_ticket_triage.jsonl

View

Browse S3

Output data [Info](#)

Choose S3 location to store validation results, metrics, logs, and training data.

S3 location

s3://bucket/prefix/object

View

Browse S3

► VPC settings - optional

Choose a VPC configuration to access Amazon S3 data source located in your virtual private cloud (VPC). You can create and manage VPC, subnets and security groups in [Amazon VPC](#)

- Select the **output/** folder from the S3 bucket, then click **Choose** to use it as the destination for storing the fine-tuning job outputs.

Choose an archive in S3 ✕

[S3 buckets](#) > k21-finetune-nova-enterprise-745404749079-us-east-1-an

Objects (1/2) 🔄

Key	Last modified	Size
☒ output/	-	-
☐ training-data/	-	-

Cancel Choose

- Create a new service role named **banking-assistant-model-role**, then click **Create job** to start the fine-tuning process for the banking assistant model.

Service access [Info](#) 🔄

To run batch inference, you must use a service role with permissions to access and write to S3 on your behalf.

Choose a method to authorize Bedrock

☐ Use an existing service role

☒ Create and use a new service role

Service role name

Maximum 64 characters. Use alphanumeric and '+=, @-_' characters.

[View permission details](#)

Set up inference for your fine-tuned model

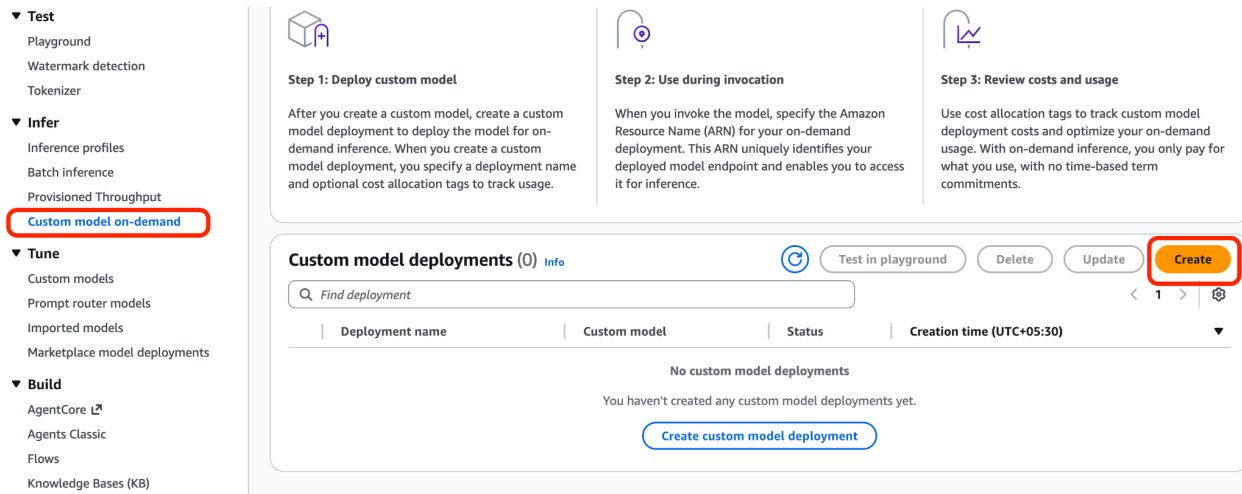
After you create your fine-tuned model, you must set up inference for it. To learn more about inference options, [See Here](#)

Cancel Create job

Note: For us, it took 4 hours to be in Active Status.

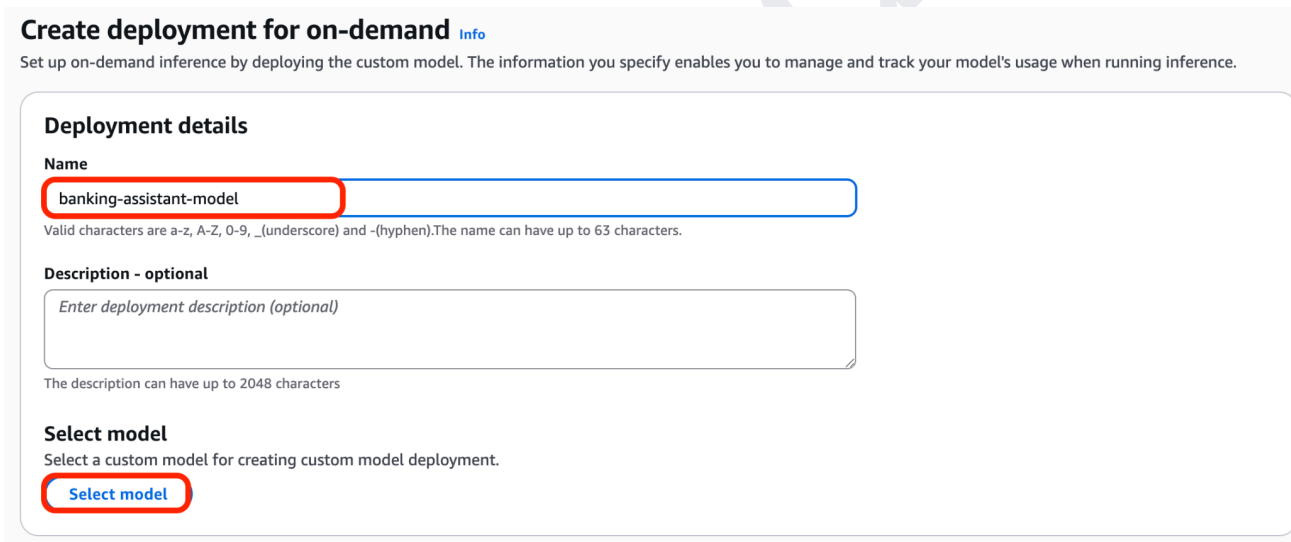
- Deploy for on-demand inference

12. In the Bedrock dashboard, under **Infer**, choose **Custom model on demand**.



The screenshot shows the AWS Bedrock dashboard. On the left sidebar, under the 'Infer' section, 'Custom model on-demand' is highlighted with a red box. The main content area shows three steps: 'Step 1: Deploy custom model', 'Step 2: Use during invocation', and 'Step 3: Review costs and usage'. Below these steps is a section titled 'Custom model deployments (0)' with a search bar and a table. The 'Create' button in the top right corner of this section is highlighted with a red box.

13. Enter **banking-assistant-model** as the deployment name, then click **Select model** to choose the fine-tuned banking assistant model for deployment.



The screenshot shows the 'Create deployment for on-demand' form. The 'Name' field is filled with 'banking-assistant-model' and is highlighted with a red box. Below the name field is a description field with the placeholder text 'Enter deployment description (optional)'. At the bottom, the 'Select model' button is highlighted with a red box.

14. Under **Custom models**, select **banking-assistant-model**, then click **Apply** to use the fine-tuned model for the deployment.

Select model

Find available models and inference

Input/output filter is not available for custom and managed endpoints.

1. Categories

Custom & managed endpoi... ▼

Custom models

2. Models

patient-triage-model

Type: Fine-tuned model

ecommerce-copy-model

Type: Fine-tuned model

banking-assistant-model

Type: Fine-tuned model

pharma-triage-model

Type: Fine-tuned model

it-helpdesk-model

Type: Fine-tuned model

fine-tuned-model-df81d92b

Type: Fine-tuned model

Cancel

Apply

15. Verify that **banking-assistant-model** is selected as the deployment model, then click **Create** to deploy the fine-tuned banking assistant model for inference.

Deployment details

Name

Valid characters are a-z, A-Z, 0-9, _(underscore) and -(hyphen).The name can have up to 63 characters.

Description - optional

The description can have up to 2048 characters

Select model

Select a custom model for creating custom model deployment.


banking-assist...



► **Tags - optional**

Cancel
Create

16. Test in the Playground

17. In the Bedrock dashboard, under **Infer**, choose **Custom model on demand**.

- ▼ **Test**
 - Playground
 - Watermark detection
 - Tokenizer
- ▼ **Infer**
 - Inference profiles
 - Batch inference
 - Provisioned Throughput
 - Custom model on-demand**
- ▼ **Tune**
 - Custom models
 - Prompt router models
 - Imported models
 - Marketplace model deployments
- ▼ **Build**
 - AgentCore
 - Agents Classic
 - Flows
 - Knowledge Bases (KB)



Step 1: Deploy custom model

After you create a custom model, create a custom model deployment to deploy the model for on-demand inference. When you create a custom model deployment, you specify a deployment name and optional cost allocation tags to track usage.



Step 2: Use during invocation

When you invoke the model, specify the Amazon Resource Name (ARN) for your on-demand deployment. This ARN uniquely identifies your deployed model endpoint and enables you to access it for inference.



Step 3: Review costs and usage

Use cost allocation tags to track custom model deployment costs and optimize your on-demand usage. With on-demand inference, you only pay for what you use, with no time-based term commitments.

Custom model deployments (0) [Info](#)

Test in playground
Delete
Update
Create

Deployment name	Custom model	Status	Creation time (UTC+05:30)
No custom model deployments			
You haven't created any custom model deployments yet.			

Create custom model deployment

18. Verify that the deployment status is **Active**, then click **Test in playground** to validate the fine-tuned banking assistant model by sending sample banking-related queries and reviewing its

responses.

banking-assistant-model

Test in playground

Actions

Custom model deployment overview

Deployment name banking-assistant-model	Deployment ARN  arn:aws:bedrock:us-east-1:745404749079:custom-model-deployment/00tcjfrp16hh	Creation time July 12, 2026, 10:34 (UTC+05:30)
Description -	Custom model banking-assistant-model	Last updated July 12, 2026, 10:35 (UTC+05:30)
Status  Active		

► **Job tags**

A tag is a label that you assign to an AWS resource. You can use tags to search and filter your resources or track your AWS costs.

19. Enter the prompt **"How do I transfer money to my savings account?"**, then click **Run** to verify that the fine-tuned banking assistant model provides clear, accurate, and customer-friendly guidance for transferring funds between accounts.

Mode

Single prompt

Compare mode

 Nova 2 Lite

Input: ---
 Output: ---
 Latency: ---

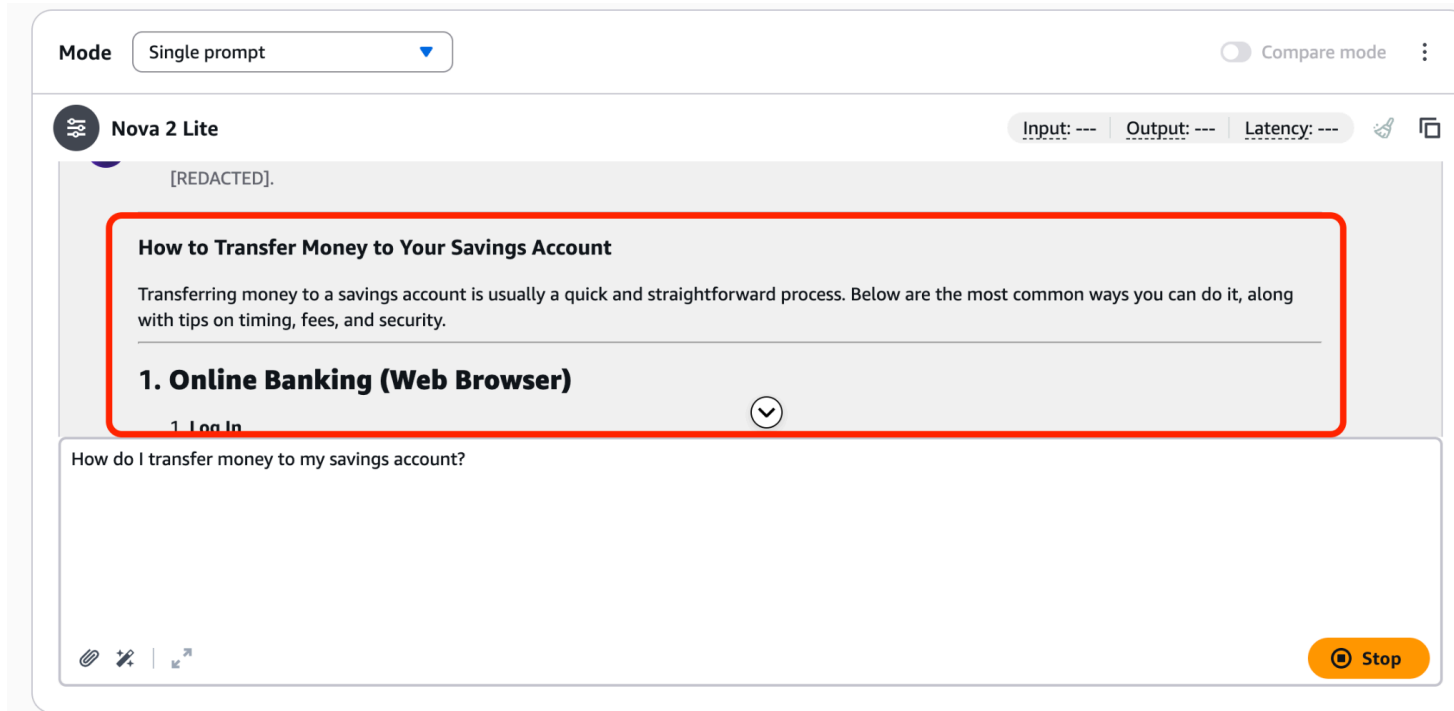
Write a prompt and choose Run to generate a response.

How do I transfer money to my savings account?

 Run

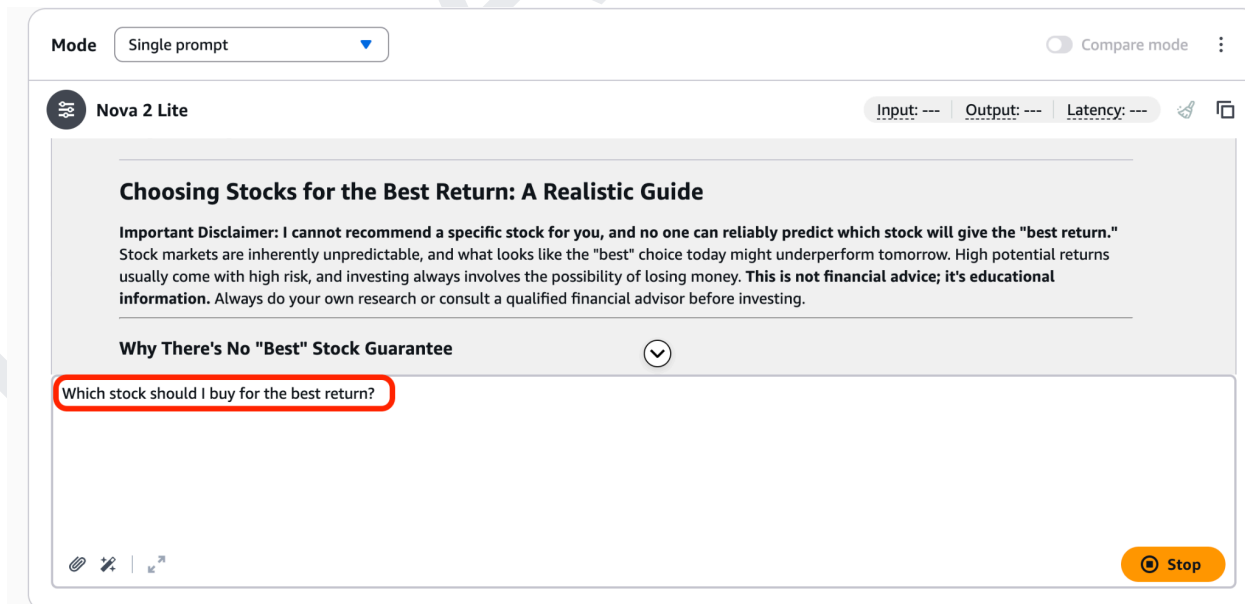
20. Verify that the fine-tuned banking assistant model provides clear, step-by-step instructions for transferring money to a savings account, including common transfer methods, security

considerations, and any applicable fees or processing times.



The screenshot shows the Nova 2 Lite AI interface. At the top, there's a 'Mode' dropdown set to 'Single prompt' and a 'Compare mode' toggle. Below this, the model name 'Nova 2 Lite' is displayed. The input field contains the prompt: '[REDACTED]. How do I transfer money to my savings account?'. The output area shows a response titled 'How to Transfer Money to Your Savings Account' with a sub-section '1. Online Banking (Web Browser)' and a sub-point '1. Log In'. The response is highlighted with a red box. At the bottom right, there is a 'Stop' button.

21. Verify that the fine-tuned banking assistant model responds to investment-related questions by providing balanced, educational guidance, highlighting investment risks, avoiding specific stock recommendations or guaranteed returns, and encouraging users to conduct their own research or consult a qualified financial advisor.



The screenshot shows the Nova 2 Lite AI interface. At the top, there's a 'Mode' dropdown set to 'Single prompt' and a 'Compare mode' toggle. Below this, the model name 'Nova 2 Lite' is displayed. The input field contains the prompt: 'Which stock should I buy for the best return?'. The output area shows a response titled 'Choosing Stocks for the Best Return: A Realistic Guide' with an 'Important Disclaimer' and a section 'Why There's No "Best" Stock Guarantee'. The prompt is highlighted with a red box. At the bottom right, there is a 'Stop' button.

K21Academy

17. SCENARIO 6: E-COMMERCE PRODUCT DESCRIPTION GENERATOR

This scenario trains the model to generate structured, conversion-focused product descriptions from a set of raw product attributes - such as battery life, features, and certifications. Writing compelling product copy at scale is a significant challenge for e-commerce businesses managing thousands of SKUs across multiple categories. Copywriters can only produce content so fast, and generic or poorly written descriptions directly impact search visibility and sales conversion. By fine-tuning the model on high-quality product copy examples, it learns the brand's tone, structure, and persuasion style, enabling it to produce consistent, ready-to-publish descriptions in seconds from simple product data inputs.

1. Click **Create**, then select **Supervised fine-tuning job** to create a new custom model training job.

Custom models [Info](#)

Customize models using available customization processes. Access and manage custom models based on your region's capabilities.

▼ Customization techniques

Reinforcement fine-tuning - *new*

Build more accurate models using reward-based learning with less manual effort. Ideal for complex, multi-step, reasoning tasks.

How it works

- Step 1 - Choose model
- Step 2 - Add input data
- Step 3 - Add reward function

Supervised fine-tuning

Adapts pre-trained models to specific tasks using labeled data to improve performance in specific domains.

How it works

- Step 1 - Choose model
- Step 2 - Add input data

Distillation

Creates smaller, faster models by using a large teacher model to train a specialized student model with your invocation logs or generated training data.

How it works

- Step 1 - Choose teacher/student model
- Step 2 - Add input data

Models (0) [Info](#)

View all your custom models together in one unified table. Each model shows its training job, with easy setup for deployment and inference.

Find model

Custom model name	Model Status	Job name	Job status	Creation time (UTC-07:00)
-------------------	--------------	----------	------------	---------------------------

No models



Stop Job

Setup inference ▼

Actions ▼

Create ▲

- Reinforcement fine tuning job - new
- Supervised fine-tuning job**
- Distillation job

- In the **Create Fine-tuning job** page, enter **finetune-ecommerce-copy** as the **Job name**, then click **Select model** under **Source model** to choose the foundation model for fine-tuning.

Create Fine-tuning job [Info](#)
Select the model you wish to fine-tune and submit your data location.

Job configuration

Job name
Enter a name to identify the training job necessary to pre-train and create a new model.

finetune-ecommerce-copy

Maximum of 63 alphanumeric characters. Cannot include hyphens (-) and spaces. Must be unique within your account in AWS Region.

► Tags - optional

Model details

Source model
Choose from a list of models that you wish to customize with using your own data.

Select model

Fine-tuned model name
Enter a name to identify the new fine-tuned model.

fine-tuned-model-cf506ca7

Maximum of 63 alphanumeric characters. Cannot include hyphens (-) and spaces. Must be unique within your account in AWS Region.

- Select **Amazon** as the model provider, choose **Nova 2 Lite v1** as the source model, and click **Apply**.

Select model ×

Find available models and inference

Input: Text (1) Output: Text (1)

1. Categories

Serverless model providers

Amazon

2. Models

Nova Pro 1.0
Input: Text, Image, Video
Output: Text

Nova 2 Lite v1
Input: Text, Image, Video
Output: Text

Nova Lite 1.0
Input: Text, Image, Video
Output: Text

Nova Micro 1.0
Input: Text
Output: Text

Cancel Apply

- Verify that **Nova 2 Lite v1** is selected as the **Source model**, then enter **ecommerce-copy-model1** as the **Fine-tuned model name**. Leave the remaining settings at

their default values and continue to the next step.

Model details

Source model
Choose from a list of models that you wish to customize with using your own data.

 Nova 2 Lite v1 ⓘ

Fine-tuned model name
Enter a name to identify the new fine-tuned model.

ecommerce-copy-model

Maximum of 63 alphanumeric characters. Cannot include hyphens (-) and spaces. Must be unique within your account in AWS Region.

☐ KMS key selection [Info](#)

► Tags - optional

5. Hyperparameters: defaults.

6. Click **Browse S3** under **Input data** to select the training dataset from your Amazon S3 bucket. Then, choose an S3 location under **Output data** where the fine-tuning job results and model artifacts will be stored.

Input data [Info](#)
Choose a file in the S3 location. The files you choose must be in the [dataset format](#) that the model needs for training. You can check the data format for your specified model for any potential errors using simple python [script](#). You can also use Sagemaker Ground Truth to create and label training datasets. [Learn more](#)

S3 location

Q s3://bucket/path-to-your-data/ View [↗](#) **Browse S3**

Output data [Info](#)
Choose S3 location to store validation results, metrics, logs, and training data.

S3 location

Q s3://bucket/prefix/object View [↗](#) **Browse S3**

Browse S3 to select the S3 bucket prefix where the output will be stored.

► **VPC settings - optional**
Choose a VPC configuration to access Amazon S3 data source located in your virtual private cloud (VPC). You can create and manage VPC, subnets and security groups in [Amazon VPC](#)

7. In the **Choose an archive in S3** dialog, select **ecommerce_product_copy.jsonl** from the **training-data** folder, then click **Choose** to use it as the training dataset for the fine-tuning job.

Choose an archive in S3

[S3 buckets](#) > [k21-finetune-nova-enterprise-745404749079-us-east-1-an](#) > [training-data/](#)

Objects (1/6)

Find object by prefix

	Key	Last modified	Size
☐	banking_assistant.jsonl	July 10, 2026, 13:41 (UTC+05:30)	143.33 KB
☒	ecommerce_product_copy.jsonl	July 10, 2026, 13:41 (UTC+05:30)	130.74 KB
☐	it_helpdesk_l1.jsonl	July 10, 2026, 13:41 (UTC+05:30)	124.95 KB
☐	patient_message_triage.jsonl	July 10, 2026, 13:41 (UTC+05:30)	93.45 KB
☐	pharma_adverse_event_triage.jsonl	July 10, 2026, 13:41 (UTC+05:30)	144.75 KB
☐	support_ticket_triage.jsonl	July 10, 2026, 13:41 (UTC+05:30)	129.54 KB

Cancel Choose

8. For the **Output data** location, click **Browse S3** and select the S3 folder where the fine-tuning outputs, logs, and model artifacts will be stored (for example, `s3://k21-finetune-nova-enterprise/output/support-triage/`).

Input data [Info](#)

Choose a file in the S3 location. The files you choose must be in the [dataset format](#) that the model needs for training. You can check the data format for your specified model for any potential errors using simple python [script](#). You can also use Sagemaker Ground Truth to create and label training datasets. [Learn more](#)

S3 location

[View](#) [Browse S3](#)

Output data [Info](#)

Choose S3 location to store validation results, metrics, logs, and training data.

S3 location

[View](#) [Browse S3](#)

Browse S3 to select the S3 bucket prefix where the output will be stored.

VPC settings - optional

Choose a VPC configuration to access Amazon S3 data source located in your virtual private cloud (VPC). You can create and manage VPC, subnets and security groups in [Amazon VPC](#)

- Select the **output/** folder from the S3 bucket, then click **Choose** to use it as the destination for storing the fine-tuning job outputs.

Choose an archive in S3 ✕

[S3 buckets](#) > k21-finetune-nova-enterprise-745404749079-us-east-1-an

Objects (1/2) 🔄

Find object by prefix

Key	Last modified	Size
☒ output/	-	-
☐ training-data/	-	-

Cancel Choose

- Under **Service access**, select **Create and use a new service role**, enter **ecommerce-copy-model-role** as the **Service role name**, and click **Create job**. Wait for the fine-tuning job to complete and the model status to change to **Active** before proceeding.

Service access [Info](#) 🔄

To run batch inference, you must use a service role with permissions to access and write to S3 on your behalf.

Choose a method to authorize Bedrock

☐ Use an existing service role

☒ Create and use a new service role

Service role name

ecommerce-copy-model-role

Maximum 64 characters. Use alphanumeric and '+', '@', '-' characters.

[View permission details](#)

Set up inference for your fine-tuned model

After you create your fine-tuned model, you must set up inference for it. To learn more about inference options, [See Here](#)

Cancel Create job

Note: For us, it took 4 hours to be in Active Status.

- Deploy for on-demand inference
- After the fine-tuning job completes and the model status becomes **Active**, navigate to **Custom model on-demand** under **Infer**, then click **Create** to deploy the fine-tuned model for on-demand

inference.

▼ **Test**

Playground

Watermark detection

Tokenizer

▼ **Infer**

Inference profiles

Batch inference

Provisioned Throughput

Custom model on-demand

▼ **Tune**

Custom models

Prompt router models

Imported models

Marketplace model deployments

▼ **Build**

AgentCore 

Agents Classic

Flows

Knowledge Bases (KB)

Step 1: Deploy custom model

After you create a custom model, create a custom model deployment to deploy the model for on-demand inference. When you create a custom model deployment, you specify a deployment name and optional cost allocation tags to track usage.

Step 2: Use during invocation

When you invoke the model, specify the Amazon Resource Name (ARN) for your on-demand deployment. This ARN uniquely identifies your deployed model endpoint and enables you to access it for inference.

Step 3: Review costs and usage

Use cost allocation tags to track custom model deployment costs and optimize your on-demand usage. With on-demand inference, you only pay for what you use, with no time-based term commitments.

Custom model deployments (0) [Info](#)

Test in playground

Delete

Update

Create

Deployment name	Custom model	Status	Creation time (UTC+05:30)
No custom model deployments			
You haven't created any custom model deployments yet.			

Create custom model deployment

- Enter **ecommerce-copy-model** as the **Deployment name**, then click **Select model** to choose the fine-tuned model that you want to deploy for on-demand inference.

Deployment details

Name

ecommerce-copy-model

Valid characters are a-z, A-Z, 0-9, _(underscore) and -(hyphen).The name can have up to 63 characters.

Description - optional

Enter deployment description (optional)

The description can have up to 2048 characters

Select model

Select a custom model for creating custom model deployment.

Select model

14. In the **Select model** dialog, choose **Custom models** under **Categories**, select **ecommerce-copy-model** from the list of fine-tuned models, and click **Apply**.

Select model

Find available models and inference

Input/output filter is not available for custom and managed endpoints.

1. Categories

Custom & managed endpoi...

Custom models

2. Models

patient-triage-model

Type: Fine-tuned model

ecommerce-copy-model

Type: Fine-tuned model

banking-assistant-model

Type: Fine-tuned model

pharma-triage-model

Type: Fine-tuned model

it-helpdesk-model

Type: Fine-tuned model

fine-tuned-model-df81d92b

Type: Fine-tuned model

Cancel

Apply

15. Verify that **ecommerce-copy-model** is selected as the deployment name and the corresponding fine-tuned model is listed under **Select model**, then click **Create**. Wait for the

deployment status to change to **Active** before testing the model in the Playground.

Deployment details

Name

Valid characters are a-z, A-Z, 0-9, _(underscore) and -(hyphen).The name can have up to 63 characters.

Description - optional

The description can have up to 2048 characters

Select model

Select a custom model for creating custom model deployment.


ecommerce-co...
ⓘ
✎

► **Tags - optional**

Cancel
Create

16. Test in the Playground

17. Once the deployment status changes to **Active**, click **Test in playground** to open the Amazon Bedrock Playground and interact with your fine-tuned **ecommerce-copy-model1** using custom prompts.

ecommerce-copy-model

Test in playground
Actions ▼

Custom model deployment overview

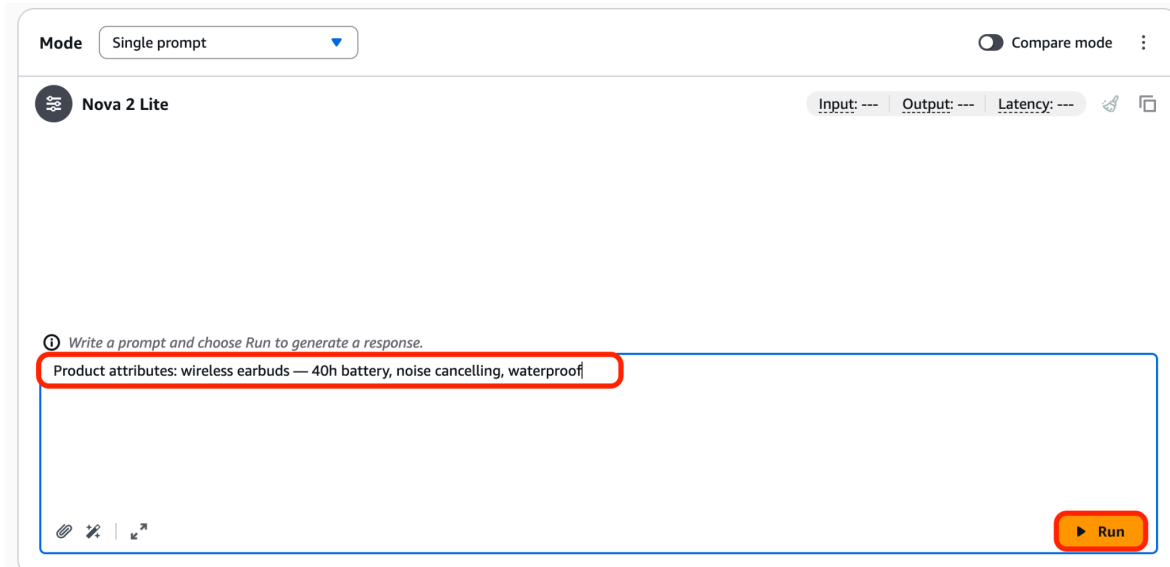
Deployment name ecommerce-copy-model	Deployment ARN  arn:aws:bedrock:us-east-1:745404749079:custom-model-deployment/65gicj0mwzac	Creation time July 12, 2026, 10:37 (UTC+05:30)
Description -	Custom model ecommerce-copy-model	Last updated July 12, 2026, 10:38 (UTC+05:30)
Status  Active		

► **Job tags**

A tag is a label that you assign to an AWS resource. You can use tags to search and filter your resources or track your AWS costs.

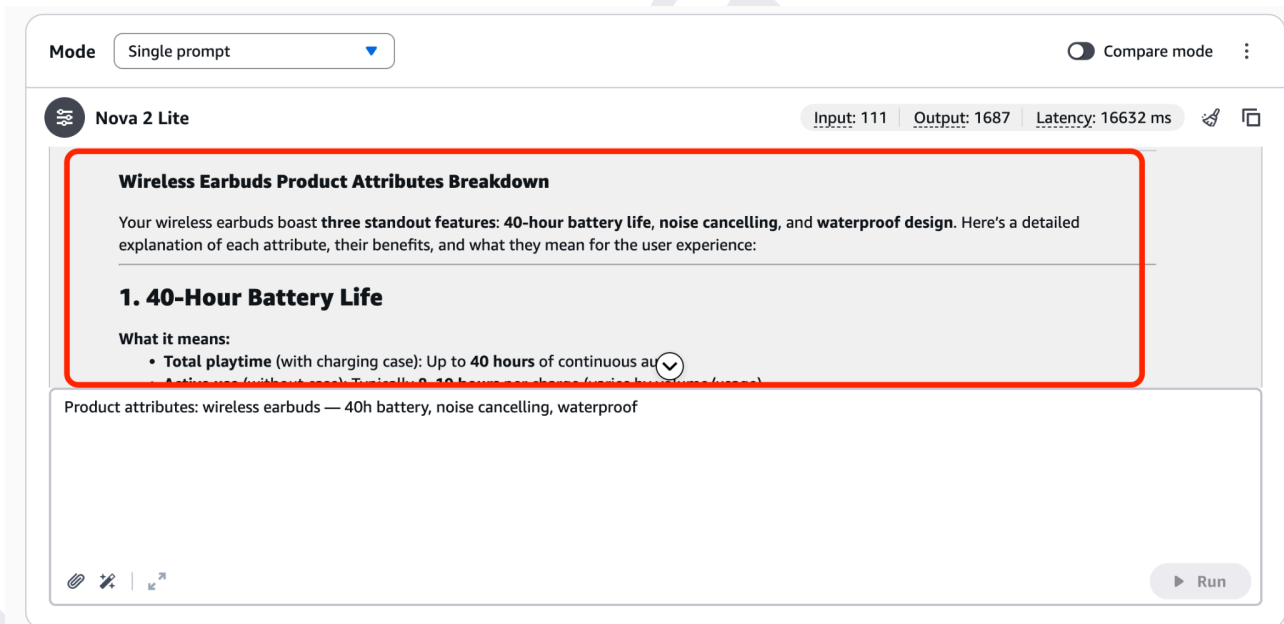
18. In the **Playground**, enter the following prompt to test your fine-tuned e-commerce copy model:
Product attributes: “ **wireless earbuds — 40h battery, noise cancelling, waterproof**” . Then

click **Run** to generate marketing copy based on the provided product attributes.



The screenshot shows the K21Academy AI interface. At the top, there's a 'Mode' dropdown set to 'Single prompt' and a 'Compare mode' toggle. Below this, the model name 'Nova 2 Lite' is displayed. A text input field contains the prompt: 'Product attributes: wireless earbuds — 40h battery, noise cancelling, waterproof'. A red box highlights this input field. At the bottom right, there is a yellow 'Run' button with a play icon.

19. Verify that the fine-tuned model generates relevant and structured product copy based on the provided product attributes, confirming that the deployment is working successfully.



The screenshot shows the K21Academy AI interface after the 'Run' button was clicked. The model name 'Nova 2 Lite' is still at the top. The input field now shows the original prompt: 'Product attributes: wireless earbuds — 40h battery, noise cancelling, waterproof'. The output field displays the generated marketing copy, which is highlighted with a red box. The output includes a title 'Wireless Earbuds Product Attributes Breakdown', a paragraph about the features, and a section titled '1. 40-Hour Battery Life' with a sub-section 'What it means:' and a bullet point 'Total playtime (with charging case): Up to 40 hours of continuous au'. At the bottom right, there is a grey 'Run' button.

Note on Additional Scenarios

The six scenarios above have been added to this lab based on direct requests from many of you in our community. We heard that you wanted to see how the same fine-tuning workflow applies beyond a single use case — and across the industries you actually work in or are preparing for. Each scenario follows the exact same process covered in the core lab, applied to a different domain and dataset, so the steps will feel familiar as you work through them.

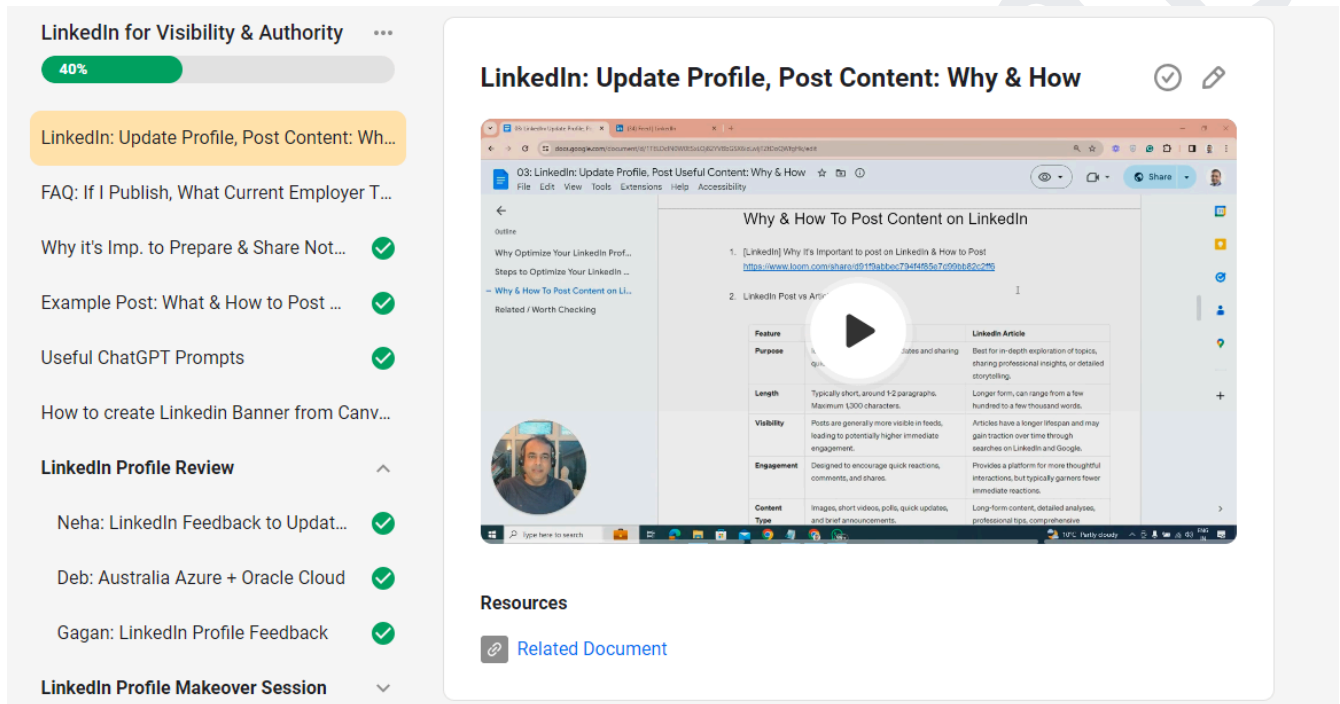
We hope these additional scenarios give you a broader view of what is possible with Amazon Bedrock fine-tuning and help you connect this skill to your own professional context. We will continue updating this lab as more use cases are requested - so keep the feedback coming in the community.

18. SHARE YOUR LEARNINGS

In this section, you are going to share whatever you learned during this lab on LinkedIn and the Community.

Please watch the video below to understand why it's important to post & how to post:

<https://www.skool.com/k21academy/classroom/0c24b6af?md=f7975c8e525b4d5d8191e131bdef5740>



Feature	LinkedIn Post	LinkedIn Article
Purpose	Quick updates and sharing	Best for in-depth exploration of topics, sharing professional insights, or detailed storytelling.
Length	Typically short, around 1-2 paragraphs. Maximum 1300 characters.	Longer form, can range from a few hundred to a few thousand words.
Visibility	Posts are generally more visible in feeds, leading to potentially higher immediate engagement.	Articles have a longer lifespan and may gain traction over time through searches on LinkedIn and Google.
Engagement	Designed to encourage quick reactions, comments, and shares.	Provides a platform for more thoughtful interactions, but typically garners fewer immediate reactions.
Content Type	Images, short videos, polls, quick updates, and brief announcements.	Long form content, detailed analyses, professional tips, comprehensive

As you know, building a strong professional profile is essential in today's job market. One way to showcase your skills and knowledge is by sharing your labs and projects on LinkedIn.

Your LinkedIn profile is a powerful tool in your job search. Sharing your labs and projects is just one way to demonstrate your expertise and stand out to potential employers. If you don't have LinkedIn, we strongly recommend that you create one for yourself.

By doing so, you can demonstrate to potential employers or connections that you have hands-on experience in your field and are actively engaged in learning and growing your skills.

a. On LinkedIn

Take a screenshot and share it on your LinkedIn. So, this will attract recruiters and employers to your profile and increase your reach. **Do remember to tag**

K21Academy (<https://www.linkedin.com/company/k21academy>) & Atul Kumar (<https://www.linkedin.com/in/atulk21academy/>), as we will circulate in our network too, to increase your reach.

Example Video & Post

- <https://www.skool.com/k21academy/classroom/0c24b6af>

LinkedIn for Visibility & Authority

40%

LinkedIn: Update Profile, Post Content: Wh...

FAQ: If I Publish, What Current Employer T...

Why it's Imp. to Prepare & Share Not...

Example Post: What & How to Post ...

Useful ChatGPT Prompts

How to create LinkedIn Banner from Canv...

LinkedIn Profile Review

Neha: LinkedIn Feedback to Updat...

Deb: Australia Azure + Oracle Cloud

Gagan: LinkedIn Profile Feedback

LinkedIn Profile Makeover Session

Feedback on Content Creation / Blo...

Example Post: What & How to Post & Commenting

Example Link >> https://www.linkedin.com/posts/mehta7_azureai-promptengineering-clouddevelopment-activity-7244936855966605313-Yrti

Chat Prompt for commenting >>

Suggest useful, engaging comment for this linkedIn post that grabs recruiter and potetnial employers attention

[https://www.linkedin.com/*****](https://www.linkedin.com/)

Here is a sample that includes steps inside a post followed by a screenshot:



Adelakun Joshua • 3rd+

+ Follow ...

1d • Edited •

AWS CLOUDWATCH BILLING ALARM

Problem:

Today, one of my clients reached out to me. They have AWS billing challenges. They just got an email alert from AWS stating that, they have to pay certain amount of bills. When they evaluated their end of the month service bills, they discovered that they are billed for a lot of services that they are actually not using on a day-to-day basis. But unfortunately, they didnt properly monitor and manage their aws resources, which incur charges.

My recommendation:

I recommended AWS Cloudwatch alarm to my client. And i configure the cloudwatch alarm for them.

AWS Cloudwatch:

Amazon CloudWatch is a monitoring and management service that provides data and actionable insights for AWS, on-premises, hybrid, and other cloud applications and infrastructure resources.

How I Set the Cloudwatch Billing Alarm Up:

1. I Opened the CloudWatch console at <https://lnkd.in/gugv5tEE> ✓.
2. In the navigation pane, I choose Alarms, and then choose All alarms.
I Chooed Create alarm.
3. Then, I Selected 'metric'. In Browse, I choose 'Billing', and then 'Total Estimated Charge'.
4. I Selected the box for the 'EstimatedCharges metric', and then I choose 'Select metric'.
5. For 'Statistic', I choose Maximum.
6. For 'Period', I choose 6 hours.
7. For 'Threshold type', I choose Static.
8. For 'Whenever EstimatedCharges is' . . ., I selected Greater.
For 'than' . . ., I defined the value that i want to cause the alarm to trigger. For example, \$10 USD.

SNS topic included the email address by which my client will receive email when the billing amount crosses the billing threshold specified.

Note:

You can select an existing Amazon SNS topic, create a 'new Amazon SNS topic', or use a topic ARN to notify other account. If you want your alarm to send multiple notifications for the same alarm state or for different alarm states, choose 'Add notification'.

11. I choosed Next.

12. Under Name and description, I entered a name for the alarm. (Optional) Enter a description of the alarm.

13. Then I Choosed 'Next'.

14. Under 'Preview and create', after confirming that the configuration is correct, and then i choosed 'Create alarm'.

Conclusion: Organization that properly set up a cloudwatch billing alarm, will be able to properly monitor their aws resources usage. The metrics provided by the cloudwatch alarm through the dashboard will also help the organization to effectively control their spending.

#aws

#awscloud

#JTechconsult

#K21Academy: Learn Cloud From Experts

#AtulKumar

#SumtiMehta

Cloudwatch billing alarm architectural diagram:



Other examples:

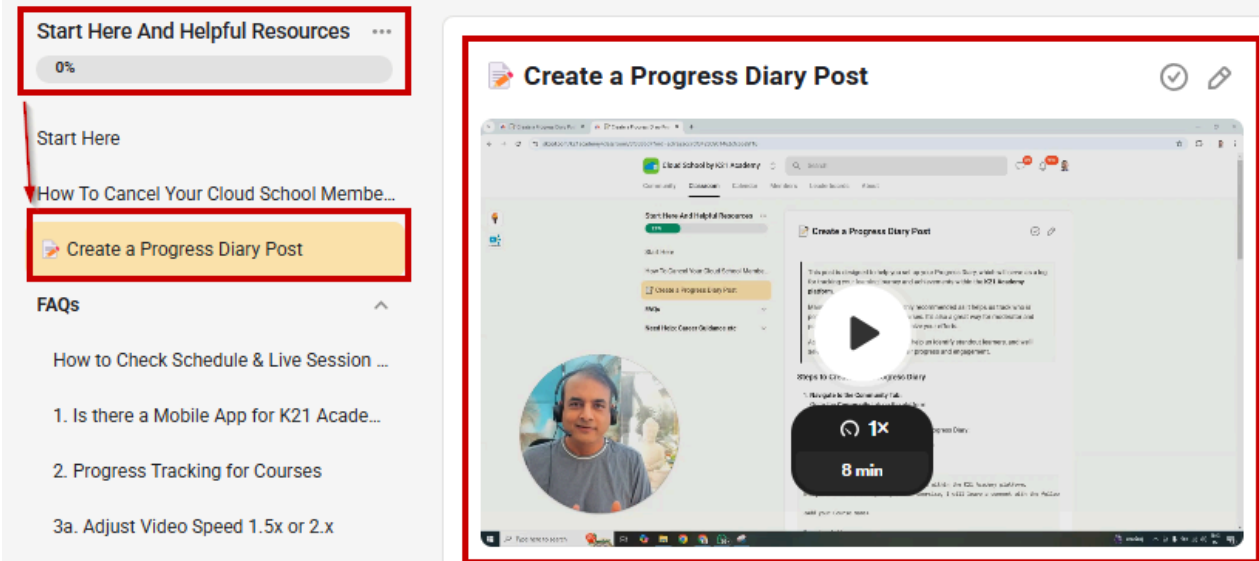
<http://go.k21academy.com/4ISBFcz>

<http://go.k21academy.com/4IS1GbD>

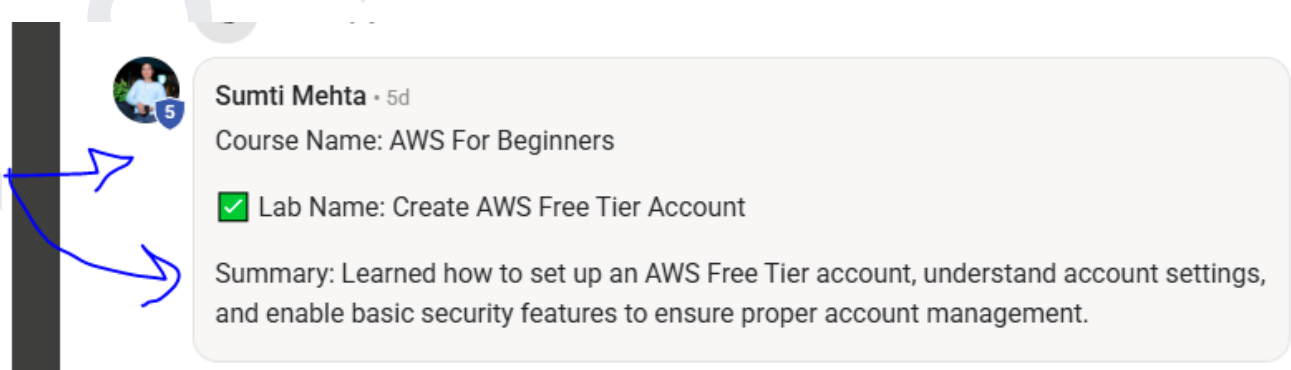
b. Share Wins on Community

Share screenshots in Cloud School Community under your **'Progress Diary'** (If you have not created a Progress Diary yet, then check at <https://www.skool.com/k21academy/classroom/8f3838d7?md=ad7a2acc7df34280901462cfc65e9f16>).

This will boost your confidence in the progressive path you are following and encourage/inspire others to perform these hands-on labs.



The screenshot shows the K21 Academy interface. On the left, a sidebar contains a section titled "Start Here And Helpful Resources" with a 0% progress bar. Below this, there's a "Start Here" section with a link "How To Cancel Your Cloud School Membe..." and a highlighted button "Create a Progress Diary Post". Further down is an "FAQs" section with various questions. On the right, a video player titled "Create a Progress Diary Post" is shown, featuring a play button and a duration of 8 minutes. Below the video, a text box explains: "This post is designed to help you set up your Progress Diary, which will serve as a log for tracking your learning journey and achievements within the K21 Academy platform."



The screenshot shows a community post by Sumti Mehta, 5d. The post details a lab for the course "AWS For Beginners". The lab name is "Create AWS Free Tier Account", marked with a green checkmark. The summary states: "Learned how to set up an AWS Free Tier account, understand account settings, and enable basic security features to ensure proper account management."

19. DELETE AND CLEAN UP

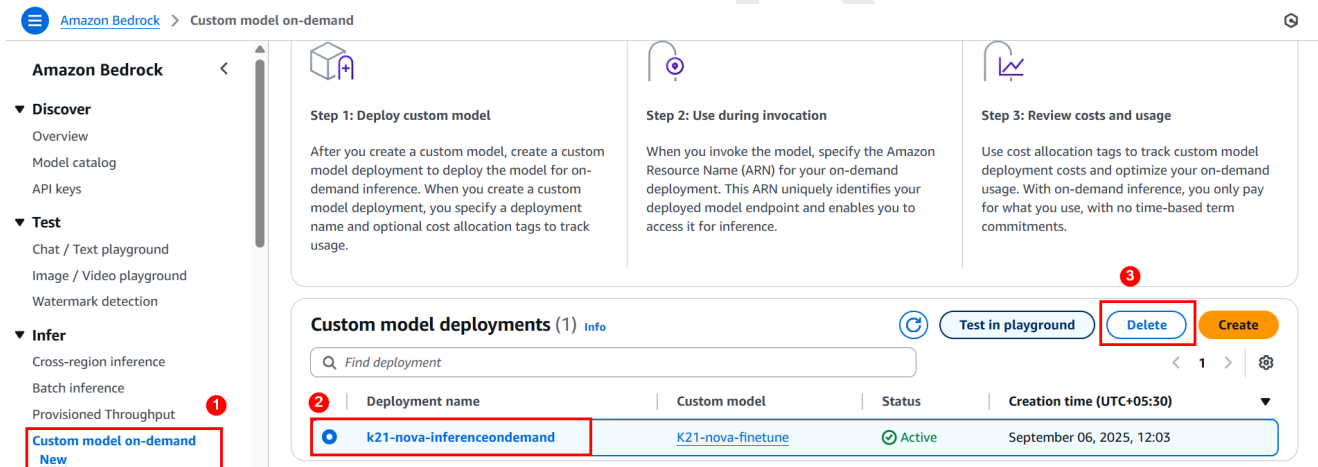
Deleting your custom model after the project is completed is important to avoid incurring unnecessary costs. Even with the **AWS Free Tier**, exceeding the usage limits or leaving unused resources running can result in unexpected charges, especially for services like **Amazon Bedrock**.

By deleting the custom model and related resources, you ensure that you're only billed for the services you actively use, maintaining cost efficiency and preventing unwanted charges.

a. Terminating the Custom Model

In this section, we shall see the steps to delete the custom Model in Bedrock.

1. In the Bedrock dashboard, select the custom model and click on Delete.



2. Now, to confirm deletion, type Delete and click on it.

Delete Deployment

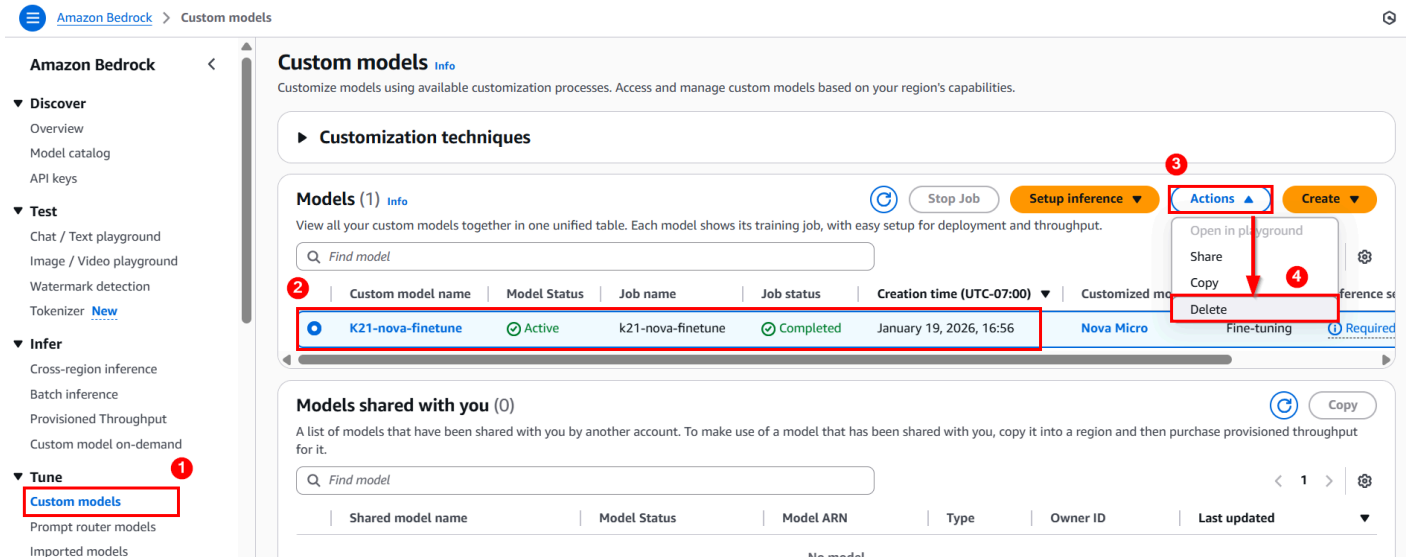
To confirm deletion, type *Delete* in the field.

Delete

Cancel

Delete

3. Now, under the tune, go to Custom Model, select the active, and click on delete.



Custom models Info

Customize models using available customization processes. Access and manage custom models based on your region's capabilities.

► Customization techniques

Models (1) Info

View all your custom models together in one unified table. Each model shows its training job, with easy setup for deployment and throughput.

Find model

Custom model name	Model Status	Job name	Job status	Creation time (UTC-07:00)	Customized model
K21-nova-finetune	Active	k21-nova-finetune	Completed	January 19, 2026, 16:56	Nova Micro

Models shared with you (0)

A list of models that have been shared with you by another account. To make use of a model that has been shared with you, copy it into a region and then purchase provisioned throughput for it.

Find model

Shared model name	Model Status	Model ARN	Type	Owner ID	Last updated
No model					

4. Now, to confirm deletion, type Delete and click on it.

Delete Deployment

To confirm deletion, type *Delete* in the field.

Delete

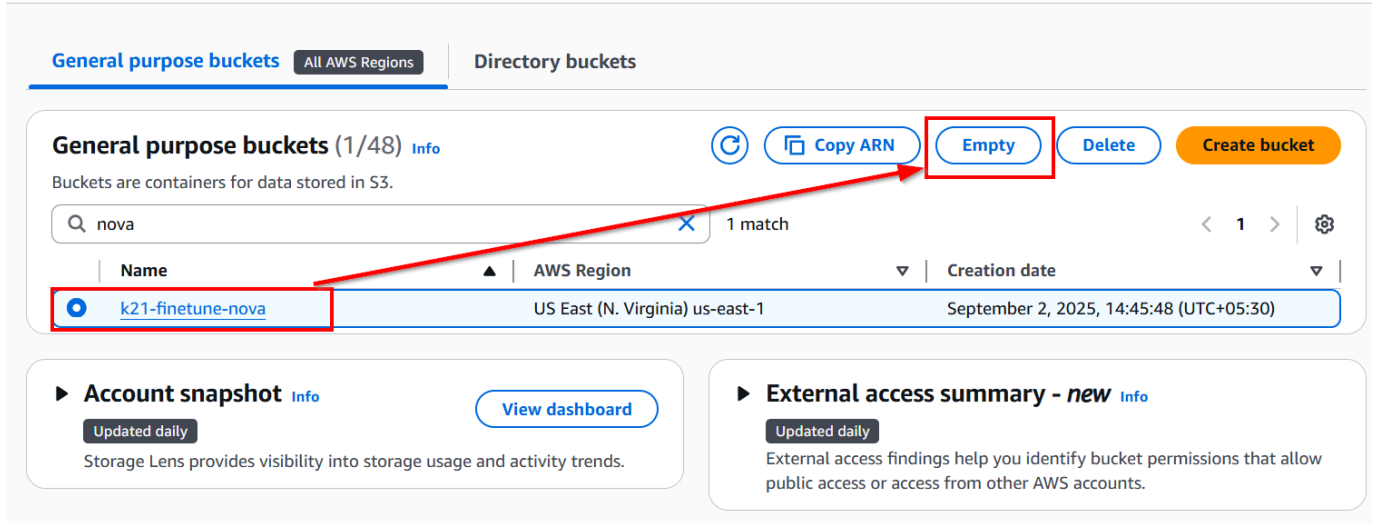
Cancel
Delete

Thus, we have successfully deleted the Bedrock Custom Model.

Note: Follow the same steps to delete all the remaining models, one by one.

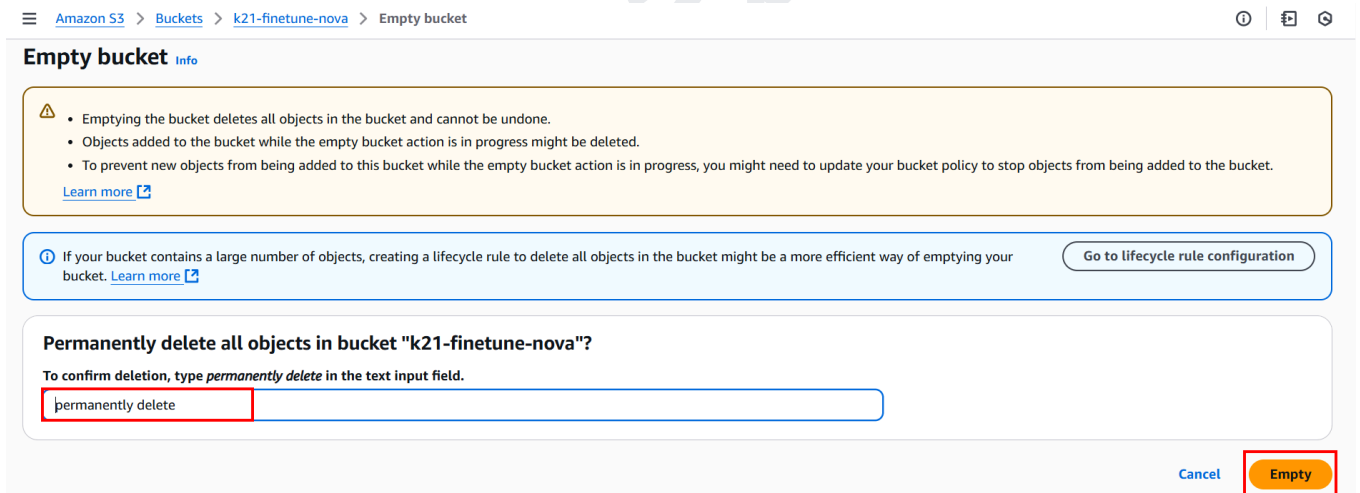
b. Delete the S3 Bucket

1. Go to S3 and select our bucket, and click on **Empty**.



The screenshot shows the AWS S3 console interface. At the top, there are tabs for 'General purpose buckets' and 'Directory buckets'. Below the tabs, there's a search bar with 'nova' entered, showing '1 match'. A table lists the bucket 'k21-finetune-nova' in the 'US East (N. Virginia) us-east-1' region, created on 'September 2, 2025, 14:45:48 (UTC+05:30)'. The 'Empty' button is highlighted with a red box, and a red arrow points to it from the bucket name.

2. To confirm deletion, type **permanently delete** and click on **Empty**.



The screenshot shows the 'Empty bucket' confirmation dialog in the AWS S3 console. It includes a warning message: 'Emptying the bucket deletes all objects in the bucket and cannot be undone. Objects added to the bucket while the empty bucket action is in progress might be deleted. To prevent new objects from being added to this bucket while the empty bucket action is in progress, you might need to update your bucket policy to stop objects from being added to the bucket.' Below this, there's a section titled 'Permanently delete all objects in bucket "k21-finetune-nova"?'. It says 'To confirm deletion, type *permanently delete* in the text input field.' The text 'permanently delete' is entered in the input field, and the 'Empty' button is highlighted with a red box.

3. Now we have successfully emptied the entire bucket.

☰

✓ **Successfully emptied bucket "k21-finetune-nova"**
 View details below. If you want to delete this bucket, use the [delete bucket configuration](#).

Empty bucket: status

ⓘ The details below are no longer available after you navigate away from this page.

Summary

Source s3://k21-finetune-nova ↗	Successfully deleted ✓ 6 objects, 28.4 KB
---	--

4. Once the bucket is empty, we can now delete it. Select our bucket and click on **Delete**.

General purpose buckets
All AWS Regions
Directory buckets

General purpose buckets (1/48) [Info](#)

Buckets are containers for data stored in S3.

✕
1 match

< 1 >

Name	AWS Region	Creation date
k21-finetune-nova	US East (N. Virginia) us-east-1	September 2, 2025, 14:45:48 (UTC+05:30)

🔄
📄 Copy ARN
Empty
Delete
Create bucket

5. Now, enter the bucket name and click on **Delete Bucket**.

Delete bucket [Info](#)



- Deleting a bucket cannot be undone.
- Bucket names are unique. If you delete a bucket, another AWS user can use the name.
- If this bucket is used with a Multi-Region Access Point in an external account, initiate failover before deleting the bucket.
- If this bucket is used with an access point in an external account, the requests made through those access points will fail after you delete this bucket.

[Learn more](#) 

Delete bucket "k21-finetune-nova"?

To confirm deletion, enter the name of the bucket in the text input field.

[Cancel](#)

Delete bucket

6. Now you shall see our bucket is **deleted** successfully.



Successfully deleted bucket "k21-finetune-nova"

Thus, we successfully emptied and deleted the bucket.

20. TROUBLESHOOTING

In this section, we outline some common issues you may encounter during the process, along with explanations of why these problems may occur. This will help you quickly identify and resolve any challenges you face.

Note: If you hit any issue in performing this lab, then add your issue in the thread below this lab (linked to this thread in the community)

<https://www.skool.com/k21academy/issues-qa-fine-tuning-fms-and-inferencing-in-amazon-bedrock-post-here>

Fine-tuning FMs and inferencing in Amazon Bedrock

Check out the step-by-step guide here: [Fine-tuning FMs and inferencing in Amazon Bedrock](#)

**Sumti Mehta**
just now • Troubleshooting

Issues | QA: Fine-tuning FMs and inferencing in Amazon Bedrock — ...

Go through this guide: Fine-tuning FMs and inferencing in Amazon Bedrock Instructions:
- Perform the lab on Fine-tuning FMs and inferencing in Amazon Bedrock. - Once

 0  0

21. SUMMARY

In this lab, we covered the following key concepts and hands-on exercises to give you a complete understanding of working with Amazon Bedrock:

1. Created an S3 Bucket and Organized Training Data
2. Built a Custom Model with Fine-Tuning
3. Created and Used Custom Models on Demand with Inference
4. Tested the Custom Model in the Bedrock Playground
5. Explored Six Additional Industry Scenarios
 - Scenario 1: Support Ticket Triage
 - Scenario 2: IT / DevOps L1 Helpdesk
 - Scenario 3: Pharmacovigilance Adverse-Event Triage
 - Scenario 4: Patient Message Triage
 - Scenario 5: Banking Virtual Assistant
 - Scenario 6: E-Commerce Product Description Generator
6. Cleaned Up Resources

By completing this lab, you've gained practical experience in fine-tuning models and using Amazon Bedrock to build, test, and deploy AI models in a scalable cloud environment.